GE Medical Systems

Technical Publications

Direction 5133316

Revision 9

Signa® EXCITE 1.5T System Upgrade from 11.x to 12.x

Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation **“damage in shipment”** written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003

GE Medical Systems

GE Medical Systems: Telex 3797371
P.O. Box 414, Milwaukee, Wisconsin 53201 U.S.A.
(Asia, Pacific, Latin America, North America)

GE Medical Systems — Europe: Telex 261794
Shortlands, Hammersmith, London W6 8BX U.K.

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST–HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST–HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

警告

・このサービスマニュアルには英語版しかありません。
・ＧＥＭＳ以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
・このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
1	Sep 30, 2004	Initial Release.
2	Apr 27, 2005	Sec 2 – Updated the upgrade fow process to reflect changes in Sections 6 and 8. Added IIQ Kit check, “saveINFO”, and ACGD to HFD–S Upgrade procedure. Sec 4 – Updated the “added” cable maps by adding optional remote MRU module, cables, and connections. Changed designations of Runs 749 and 1011 from disconnect to remove (MRIhc06669). Added Non–LCC “removal” and “added” cable maps. Sec 6 – Reorgainzed section to clarify the various magnet enclosure upgrades. Added instructions for removing gradient cable foam and installing clamp blocks (MRIhc06323). Added instructions for upgrading Horizon style enclosures for LPCA and cable track (MRIhc05711). Sec 7 – Added Legacy start up instructions. Sec 8 – Added specific section for removing TNF modules and cables (MRIhc06354). Added optional location for RF Splitter box (MRIhc06149). Added information on installing CAN Terminator (MRIhc05769). Sec 10 – Added information on upgrading ACGD to HFD–S.
3	Jun 30, 2005	Sec 6 – Added de–installation of single–piece long bridge for WideOpen. Added IIQ Kit (M3090NE) installation guidance for WideOpen and S–Series magnets. Added new plastic mounting bracket for Runs 743, 744, and RF Drive Cables in Rear Pedestal. Sec 8 – Added step for moving 50 Ohm terminator from Mux1 to Mux2 J6 on RRF module.
4	Dec 6, 2005	Sec 4 –Corrected location of new cable, Run 1180, in all the “added” cable maps (MRIhc12321). Corrected GOC Power Strip connection designators. Sec 8 – Replaced photo of rear of RRF Module with one showing correct cable routing.
5	Aug 18, 2006	Sec 2 – Added “Power On” and “Host Monitor Setup and Cal” procedures. Sec 6 – Revised the fastener mounting hardware for the 16 Channel switch mounting plate installation. Added instructions to add two new mounting holes to new RF Drive cable plastic brakcet (MRIhc09538). Added alternative mounting for Ground Isolatoin Filter Boxes (MRIhc09510). Sec 8 – Revised the instructions for installing new boards in RRF Chassis and added the alternative RF–DIF board, 5250028.
6	Oct 26, 2006	Section 6: Updated PAC replacement instructions with information on removing the PAC Control Housing (MRIhc17607, MRIhc18836).
7	Mar 7, 2007	Section 6: Removed PAC 2B part number. Section 8: Revised 8 Channel installation procedures by updating MUX board part number and adding procedure for installing UTNS3, RCVR2, MUX, & EXCR2 boards (CQA 13068576). Added instructions for storing Service Cable Kit.
8	Aug 10, 2007	Section 6: Updated the Hybrid Splitter upgrade, Section 6–4–7, by adding the Netcom Hybrid Splitter which has different Isolation Filter mounting brackets than the existing Teledyne Hybrid Splitter.
9	Jun 17, 2009	Section 7: Removed instructions on installation of rating plates (PQR 13246367).

LIST OF EFFECTIVE PAGES

* This revision number/letter corresponds to the indicated document’s revision control system.

TABLE OF CONTENTS

SECTION 1 – GETTING STARTED

1–1 UPGRADE INTRODUCTION

This publication documents the mechanical installation information for upgrading a Signa® 1.5T EXCITE (Release 11.x, EXCITE II) to Signa 1.5T EXCITE HD (Release 12.x, EXCITE III). Refer to illustration 1–1 for pictorial overview of upgraded system.

WARNING!

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

1–2 SITE READY CHECK

Pre–installation work must be completed before equipment is delivered to avoid delays and confusion.

Tools and test equipment to install and calibrate the Signa 1.5T EXCITE system are listed in the *Signa 1.5T EXCITE HD Upgrade Pre–Installation* manual, Section 10, TOOLS AND TEST EQUIPMENT.

Service installation tools used in this manual that may not be included in above references lists include:

- 46–301450G1 Fiber Optic Cable Connector Repair kit

1–3 PRE–DELIVERY FDO CHECKS FOR SIGNA EXCITE

Before the shipment is made, Be sure that FDO review has been completed by consultation with the Signa 1.5T Upgrade Specialist. Also verify the Surface coil compatibility issues have also been resolved.

1–4 UPGRADE SCHEDULE CONSIDERATIONS

FMIs, periodic maintenance, installation of prerequisite upgrades and/or options will directly add to the man–hour and system down–time total. The IIS key will need to be re–generated. Be sure that all planned activity is considered when predicting the return of system to customer. Also verify Application Specialist is scheduled at the end of the installation to instruct users. Additional time must also be scheduled to instruct users following completion of upgrade.

1–5 PRODUCT DELIVERY INSTRUCTIONS AND PACKING

The “Shipping Document” lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short.

Labels that are attached to the outside of packing boxes summarize box contents. PDI (Product Delivery Instructions) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number. Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with “Technical Publication” boxes for a list of delivered documents.

1–6 DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation **“damage in shipment”** written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

1–7 DISCARDED MATERIAL RETURN POLICY

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1–7–1 MATERIAL POLICY OVERVIEW

1–7–1–1 Returned Material

Per GE Healthcare Policy 3.9: all requested material is to be returned, as directed, intact and promptly either to GoldSeal or Renewable Resources. Do not remove or swap any parts from the material to be returned.

1–7–1–2 Disposed Material

Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items in the per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.

1–7–2 MATERIAL TO BE RETURNED

Return all parts, cables, modules, and equipment deinstalled for the EXCITE Release 12.x upgrade to GoldSeal which includes:

- Driver Module
- SRI 2
- 8 Channel Switch
- LPCA
- Front and Rear Split Bridge
- Cable Take–Up
- All replaced Circuit Boards
- RF Body Coil (TRM only)

1–7–3 RETURN KIT

Packaging and/or transportation details can be obtained through the local Installation Specialist. If unavailable, call GoldSeal. The following material may have been provided to facilitate return of material.

- Shipping labels
- Waybill
- Address labels
- Reuse appropriate containers that new hardware and modules were shipped to site for return of above material.
- Packing material

1–7–4 DEINSTALLATION PROCEDURES

Refer to instructions in this manual for deinstallation procedures of site.

1–7–5 REGION SPECIFIC RETURN INFORMATION

1–7–5–1 Sites in Americas

A GoldSeal return pre–printed BAX waybill and mailing label(s) may be included with the upgrade kit. The upgrade returns kit (p/n 2352804) consists of:

- 2358604 – Waybill
- 2357240 – GoldSeal Return Labels
- 2357241– Upgrade Return Envelope (waybill and return labels)

Gold Seal Address and contact for coordination of transportation arrangements and packaging:

GE Medical Systems – GoldSeal
3114 North Grandview Blvd
Waukesha, WI 53188
Phone: 1–877–251–4468 or 262–513–4147

1–7–5–1 Sites in Americas (Continued)

For non–GoldSeal Returns, utilize GEMS Renewable Resources. For coordination of transportation arrangements and packaging:

Phone: (414)–747–6997 or Dial Comm 8*579–6997
Fax: (414)–747–6855
Field ASPEN: 1–800–525–1516, Box #26041

1–7–5–2 Sites in Asia

At this time, system components are not returned for systems in Asia, but whole systems must be returned from complete system (forklift) upgrades.

Return procedure for GEMS–Asia countries:

Please contact your Regional Service Headquarters to get instructions.

1–7–5–3 Sites in Europe

The GoldSeal Returns Coordinator contacts each site individually and reviews the material to be returned and the logistics for the return. A list of material and a printable shipping label is sent to the site's primary FE, service providers, and forwarders.

GoldSeal Europe return information:

GEMS–E – GOLD SEAL OPERATION & RECYCLING CENTER
GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL
Quai 45 à 49 – 91090 LISSES – FRANC
Tel: (+33) 1–60–91–57–35 or (+33) 1 64 97 55 41
Fax: (+33) 1–60–86–12–31

For non–Gold Seal returns in Europe:

European Renewable Resources Center
GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL2
Quai 45 à 49 – 91090 LISSES – FRANCE
TEL = (+33)–1–64–97–55–41 FAX = (+33)–1–60–86–12–31

1–7–6 PRODUCT LOCATOR INORMATION

Process product locator information for all serialized components removed from the system according to policy. Refer to Section 1–8.

1–8 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans–shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at:
   http://gein2.med.ge.com/gib
   The FE Site Verification consists of three components that are available on the web from the main menu. They are:
   - Install/deinstall product locator model and serial numbers
   - Add/modify ship to address information
   - Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: Product Locator Support Central Page – http://supportcentral.ge.com/15563

2. One "Shipping Card" is filled out and submitted when shipped (extra cards are supplied for trans–shipments between storage and distribution points), and the "Installation Card" and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1–9 UPGRADE SEQUENCE

Follow the system Upgrade Flow chart in SECTION 2 for best installation efficiency.

Note

All on–site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site. However all parallel tasks are to be completed prior to the next task.

The general sequence of events is:

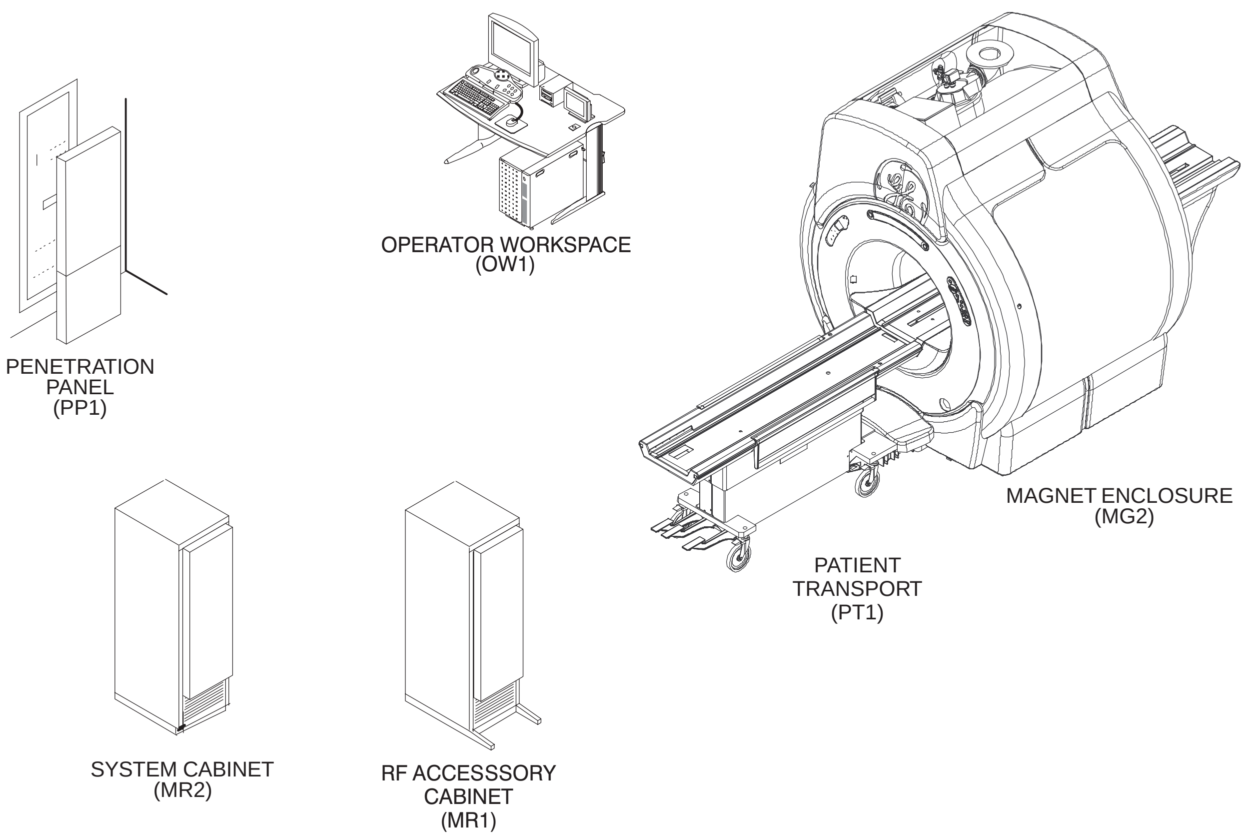
- Verify that all required upgrades have arrived complete and site is ready to shut down.
- Shut site down.
- Remove obsolete equipment and cables.

Aside: Removing and installing cables is more efficient and accurate when the equipment room is un–cluttered. It is strongly recommended that this be done after old equipment is removed and before new equipment is positioned. This house cleaning will make the installation tasks much easier.

- Route new cables
- Install new equipment and modify existing equipment.
- All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy
- Power up
- Load software.
- Perform Functional Checks, System calibrations, and System performance checks
- Replace covers, complete final tasks, and return site to customer.

1–10 UPGRADE SYSTEM CONFIGURATION

Illustration 1–1 shows the major equipment replaced and modified by Signa 1.5T EXCITE 11.x to 12.x Upgrade.



SIGNA 1.5T EXISTING HARDWARE REPLACED OR MODIFIED FOR UPGRADE TO EXCITE HD (Release 12.x)
ILLUSTRATION 1–1

SECTION 2 – UPGRADE INSTALLATION PROCESS

2–1 SITE READINESS

All on–site construction must be completed before equipment is delivered and installation starts. Attempting to upgrade the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

WARNING!

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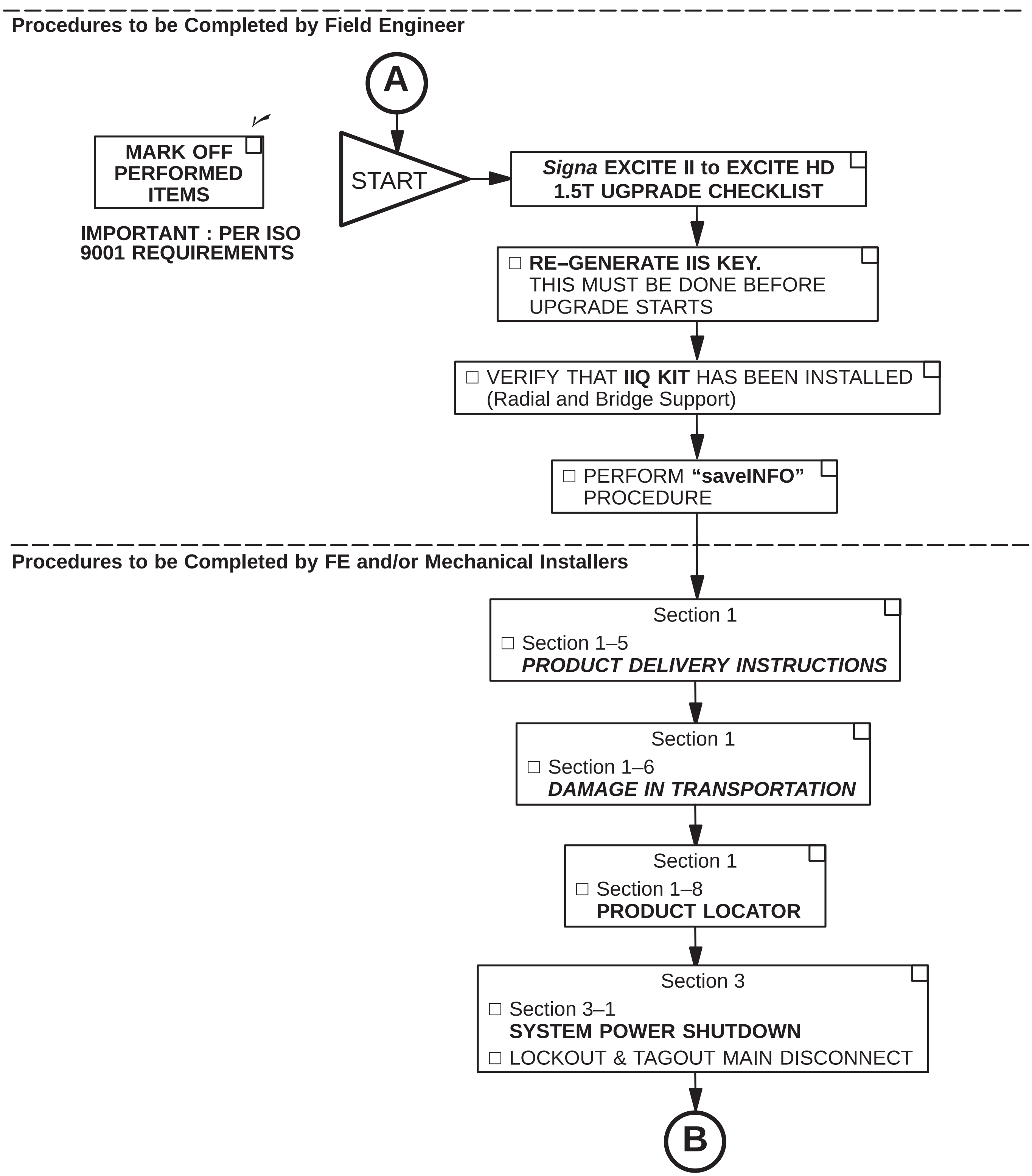
2–2 UPGRADE PROCEDURE

The following upgrade flow charts contain information for upgrading to Signa EXCITE HD 1.5T Release 12.x.

The upgrade flow chart should be followed for an orderly and efficient installation. Note that procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

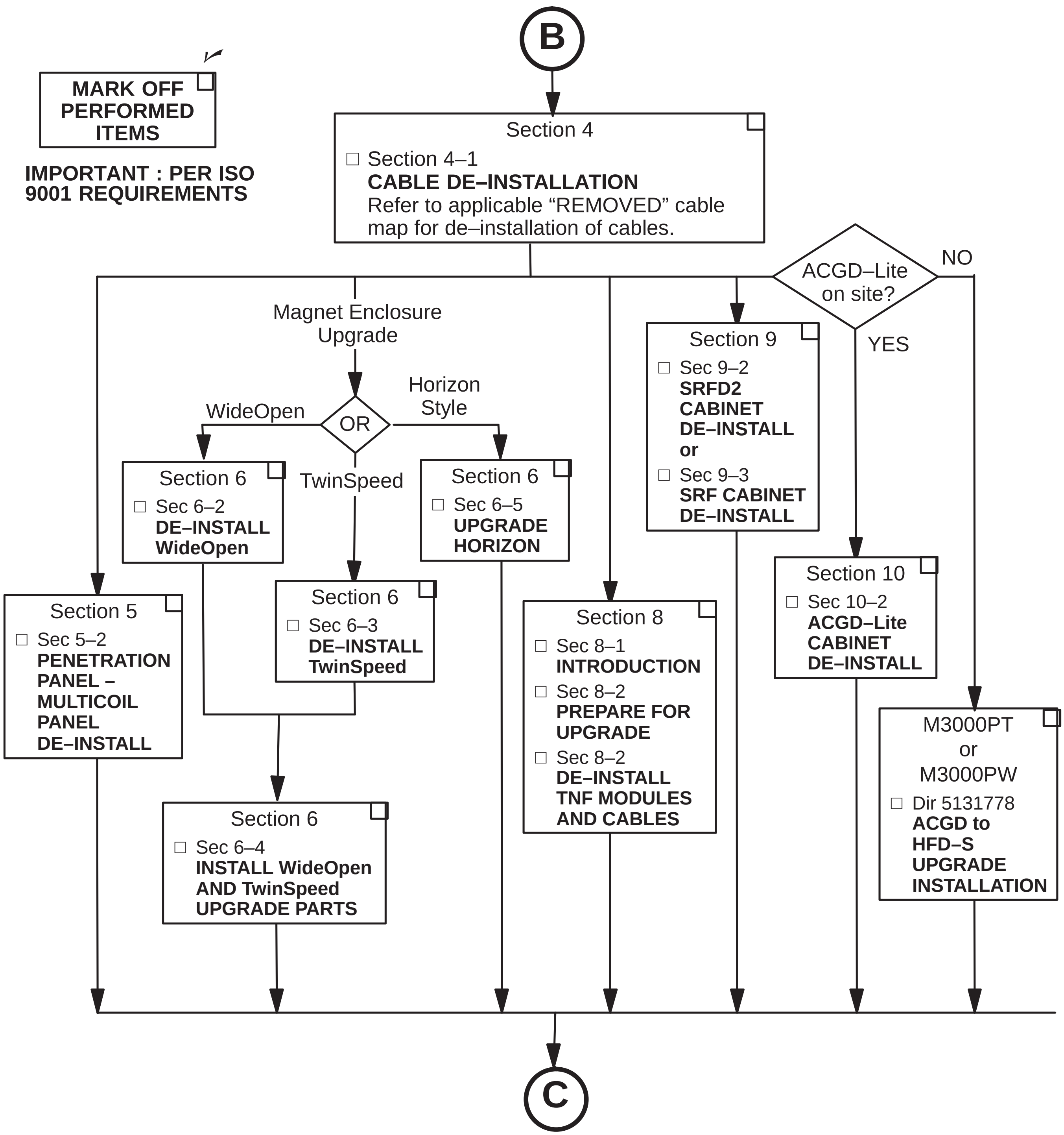
This upgrade flowchart has been developed assuming that all Signa EXCITE HD equipment has been delivered together. **Make sure that every part required is available before starting.**

2–3 SYSTEM MECHANICAL UPGRADE FLOWCHART



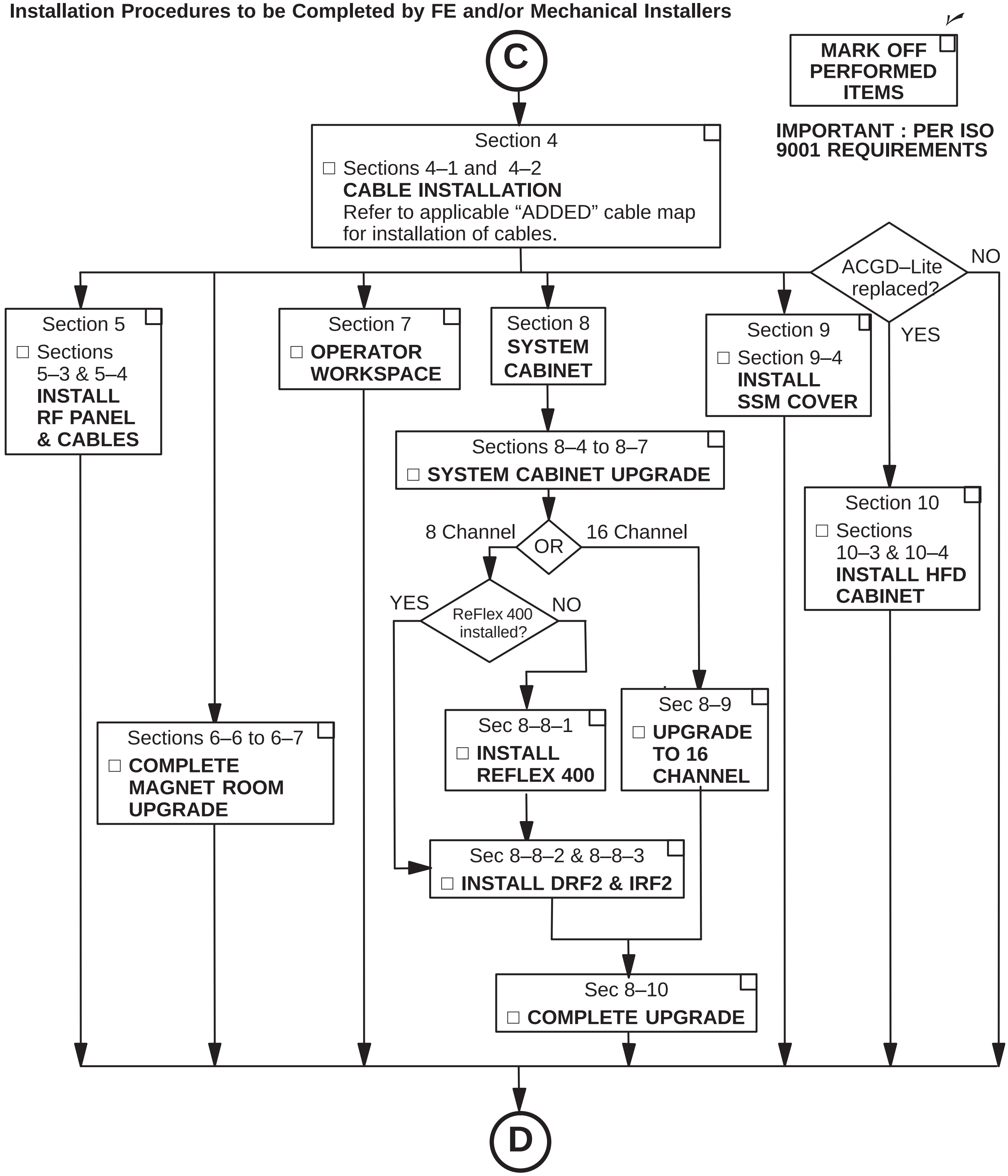
2–3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

De–Installation Procedures to be Completed by FE and/or Mechanical Installers



2–3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

Installation Procedures to be Completed by FE and/or Mechanical Installers



2–3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

Procedures to be Completed by FE and/or Mechanical Installers

MARK OFF PERFORMED ITEMS ☐

IMPORTANT : PER ISO 9001 REQUIREMENTS

D

Section 11 ☐

- ☐ Section 11–1 **MECHANICAL INSTALLATION COMPLETION CHECKLIST**
- ☐ Section 11–2 **RETURNING SHIPPING EQUIPMENT**
- ☐ Section 11–3 **MATERIAL DISPOSITION GUIDANCE**
- ☐ Section 11–4 **FINAL COORDINATION**

MECHANICAL INSTALLATION COMPLETE

Procedures to be Completed by FE and Applications

Section 11 ☐

☐ Direction 5133316>Section 11–5>**POWER ON**

or

MR Service Methods CD–ROM – Direction 5133315

☐ Adjustments/Calibrations>System Level Procedures>***System Power On***

MR Service Methods CD–ROM – Direction 5133315 ☐

☐ Adjustments/Calibrations>***Host LCD Monitor Setup and Calibration***

MR Service Methods CD–ROM – Direction 5133315 ☐

☐ Installation>EXCITE 1.5 HD Installation Calibration Flows> **EXCITE 1.5T HD System Upgrade Callibration Flow**

Applications/Training ☐

☐ **Turn site over to Applications who will instruct users.**

TURN OVER TO CUSTOMER

SECTION 3 – POWER OFF

3–1 SYSTEM POWER SHUTDOWN

DANGER!!

FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

1. Make sure that site customer has archived all patient information. Execute “Save Info”.
2. Verify that System performance baseline measurements have been obtained, especially:
   - SPT Full Test Mode
   - White Pixel Test
   - LV Shim
   - IIS
3. Record center frequency.
4. Ensure that Patient Transport is undocked with cradle in place and Head Coil carriage is at rear of travel.
5. Press the FULL OFF button on the PDU front control panel.
6. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
7. Locate and shut off main disconnect supplying power to the PDU.
8. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
9. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring

SECTION 4 – SYSTEM UPGRADE CABLES

4–1 INTRODUCTION

This section provides information on removing, relabeling and installing system cables for upgrading to Signa EXCITE 1.5T Release 12.x. Refer to the applicable “Removed” and “Added” cable map for the system that is being upgraded.

The “Affected” (removed) Cable maps are located on the following pages.

- Page 4–3 1.5T EXCITE with ACGD Cabinet
- Page 4–5 1.5T EXCITE SmartSpeed with ACGD–Lite Cabinet
- Page 4–7 1.5T EXCITE TwinSpeed with ACGD Cabinet
- Page 4–9 1.5T EXCITE Non–LCC Magnet with ACGD Cabinet

The “Affected” (added) Cable maps are located on the following pages.

- Page 4–11 1.5T EXCITE with ACGD Cabinet
- Page 4–13 1.5T EXCITE SmartSpeed with ACGD–Lite Cabinet
- Page 4–15 1.5T EXCITE TwinSpeed with ACGD Cabinet
- Page 4–17 1.5T EXCITE Non–LCC Magnet with ACGD Cabinet

Tape map in a convenient location to check off routed cable. Additional information on each cable is located in the Appendix. Upon completion of connecting cables, return fold–out map to the binder for future reference.

4–2 SORTING AND ROUTING CABLES

CAUTION

Ty–wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

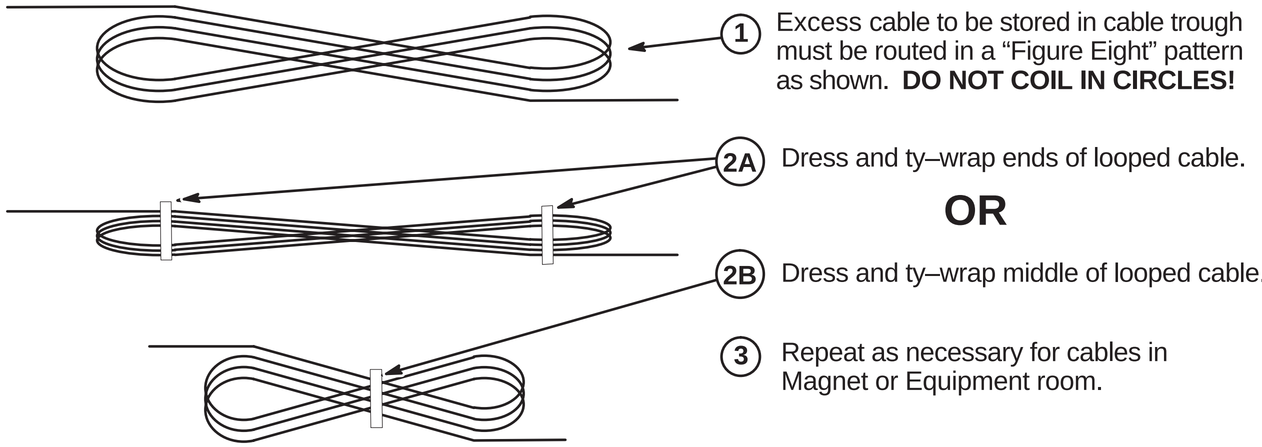
System power, ground, and data cables are color coded at each end by destination per colors shown on Cable Map.

1. Sort cables by destination. Normally, it is best to sort cables originating from the farthest cabinet or component from the PDU first. This will make it easier to pull cables through the troughs.

Note: Carefully review architectural drawings to insure that all ducts are properly installed and that all signal and power cables are accounted for before starting to route cable runs.

2. Unroll coiled cables along the intended route to insure that they will lay freely without twists or kinks. Route cables in accordance with the applicable System Interconnect Diagram.
3. Plan for storage location of coiled cable excess length if cables are to remain at delivered length. See Illustration 4–1. Some cables are usually cut to length such as gradient cables, power cables, Head and Body RF, and Fiber–optic cables.

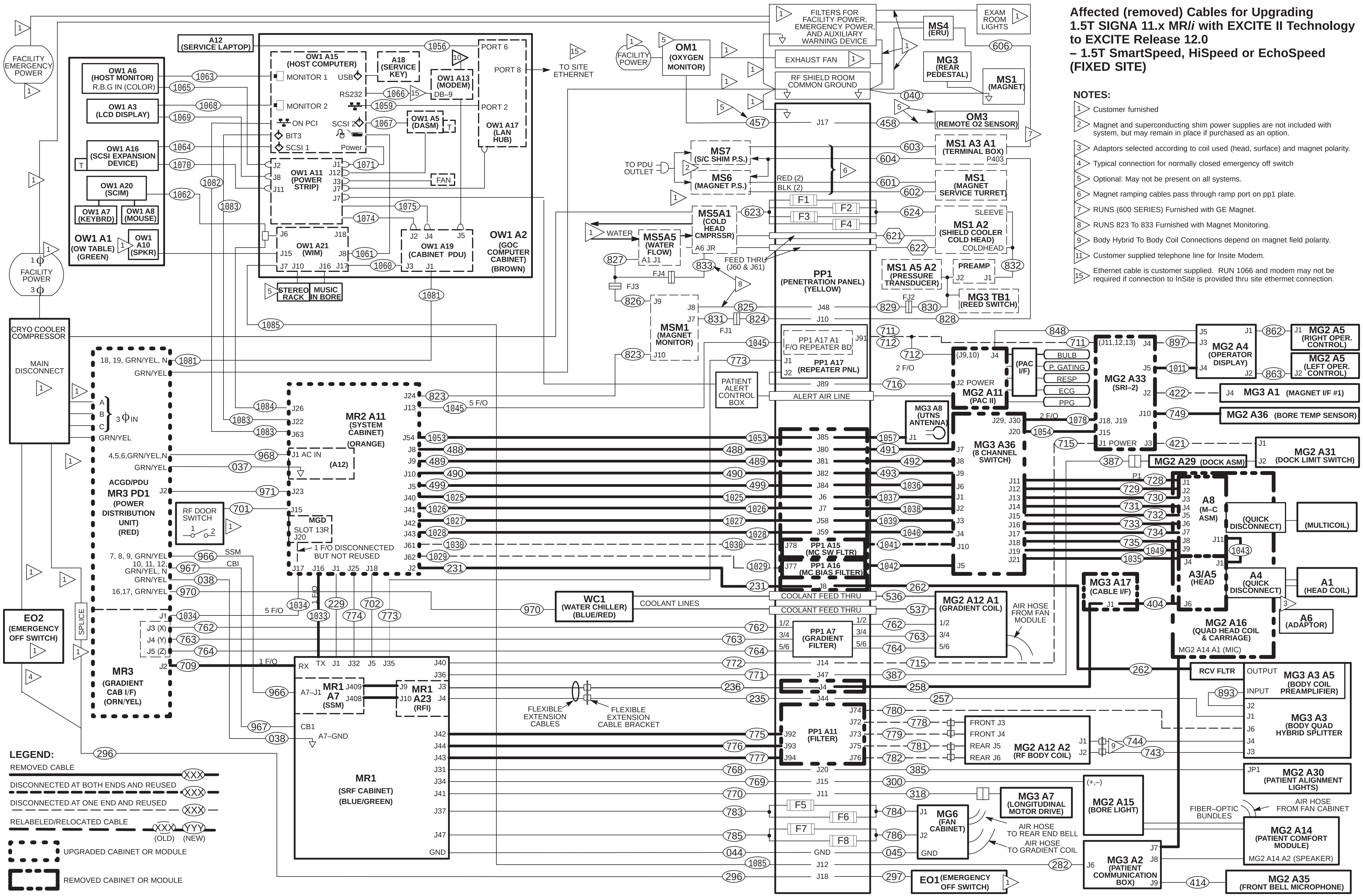
4–2 SORTING AND ROUTING CABLES (Continued)



PROPER STORAGE OF EXCESS CABLES
ILLUSTRATION 4–1

4. Place cables in provided troughs, conduit, or ducts. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans. Route cables inside exam room to Rear Pedestal area with sufficient length remaining to complete routing and connecting within Magnet Enclosure.
5. Rear Pedestal Cables – It is desired to keep the gradient cables separated from the signal and bias cables.

Affected (removed) Cables for Upgrading 1.5T SIGNA 11.x MR/*i* with EXCITE II Technology to EXCITE Release 12.0 – 1.5T SmartSpeed, HiSpeed or EchoSpeed (FIXED SITE)



NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
11. Customer supplied telephone line for Insite Modem.
15. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection.

LEGEND:

- REMOVED CABLE (XXX)
- DISCONNECTED AT BOTH ENDS AND REUSED (XXX)
- DISCONNECTED AT ONE END AND REUSED (XXX)
- RELABELED/RELOCATED CABLE (XXX) (OLD) (YYY) (NEW)
- UPGRADED CABINET OR MODULE
- REMOVED CABINET OR MODULE

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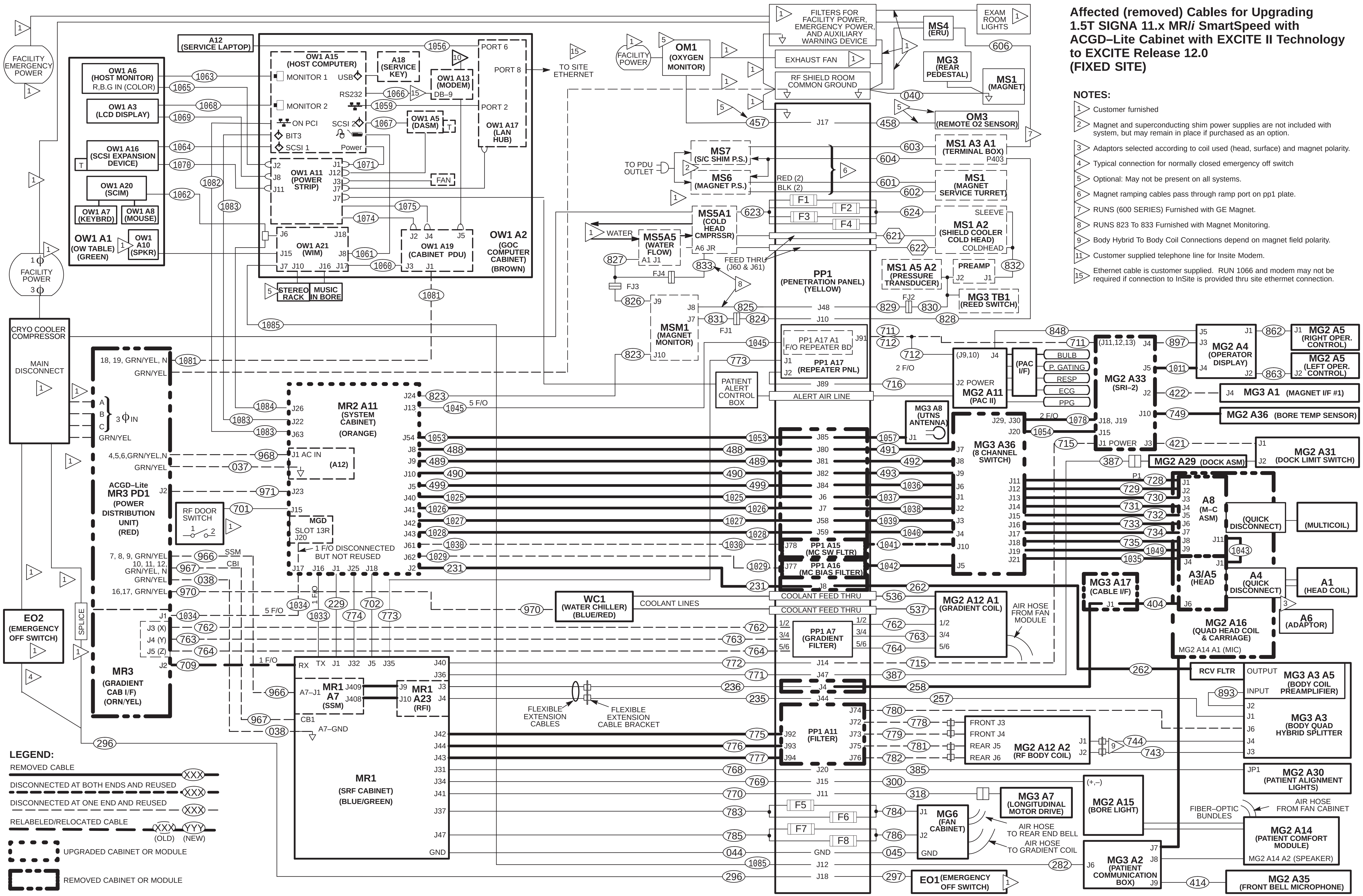
Affected (removed) Cables for Upgrading 1.5T SIGNA 11.x MR/*i* SmartSpeed with ACGD–Lite Cabinet with EXCITE II Technology to EXCITE Release 12.0 (FIXED SITE)



NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
11. Customer supplied telephone line for Insite Modem.
15. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection.

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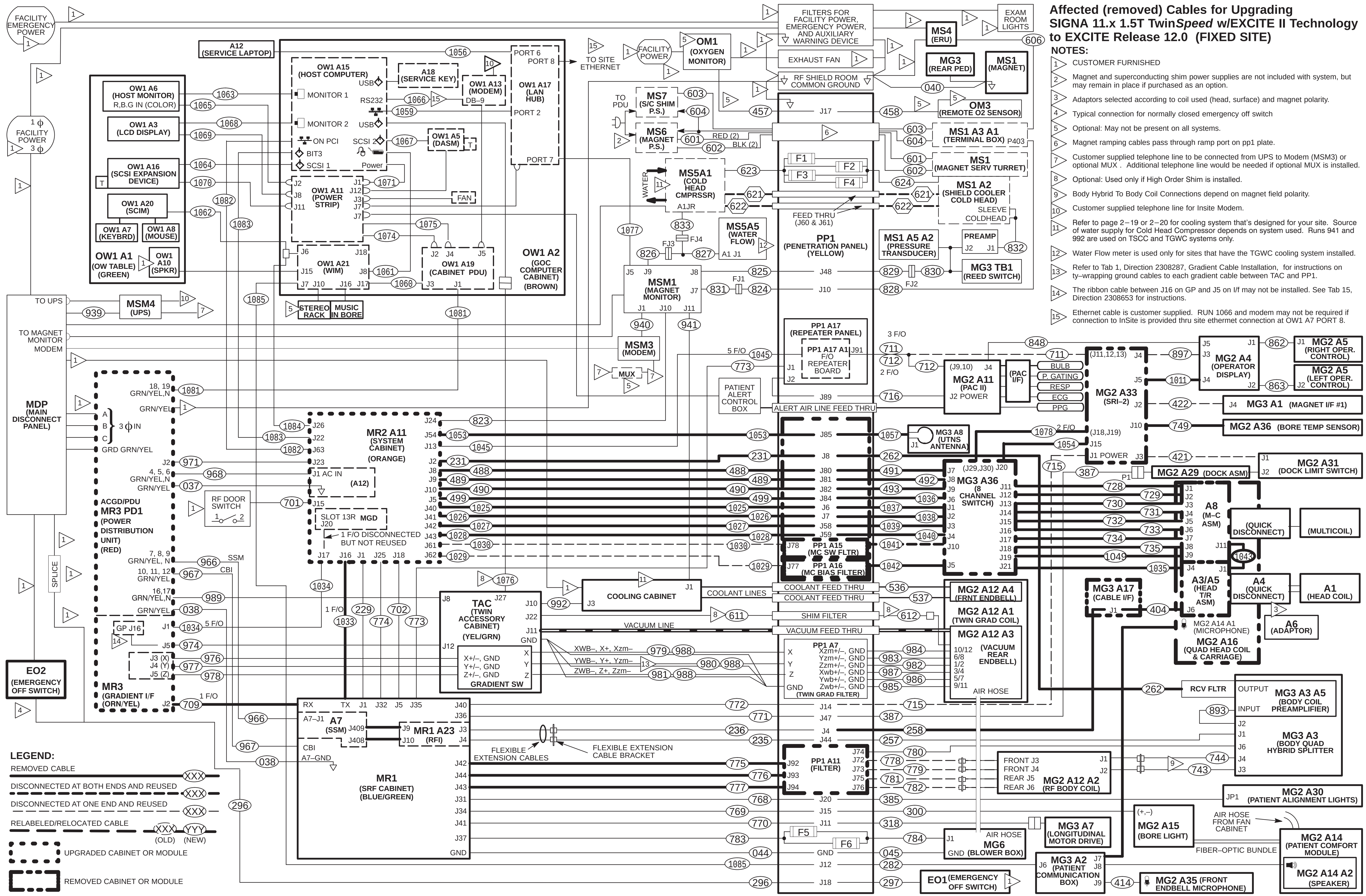
Affected (removed) Cables for Upgrading SIGNA 11.x 1.5T Twin*Speed* w/EXCITE II Technology to EXCITE Release 12.0 (FIXED SITE)

NOTES:

1. CUSTOMER FURNISHED
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. Customer supplied telephone line to be connected from UPS to Modem (MSM3) or optional MUX . Additional telephone line would be needed if optional MUX is installed.
8. Optional: Used only if High Order Shim is installed.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. Customer supplied telephone line for Insite Modem.
11. Refer to page 2–19 or 2–20 for cooling system that's designed for your site. Source of water supply for Cold Head Compressor depends on system used. Runs 941 and 992 are used on TSCC and TGWC systems only.
12. Water Flow meter is used only for sites that have the TGWC cooling system installed.
13. Refer to Tab 1, Direction 2308287, Gradient Cable Installation, for instructions on ty–wrapping ground cables to each gradient cable between TAC and PP1.
14. The ribbon cable between J16 on GP and J5 on I/f may not be installed. See Tab 15, Direction 2308653 for instructions.
15. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection at OW1 A7 PORT 8.



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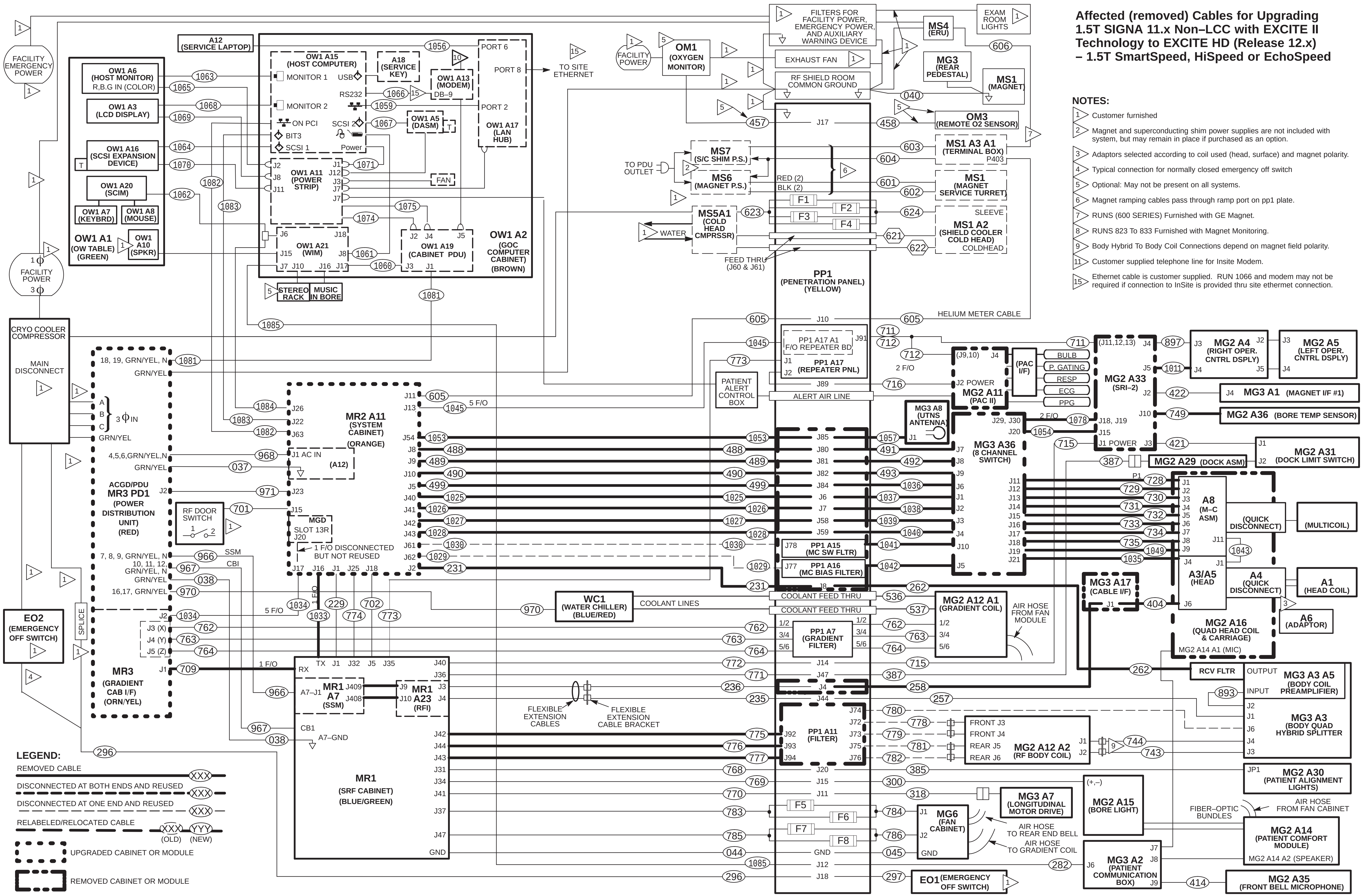
Affected (removed) Cables for Upgrading 1.5T SIGNA 11.x Non–LCC with EXCITE II Technology to EXCITE HD (Release 12.x) – 1.5T SmartSpeed, HiSpeed or EchoSpeed

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
11. Customer supplied telephone line for Insite Modem.
15. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection.



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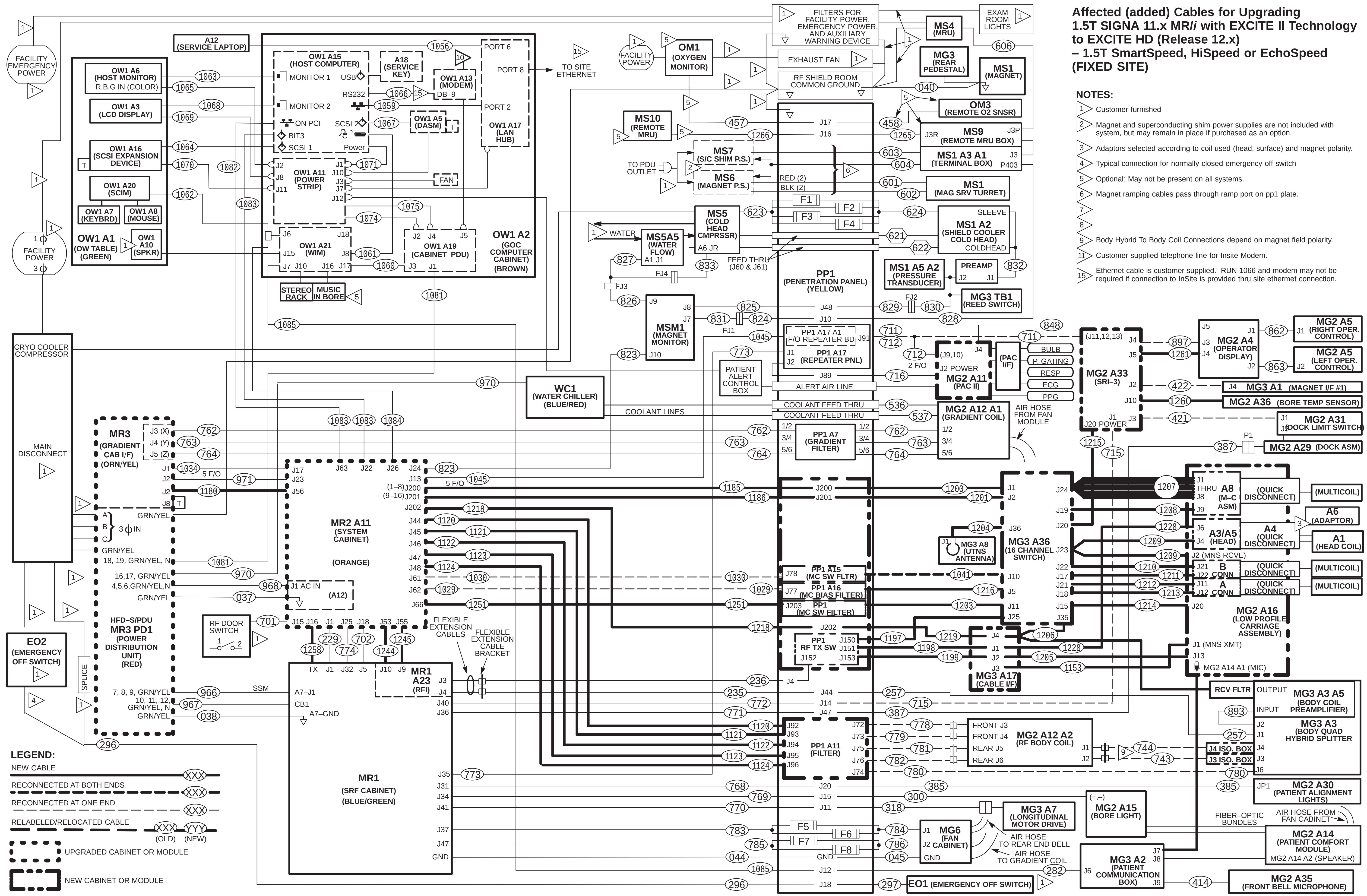
Affected (added) Cables for Upgrading 1.5T SIGNA 11.x MR/*i* with EXCITE II Technology to EXCITE HD (Release 12.x) – 1.5T SmartSpeed, HiSpeed or EchoSpeed (FIXED SITE)

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7.
8.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
11. Customer supplied telephone line for Insite Modem.
15. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection.



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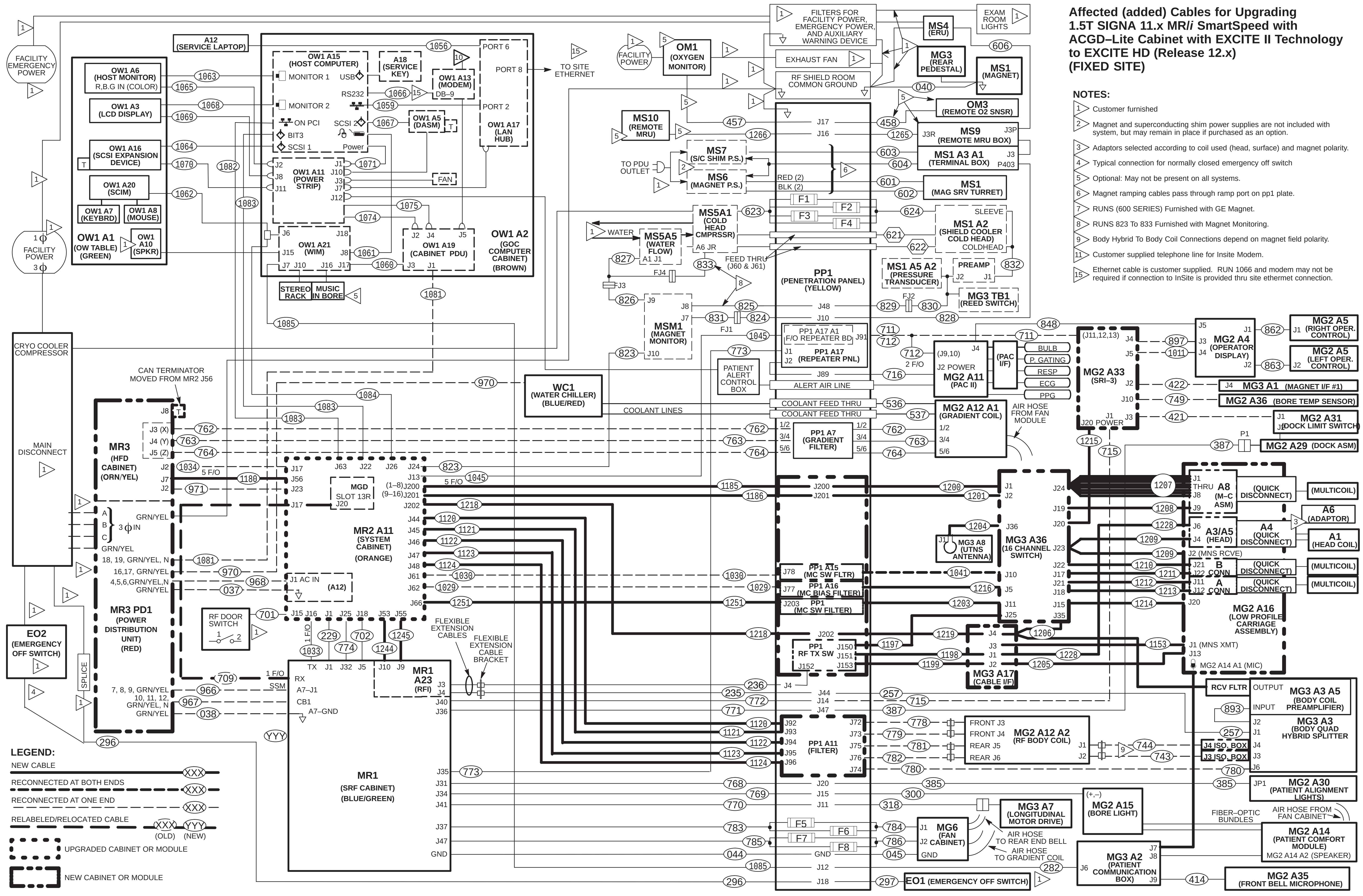
Affected (added) Cables for Upgrading 1.5T SIGNA 11.x MR/*i* SmartSpeed with ACGD–Lite Cabinet with EXCITE II Technology to EXCITE HD (Release 12.x) (FIXED SITE)

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
11. Customer supplied telephone line for Insite Modem.
15. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection.



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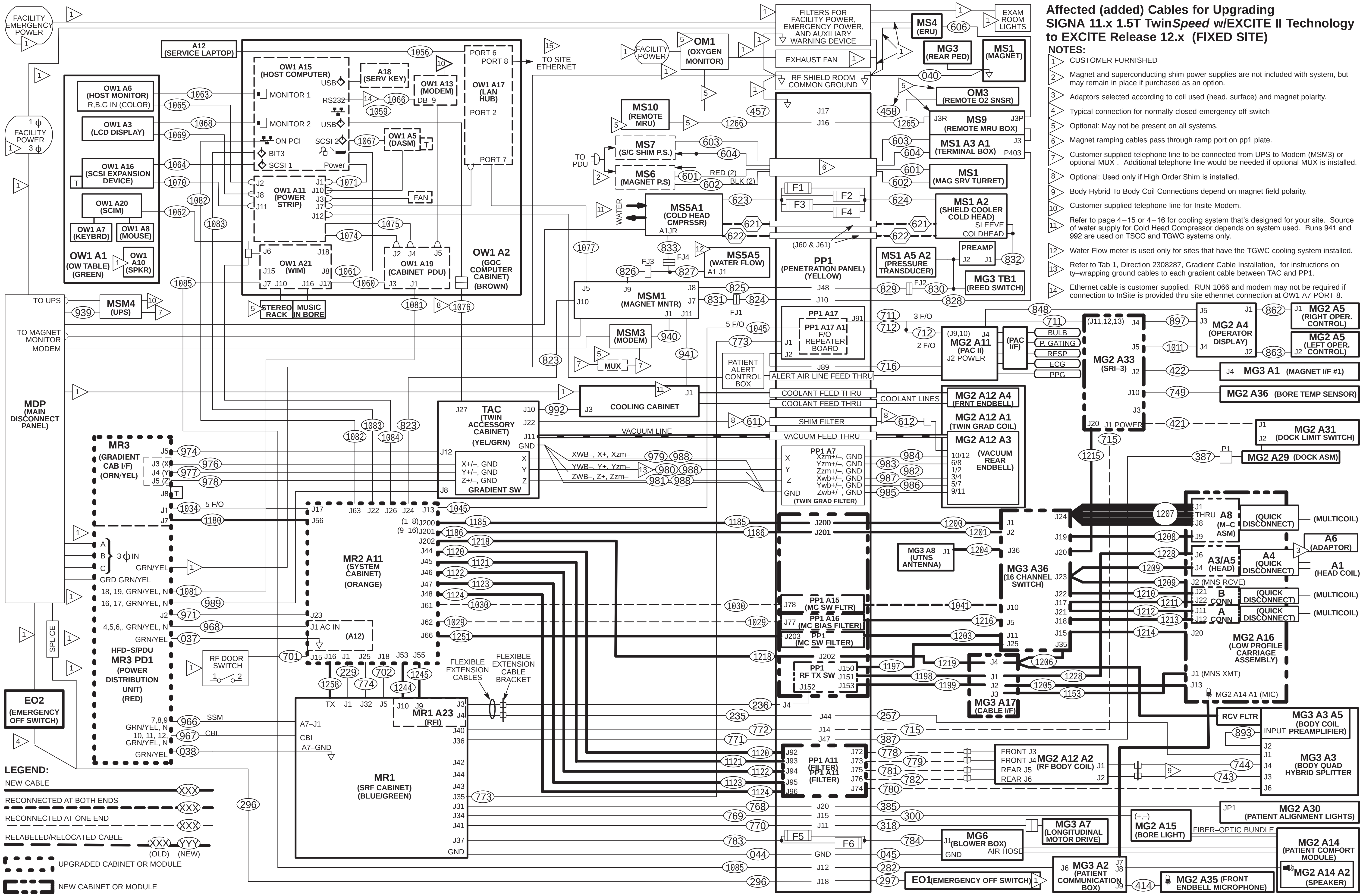
Affected (added) Cables for Upgrading SIGNA 11.x 1.5T Twin*Speed* w/EXCITE II Technology to EXCITE Release 12.x (FIXED SITE)

NOTES:

1. CUSTOMER FURNISHED
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. Customer supplied telephone line to be connected from UPS to Modem (MSM3) or optional MUX . Additional telephone line would be needed if optional MUX is installed.
8. Optional: Used only if High Order Shim is installed.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. Customer supplied telephone line for Insite Modem.
11. Refer to page 4–15 or 4–16 for cooling system that's designed for your site. Source of water supply for Cold Head Compressor depends on system used. Runs 941 and 992 are used on TSCC and TGWC systems only.
12. Water Flow meter is used only for sites that have the TGWC cooling system installed.
13. Refer to Tab 1, Direction 2308287, Gradient Cable Installation, for instructions on ty–wrapping ground cables to each gradient cable between TAC and PP1.
14. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection at OW1 A7 PORT 8.



LEGEND:

- NEW CABLE
- RECONNECTED AT BOTH ENDS
- RECONNECTED AT ONE END
- RELABELED/RELOCATED CABLE (OLD) (NEW)
- UPGRADED CABINET OR MODULE
- NEW CABINET OR MODULE

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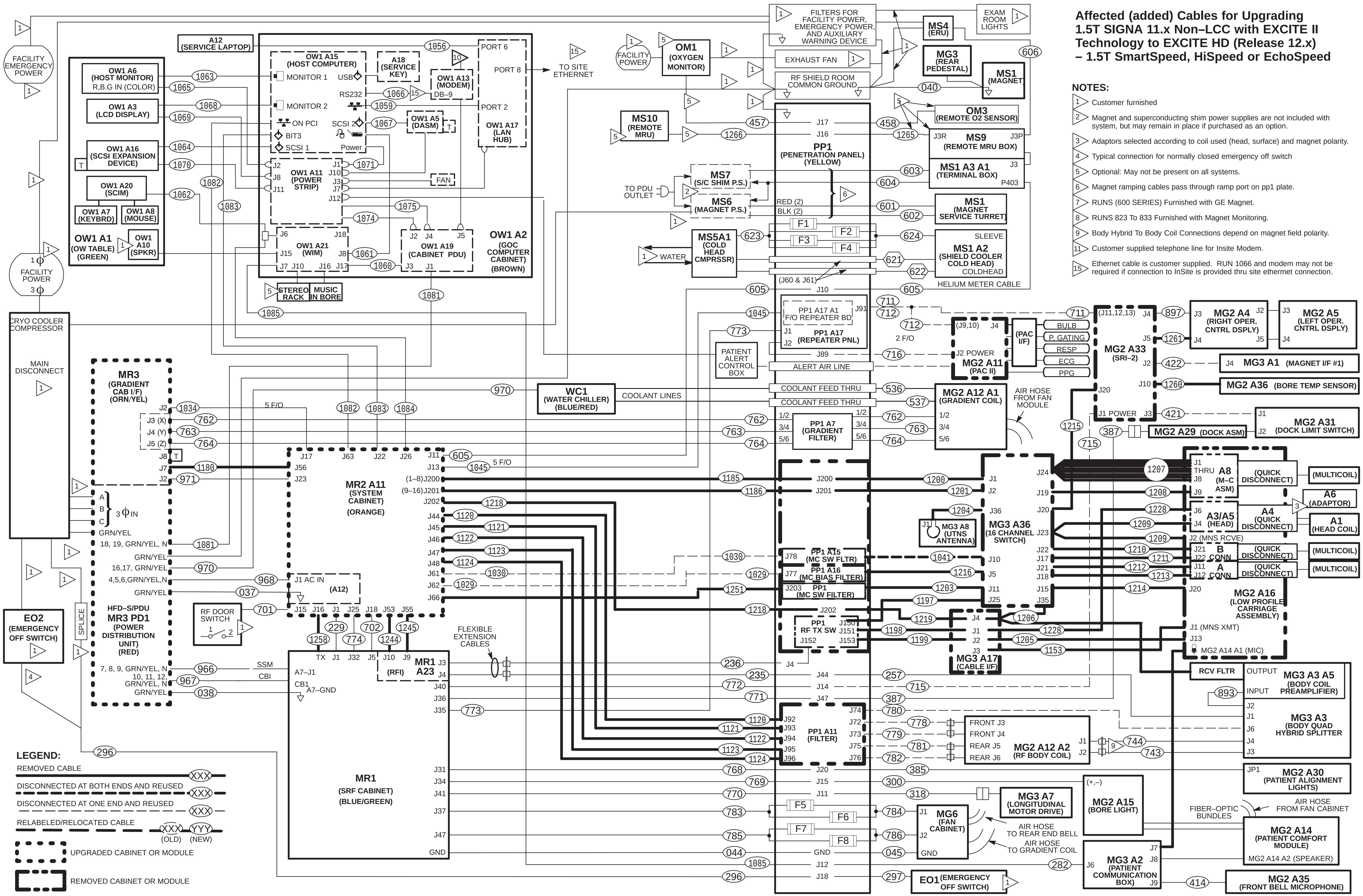
Affected (added) Cables for Upgrading 1.5T SIGNA 11.x Non–LCC with EXCITE II Technology to EXCITE HD (Release 12.x) – 1.5T SmartSpeed, HiSpeed or EchoSpeed

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
11. Customer supplied telephone line for Insite Modem.
15. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection.



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SECTION 5 – PENETRATION PANEL

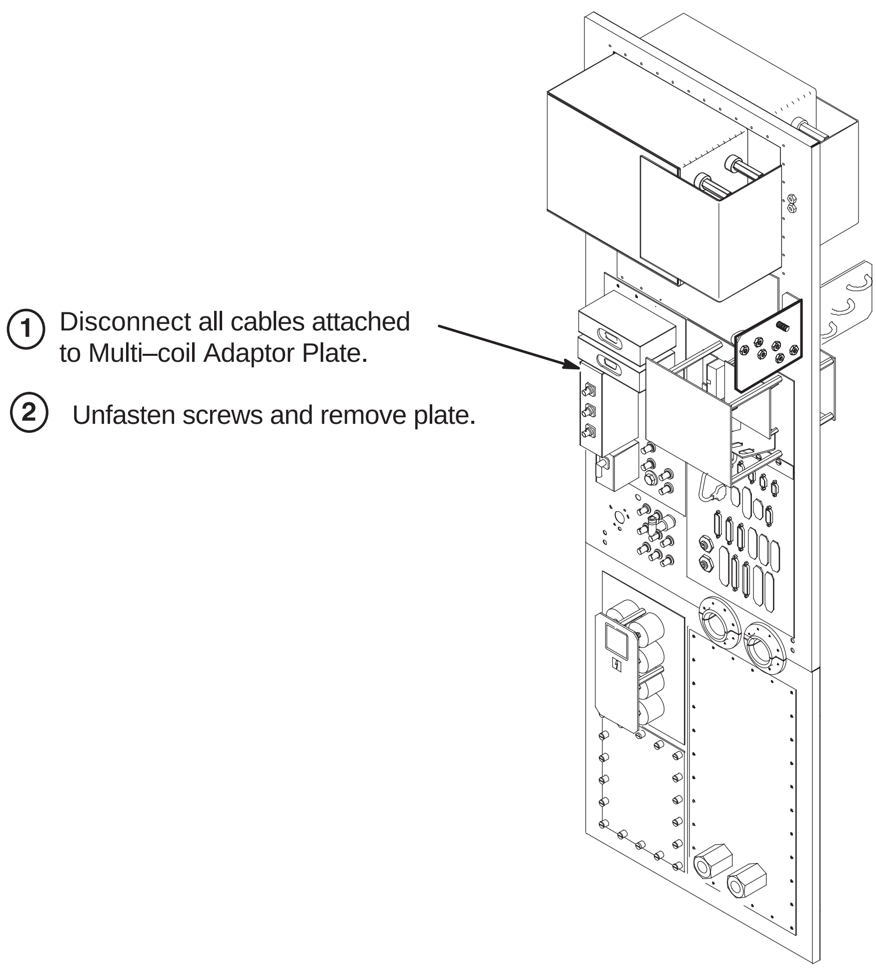
CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

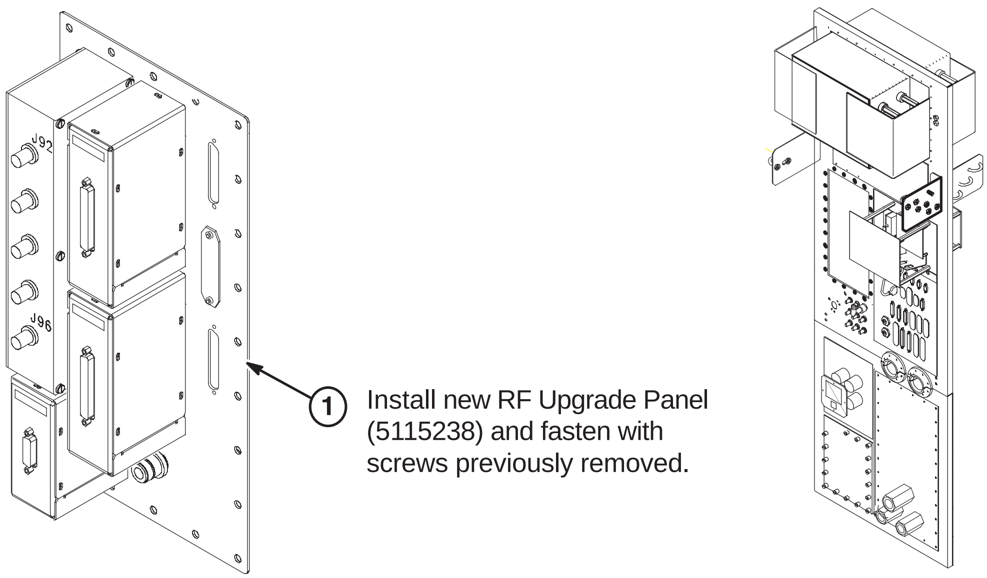
5–1 INTRODUCTION

The Multi–coil Adaptor Plate will be replaced by the RF Upgrade Panel (5115238).

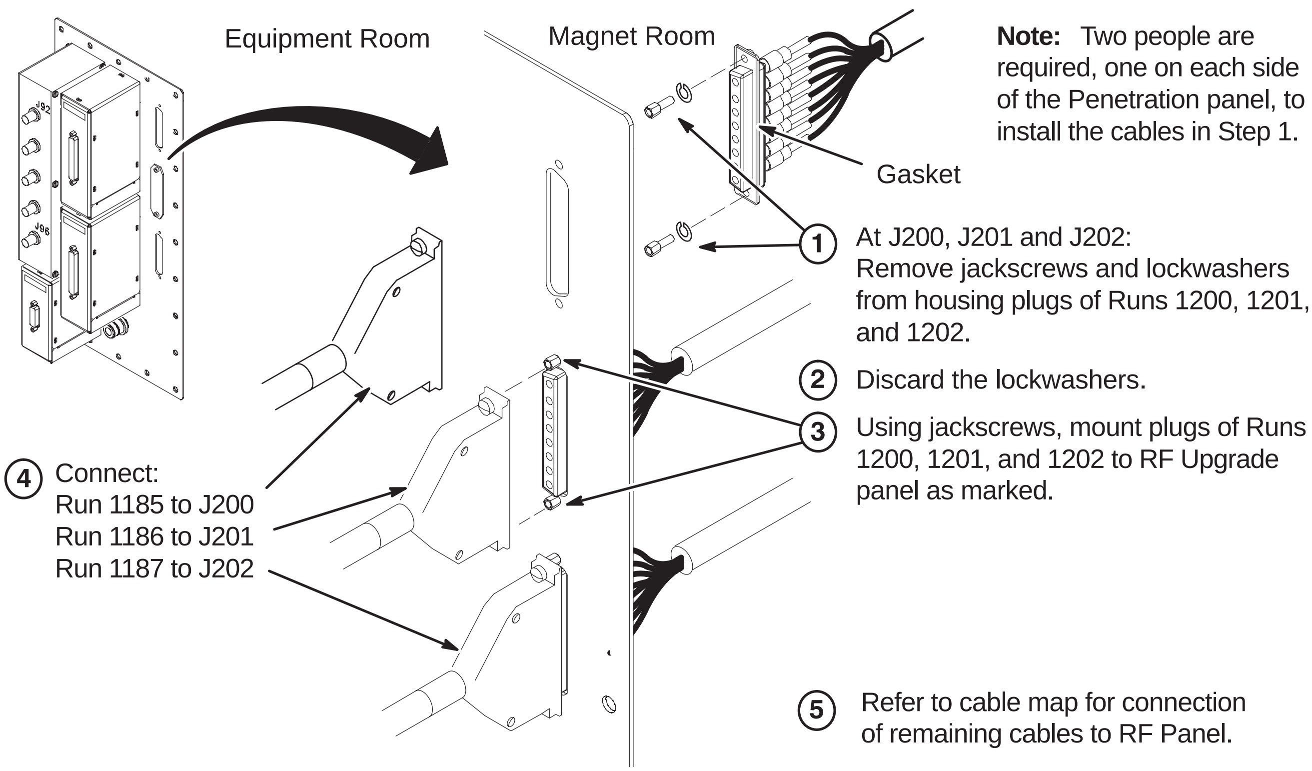
5–2 REMOVE MULTI–COIL ADAPTOR PLATE FROM PENETRATION PANEL



5–3 INSTALL NEW EXCITE RF PANEL



5–4 INSTALLATION OF CABLES AT J200, J201, AND J202



SECTION 6 – MAGNET ROOM

TABLE OF CONTENTS

(Continued on next page)

TABLE OF CONTENTS (Continued)

CAUTION

MAKE SURE POWER TO SYSTEM IS OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

6–1 INTRODUCTION

Modules on the Rear Pedestal, the front and rear Split Bridge on WideOpen style enclosures, LPCA and Cable Track, SRI, and PAC will be replaced.

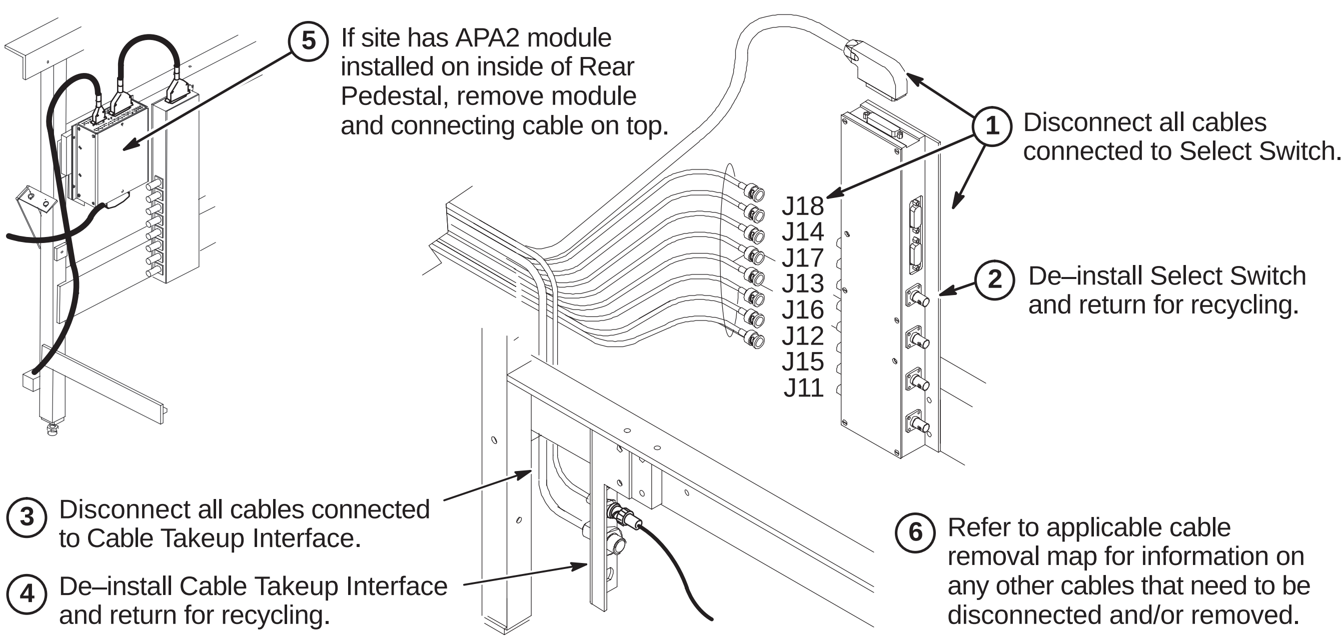
A new Dual Thermal Sensor cable, Run 1260, will be installed along with other new cables that will be routed and connected. Several other cables will also be removed and replaced with new cables. Refer to Section 4 cable maps for information on removing and installing cables

- Refer to Sections 6–2 and 6–4 for upgrading a system with a LCC magnet and a WideOpen Enclosure
- Refer to Sections 6–3 and 6–4 for upgrading a system with a LCC magnet and a TwinSpeed Enclosure
- Refer to Section 6–5 for upgrading a system with a Non–LCC magnet and a Horizon style Enclosure
- Complete Section 6–6 for all systems
- Also included are the instructions for upgrading the Cradle Release on the Patient Table. Refer to section 6–7 if this upgrade has not already been performed at this site.

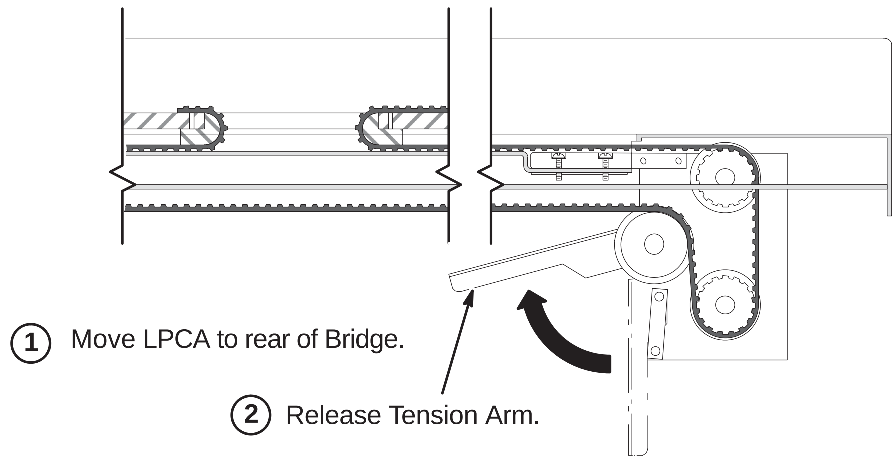
6–2 PREREQUISITE UPGRADE PROCEDURES FOR WideOpen ENCLOSURE

6–2–1 DE–INSTALL EXISTING LPCA & CABLE TRACK

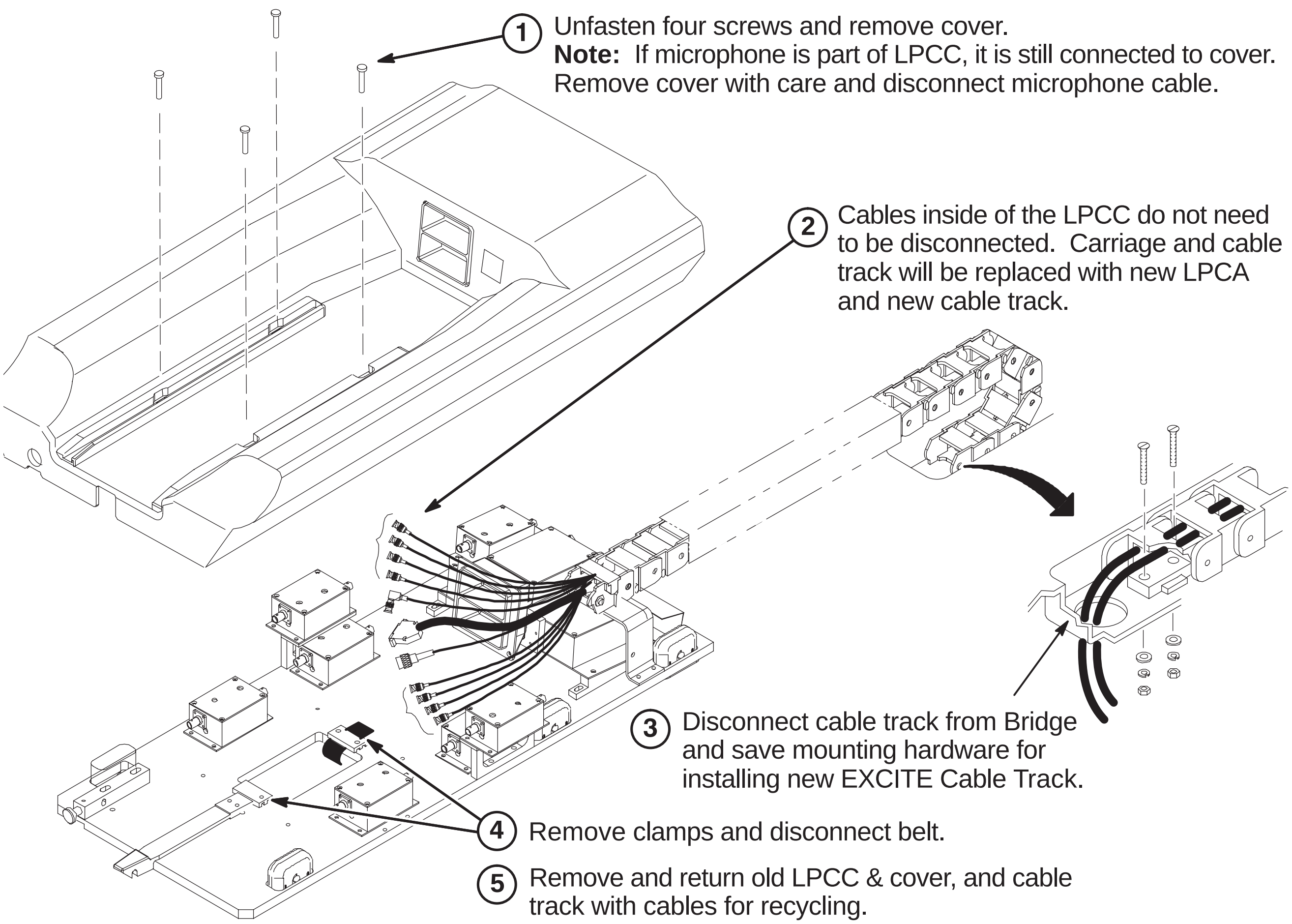
6–2–1–1 Disconnect Cables In Rear Pedestal



6–2–1–2 Relese Tension Arm Under Rear Pedestal



6–2–1–3 Disconnect Low Profile Carriage Cover & Cable Track



6–2–2 DE–INSTALL BRIDGE FOR WideOpen

Upgrading to EXCITE HD requires a new Front and Rear Split Bridge. Early production versions of the WideOpen enclosure were installed with the long single–piece Bridge. Later versions used the Split Bridge. Either existing bridge type will need to be removed for installation of the new dual–track Split Bridge. Refer to one of the following for instructions of removing the bridge at site being upgraded.

- Procedure **6–2–2–1** for the de–installation of the long single–piece bridge.
- Procedures **6–2–2–2** thru **6–2–2–4** for the de–installation of the single–track split bridge.

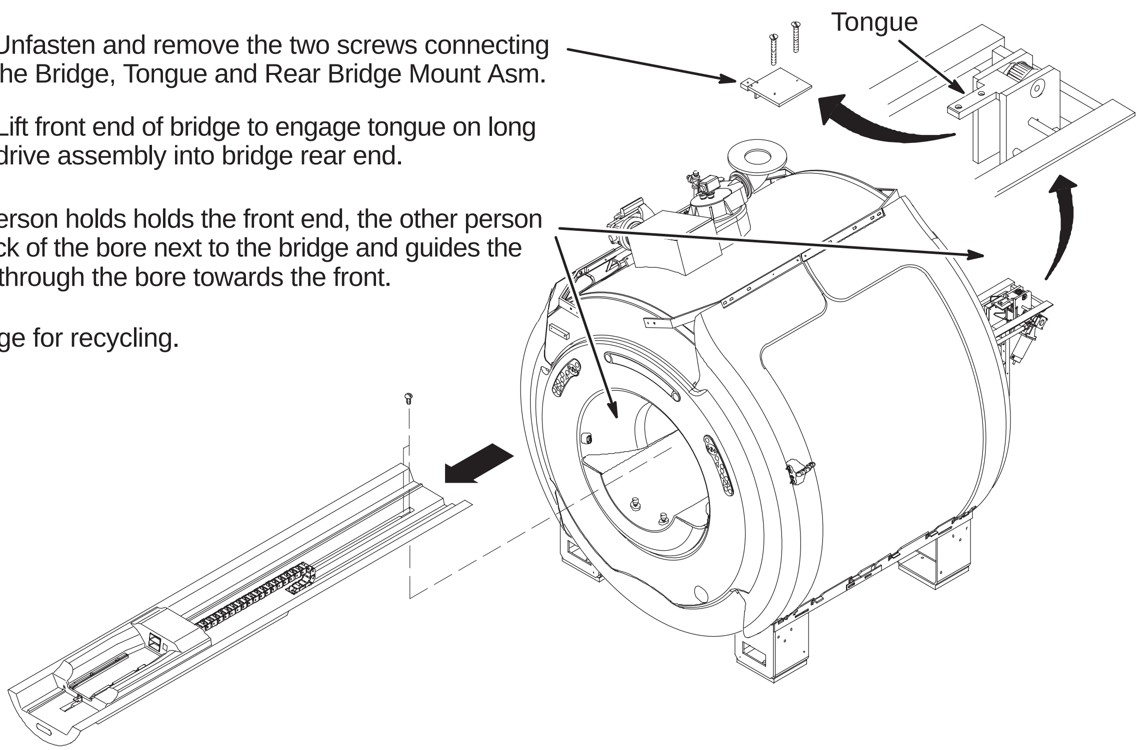
6–2–2–1 Remove Long Bridge

1 Unfasten and remove the two screws connecting the Bridge, Tongue and Rear Bridge Mount Asm.

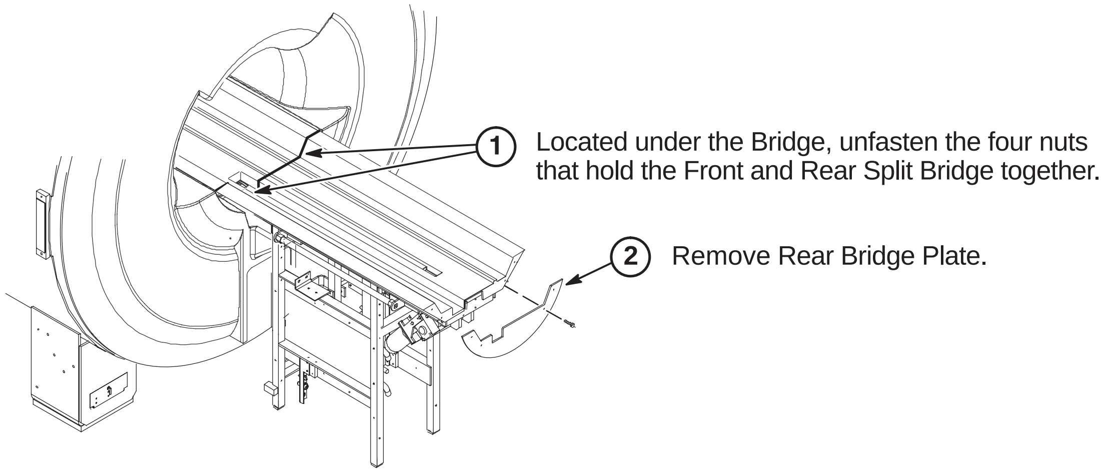
2 Lift front end of bridge to engage tongue on long drive assembly into bridge rear end.

3 While one person holds holds the front end, the other person stands in back of the bore next to the bridge and guides the Bridge Asm through the bore towards the front.

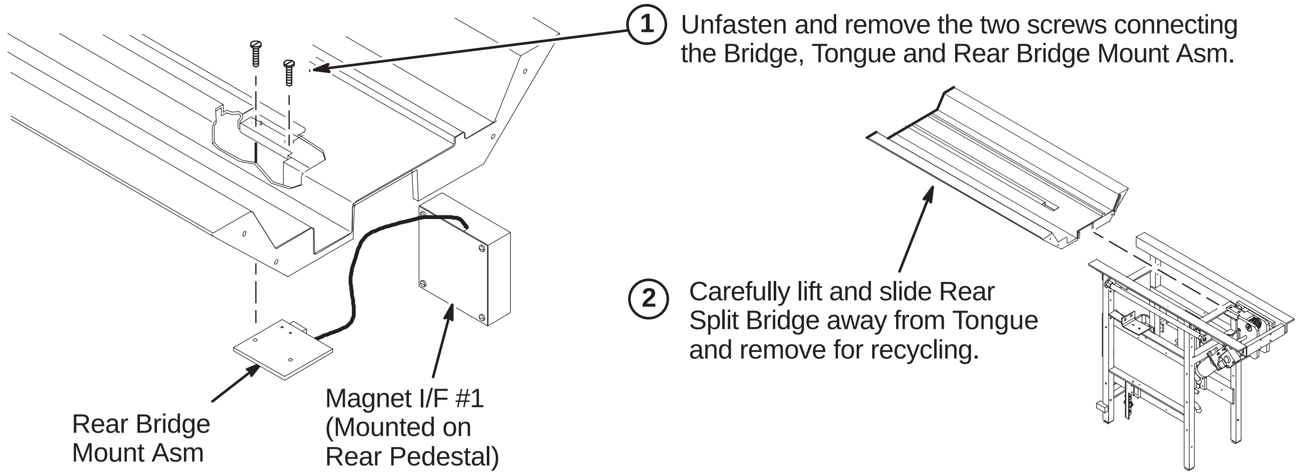
4 Return Bridge for recycling.



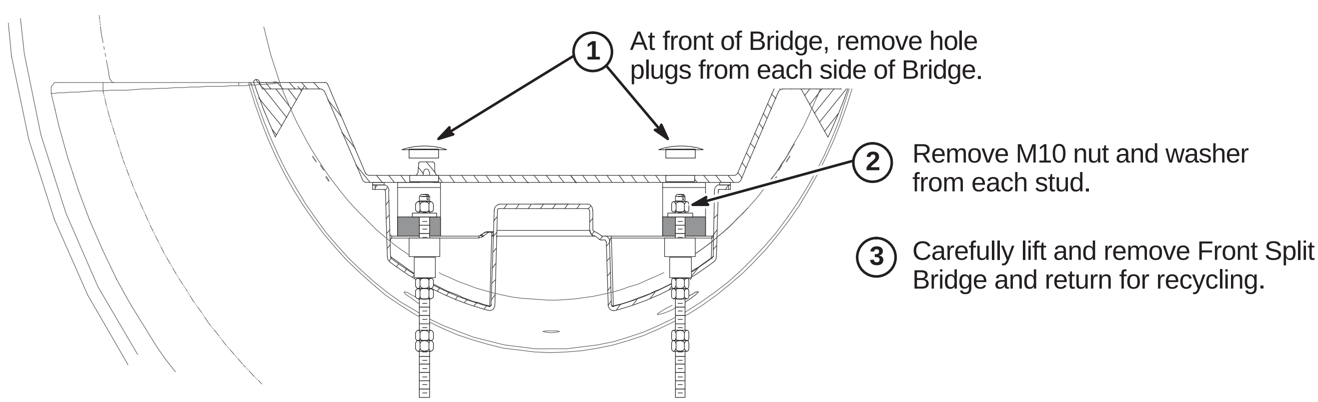
6–2–2–2 Unfasten Rear Split Bridge



6–2–2–3 Remove Rear Split Bridge

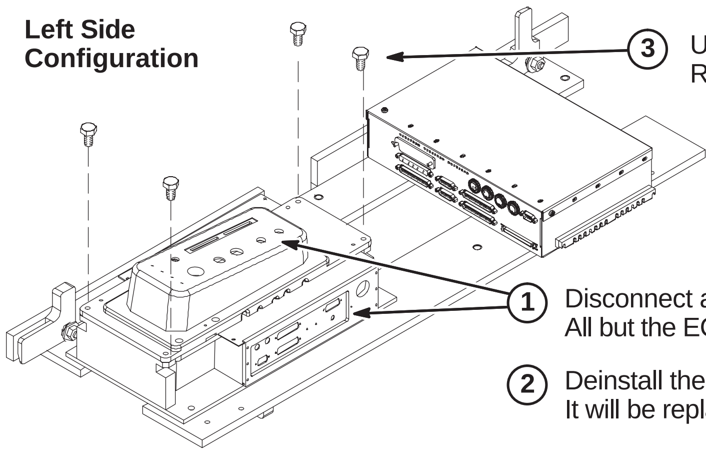


6–2–2–4 Remove Front Split Bridge



6–2–3 DE–INSTALL PAC MODULE FOR WideOpen

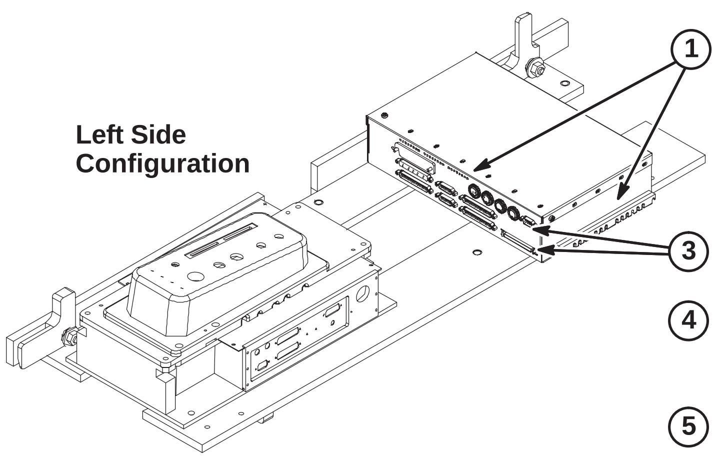
The existing PAC Module will be replaced by a new PAC Module with independent vector gating.



6–2–4 DE–INSTALL SRI MODULE FOR WideOpen

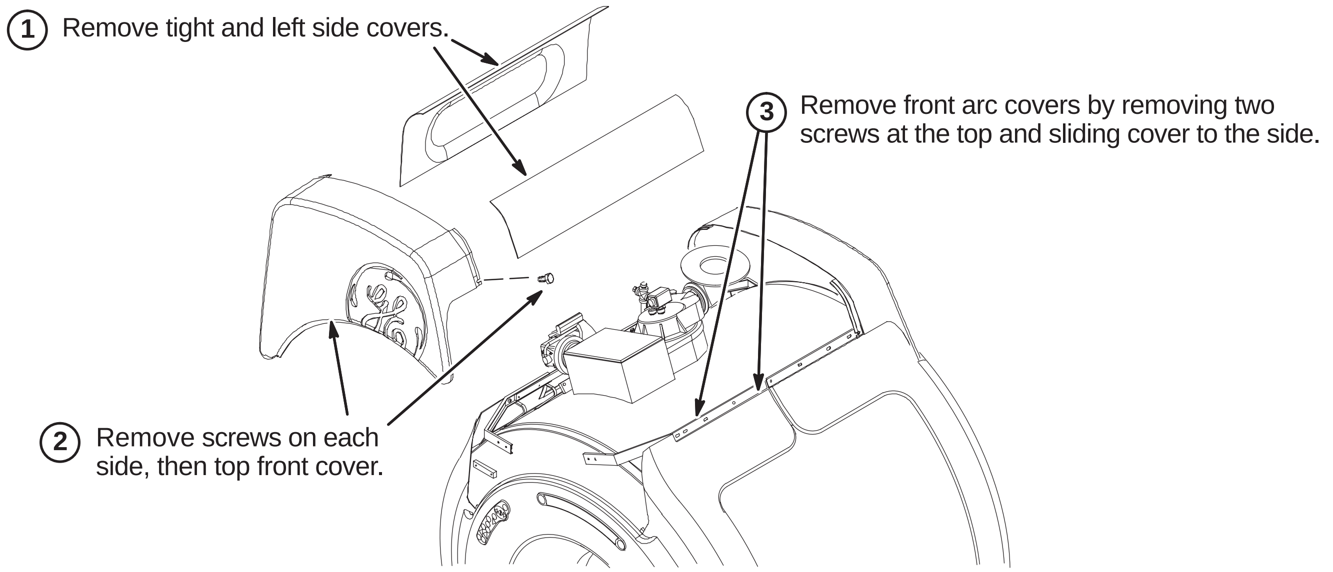
The existing SRI Module will be replaced by the SRI–3 Module.

NOTE: Existing Runs 749 and 1011 must be removed and replaced with new Runs 1260 and 1261. There is a functionality difference.

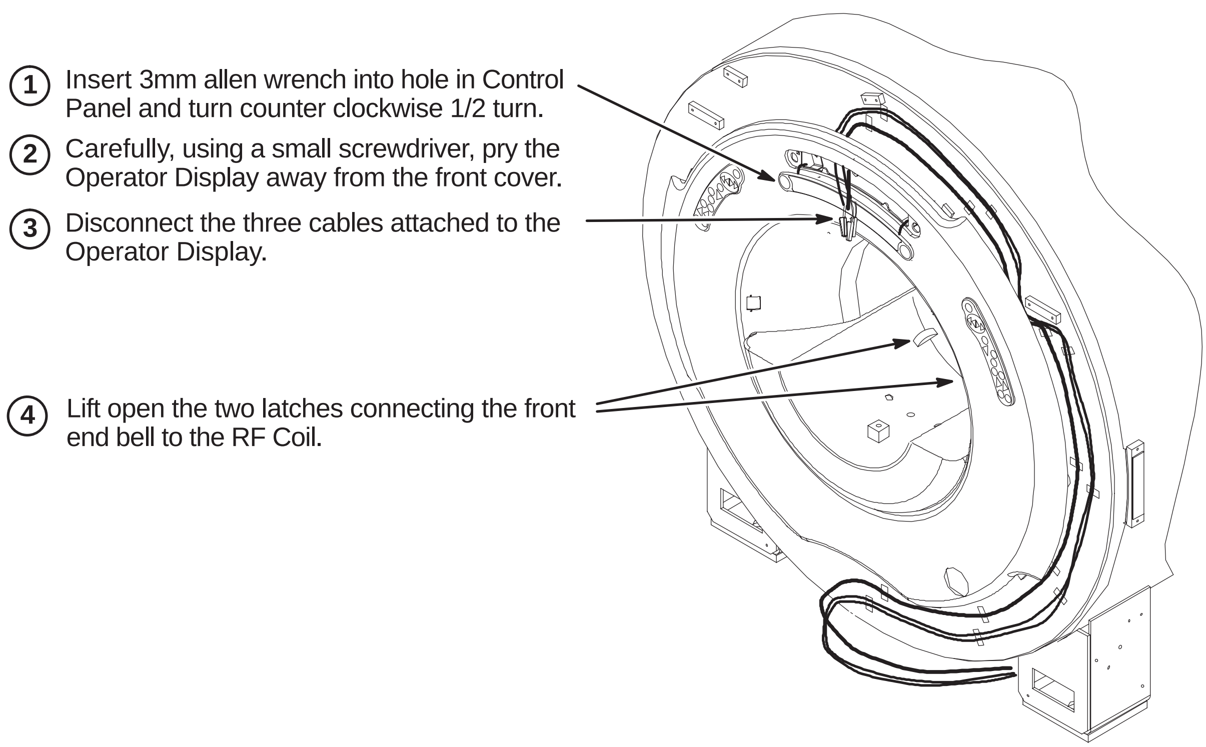


6–2–5 DE–INSTALL FRONT END BELL

6–2–5–1 Remove WideOpen Enclosure Covers



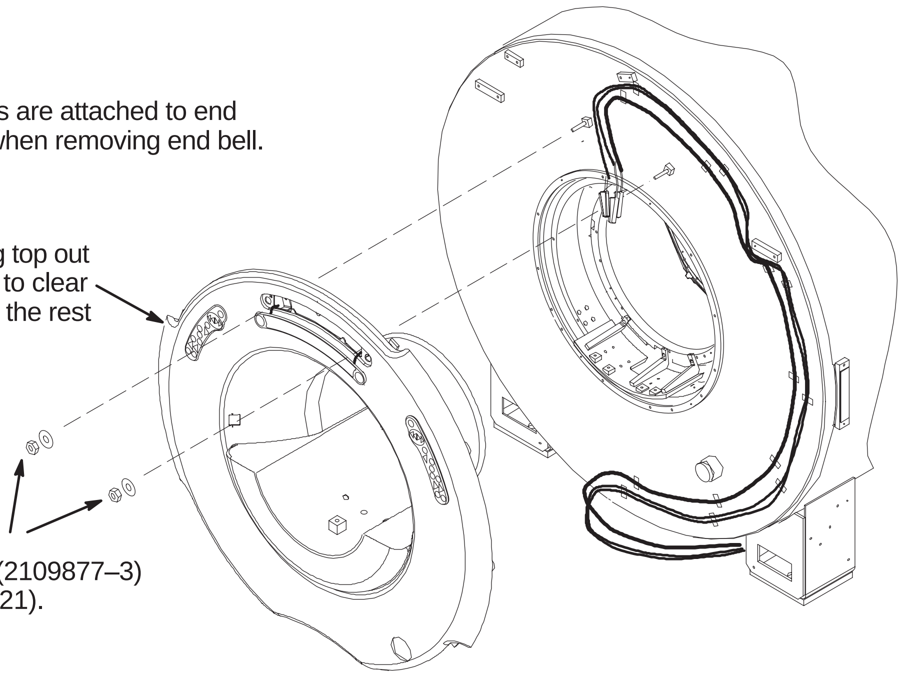
6–2–5–2 Disconnect Operator Control Panel



6–2–5–3 Remove Front End Bell

CAUTION:
Bore Light inserts are attached to end bell. Take care when removing end bell.

(2) Remove front end bell by pulling top out past studs, then lifting vertically to clear end bell inserts, then pulling out the rest of the way.



(1) Unfasten the M10 Lock Nuts (2109877–3) and Rubber Washers (2167321).

6–2–6 INSTALL IIQ KIT (IF REQUIRED)

Prior to this upgrade, the site should have been audited to determine if the IIQ Kit (M3090NE) and the RF Ground Strap (M3090NF) have been installed.

The IIQ Kit includes parts and instructions for the installation of the Gradient Radial Support and the Bridge Support. It would be required for earlier versions of the WideOpen enclosure. If this upgrade has not been performed on the existing enclosure, then M3090NE should have been ordered and must be installed at this point of the EXCITE HD ugprade.

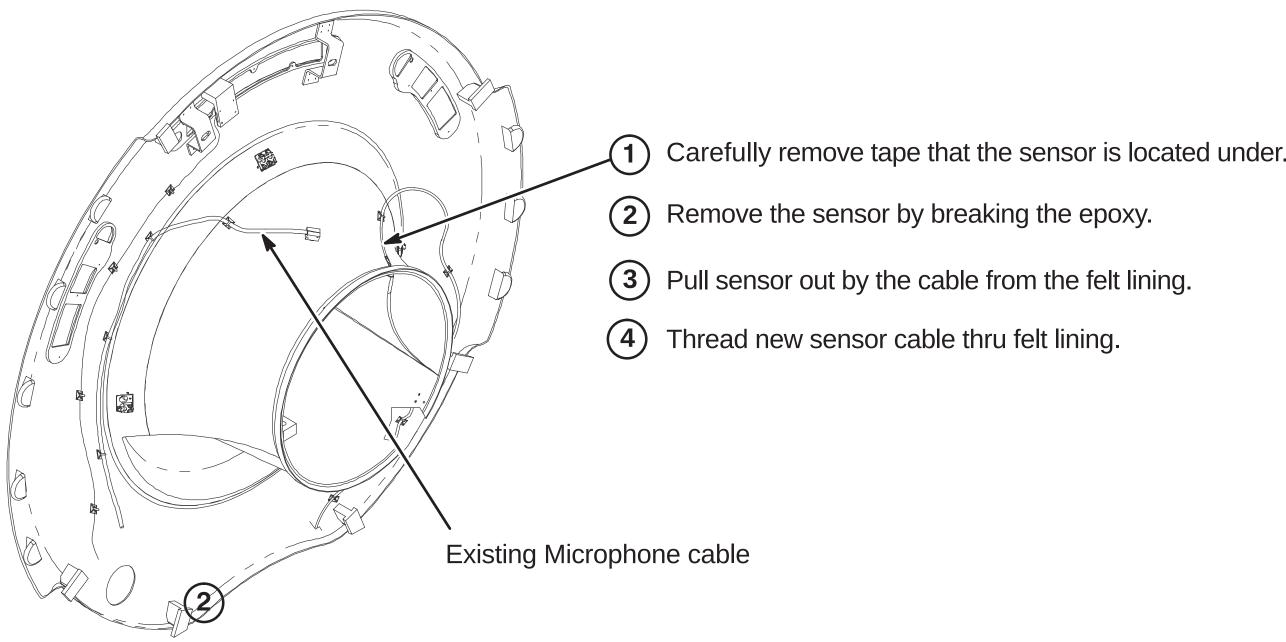
Refer to steering Direction 2401983, included in M3090NE, for guidance. Directions 2318581, *Signa Cx/K4 Magnet WideOpen Enclosure Bridge Support* and Direction 2319693, *Signa Gradient Radial Support Upgrade*, along with associated parts, are also included in M3090NE.

Also, the RF Ground Strap (M3090NF) must be installed if this upgrade hasn't been performed previously. This kit includes parts and instructions for installation of the RF Ground Strap. Refer to Direction 2319685, *BRM of CRM RF Shield Grounding Upgrade,* for instructions on installing this kit.

6–2–7 INSTALL NEW DUAL TEMPERATURE SENSOR CABLE, RUN 1260

6–2–7–1 Temperature Sensor Cable Removal

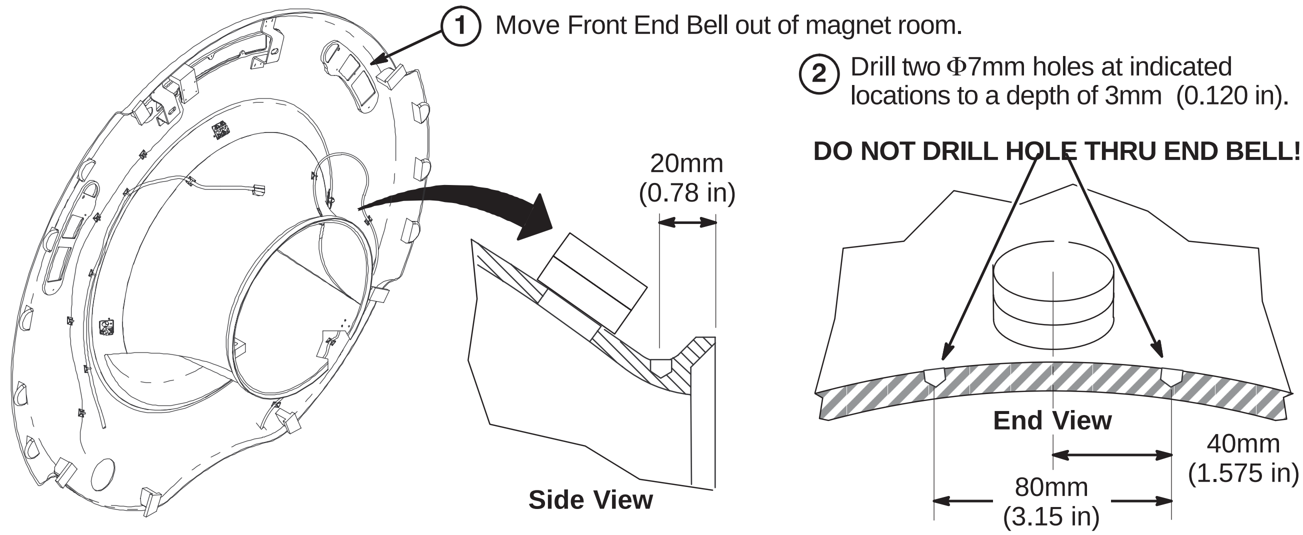
The site may have upgraded to a Dual Thermal Sensor Run 749 recently, but this cable has to be removed and replaced with new Dual Thermal Sensor cable, Run 1260 (5123812). EXCITE HD requires Run 1260 to be installed.



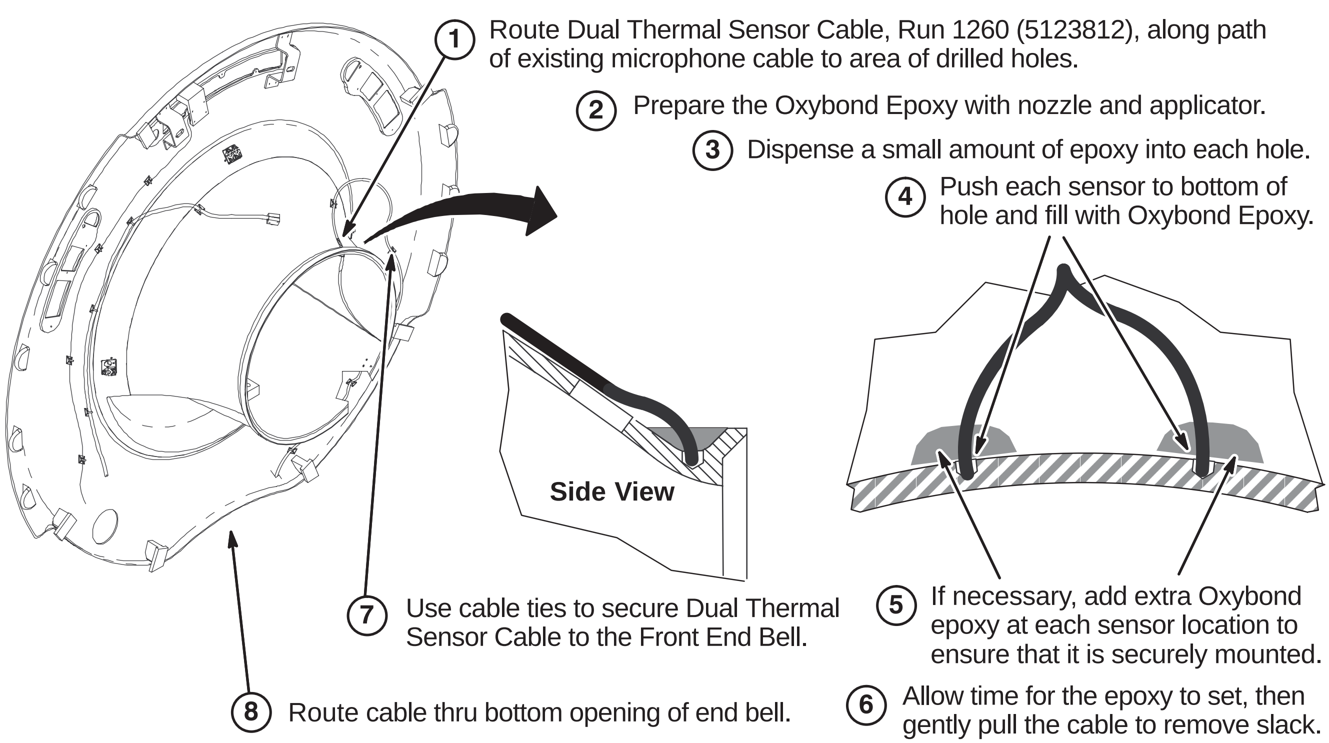
6–2–7–2 Prepare End Bell for Installation of Dual Thermal Sensor Cable

WARNING!

FERROUS MATERIAL HAZARD! DO NOT DRILL, HAMMER, CHISEL, ETC. OR USE ANY FERROUS TOOLS IN MAGNET ROOM UNLESS MAGNET IS RAMPED DOWN!



6–2–7–3 Attach Dual Thermal Sensor Cable

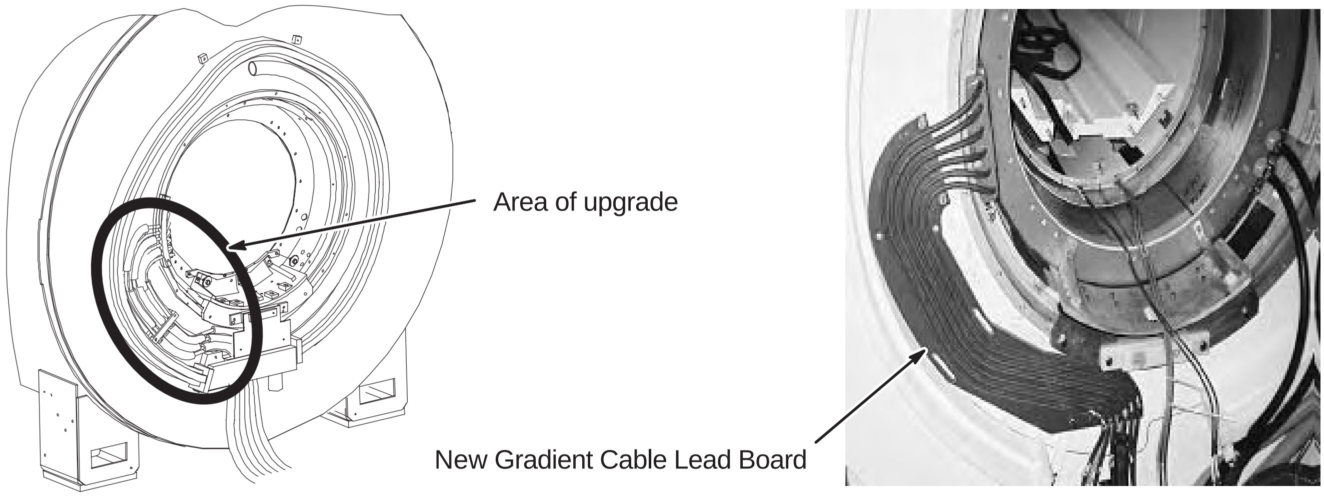


6–2–7–4 Install Front End Bell

Reverse the Front End Bell Removal procedure in this section to reinstall the front end bell.

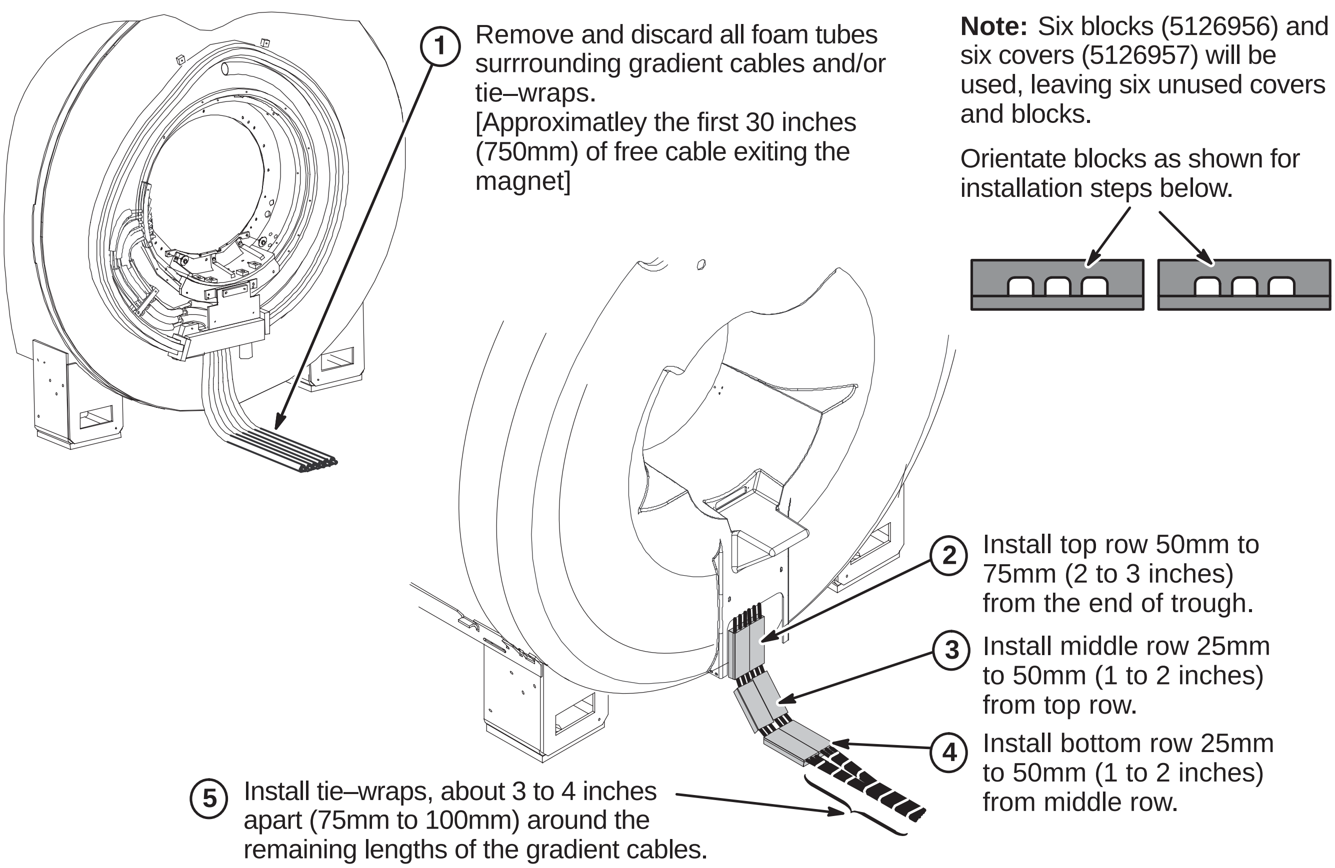
6–2–8 INSTALL BRM GRADIENT CABLE LEAD BOARD (MR/i or CV/i Systems Only)

The mounting of the gradient cables at the rear of the magnet must be upgraded. Refer to Direction 2404642, provided with Gradient Cable Lead Board collector (5110336), for instructions on how to install the new design.



6–2–9 REPLACE GRADIENT CABLE FOAM TUBES WITH CLAMP BLOCKS

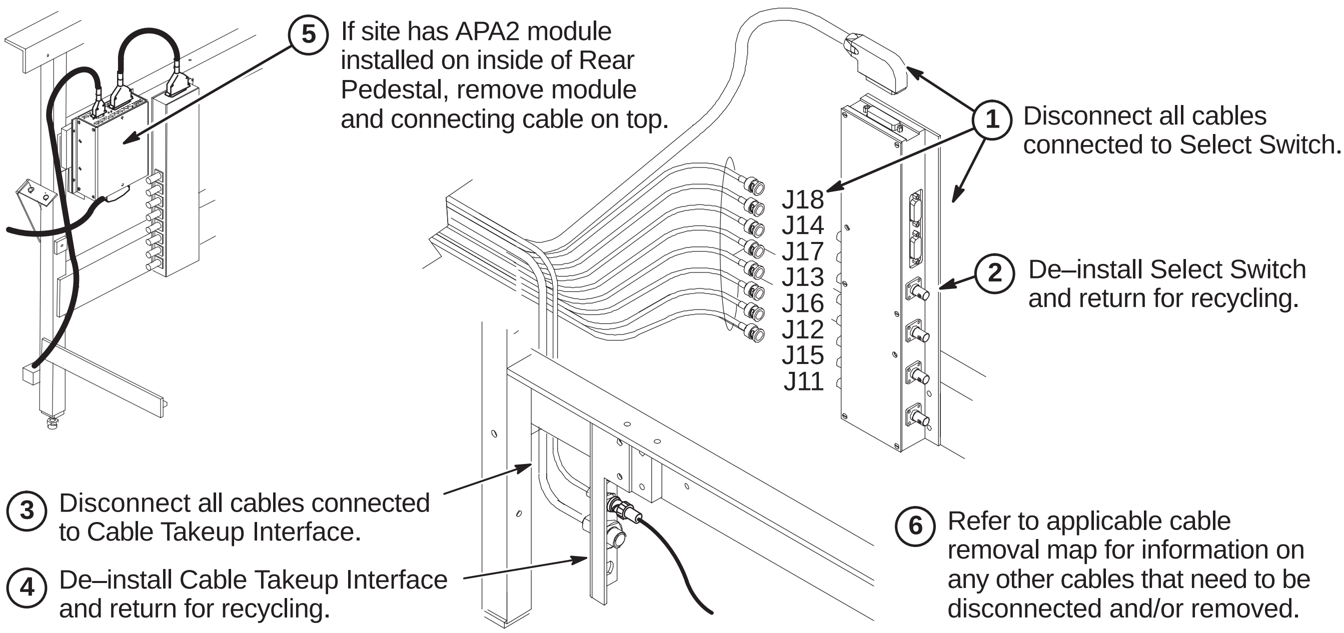
The existing foam tubes that are wrapped around the gradient cables at the rear of the magnet are to be removed and replaced with cable clamp blocks.



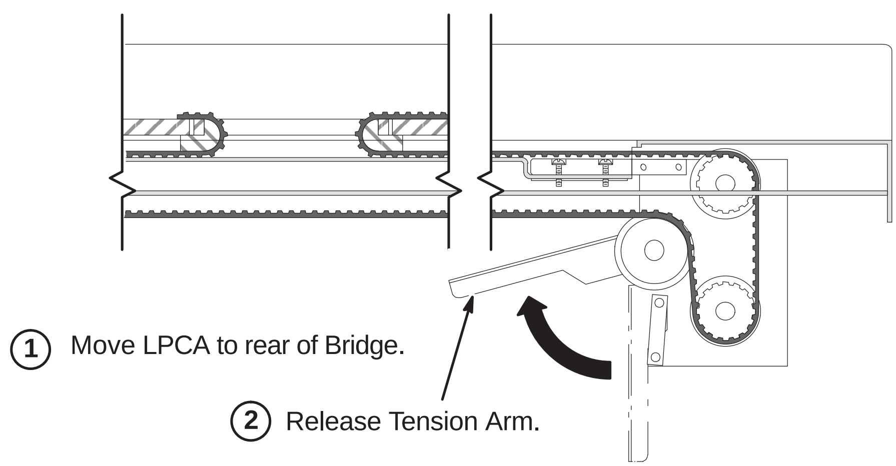
6–3 PREREQUISITE UPGRADE PROCEDURES FOR TwinSpeed ENCLOSURE

6–3–1 DE–INSTALL EXISTING LPCA & CABLE TRACK FOR TwinSpeed

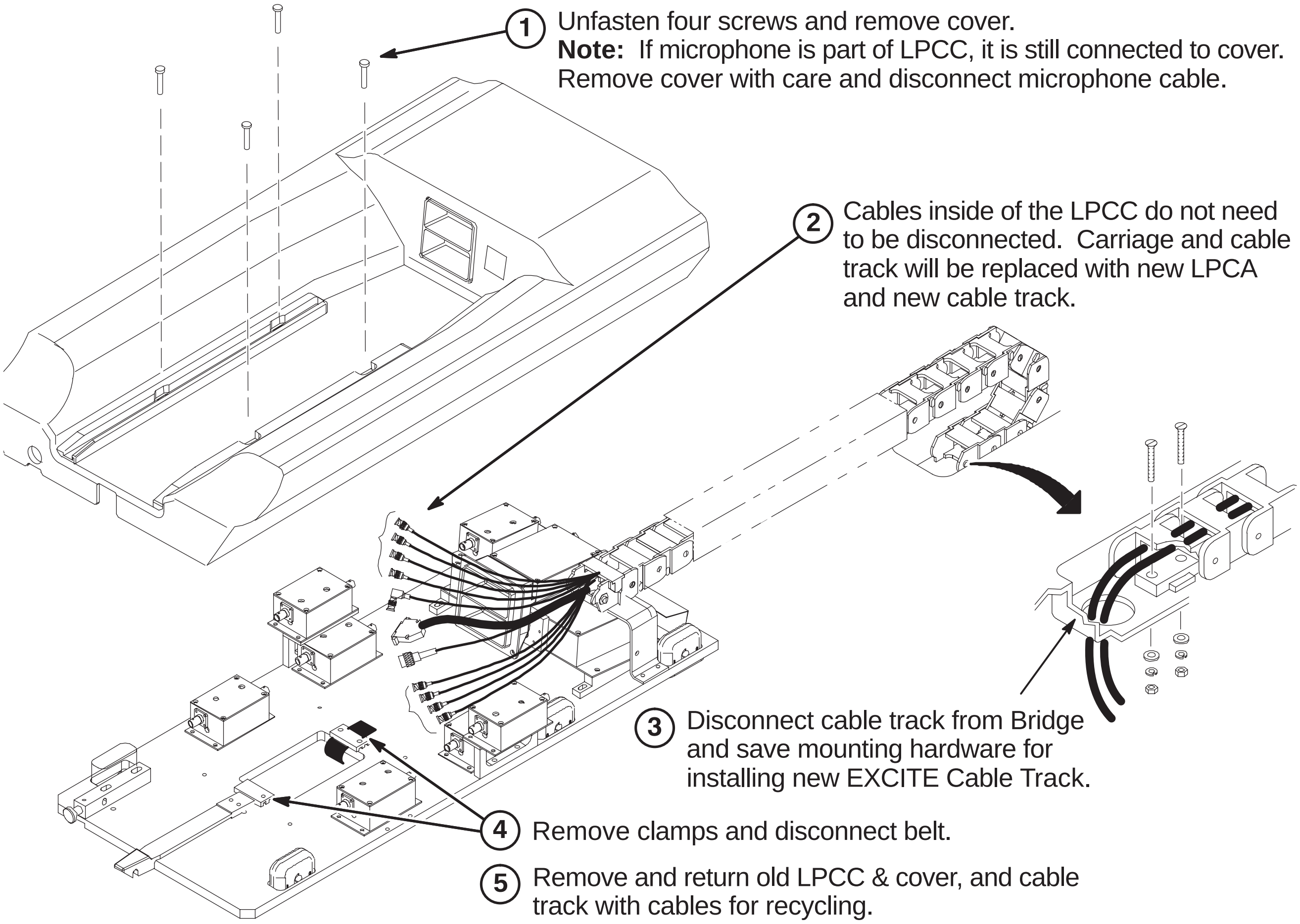
6–3–1–1 Disconnect Cables In Rear Pedestal



6–3–1–2 Relese Tension Arm Under Rear Pedestal



6–3–1–3 Disconnect Low Profile Carriage Cover & Cable Track

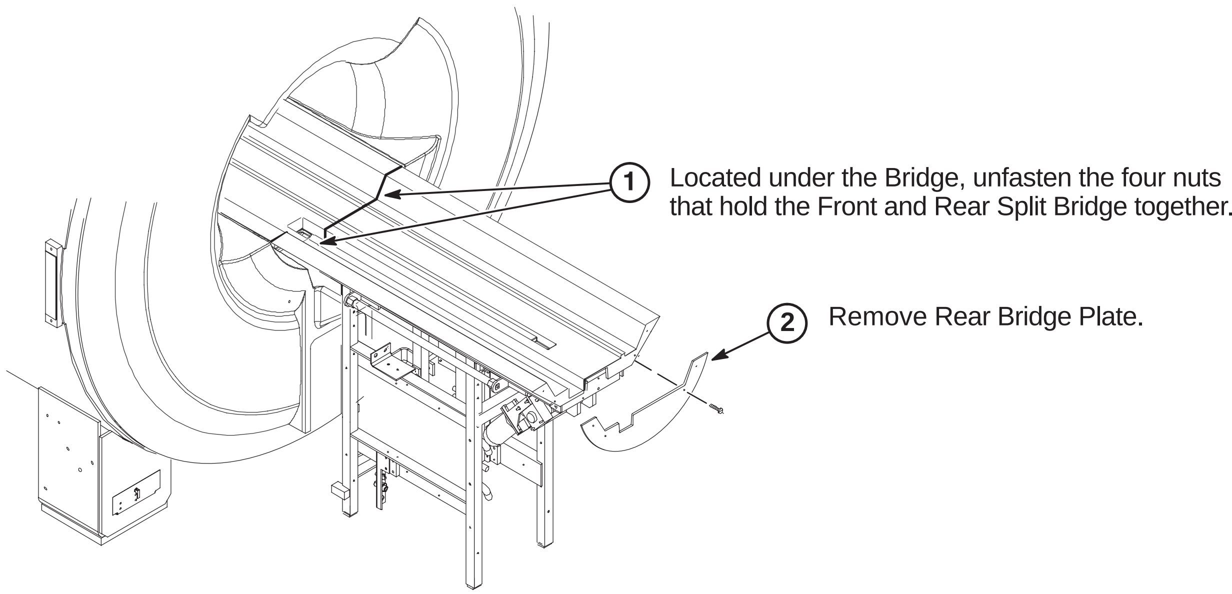


6–3–2 DE–INSTALL BRIDGE FOR TwinSpeed

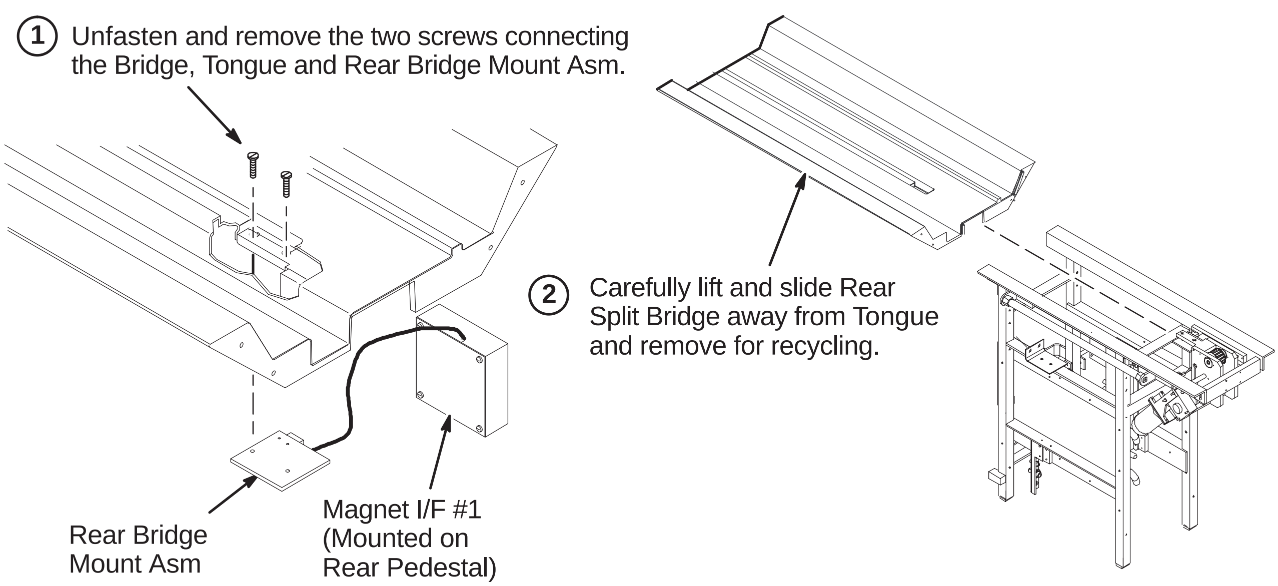
Upgrading to EXCITE TwinSpeed Release 12.x requires a new Front and Rear Split Bridge. The method of attachment at the front of the Bridge is different between a bridge that was installed for a forward production (new) installation and a Long bridge that was upgraded from a non–TwinSpeed WideOpen enclosure to TwinSpeed.

- Procedures 6–3–2–1 thru 6–3–2–3 shows the deinstallation for a forward production installed bridge.
- Procedure 6–3–2–4 thru 6–3–2–6 for the deinstallation of the upgraded bridge.

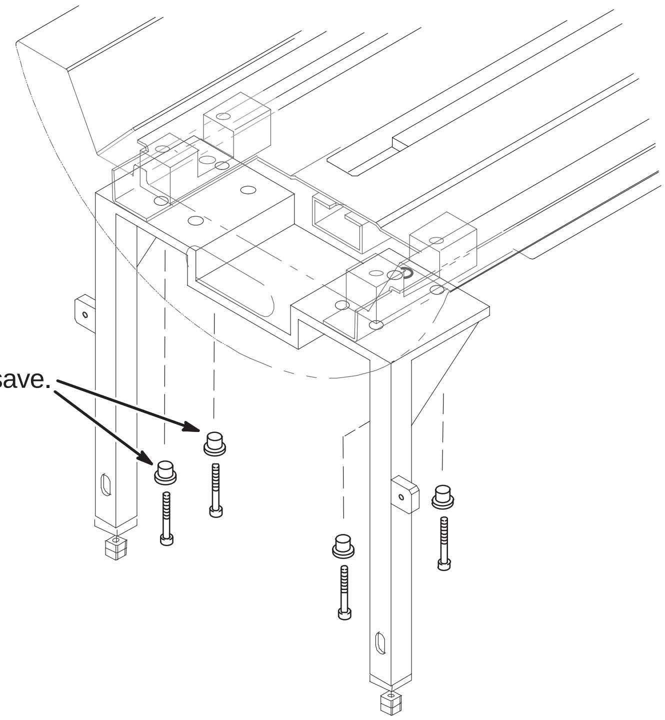
6–3–2–1 Unfasten Rear Split Bridge



6–3–2–2 Remove Rear Split Bridge

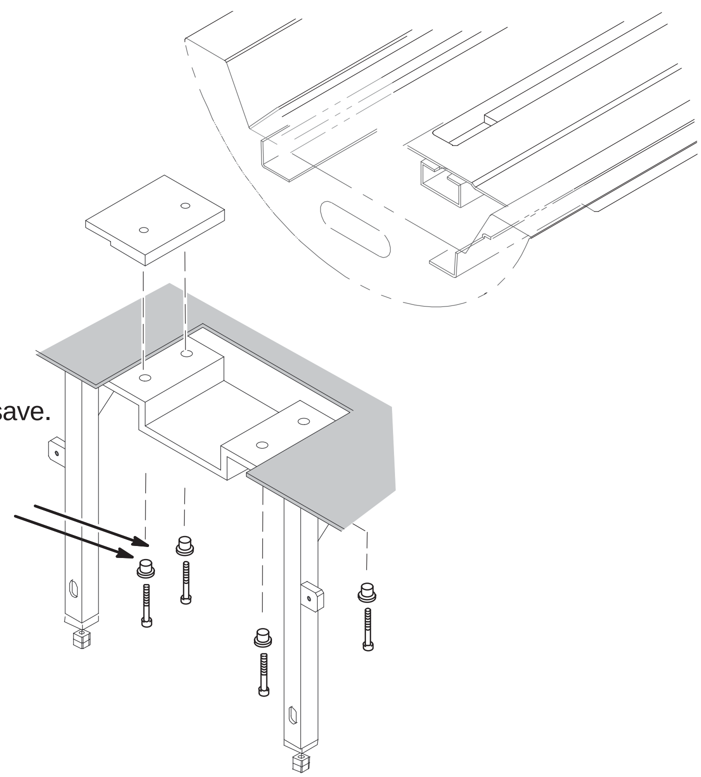


6–3–2–3 Remove Forward Production Installed Front Split Bridge



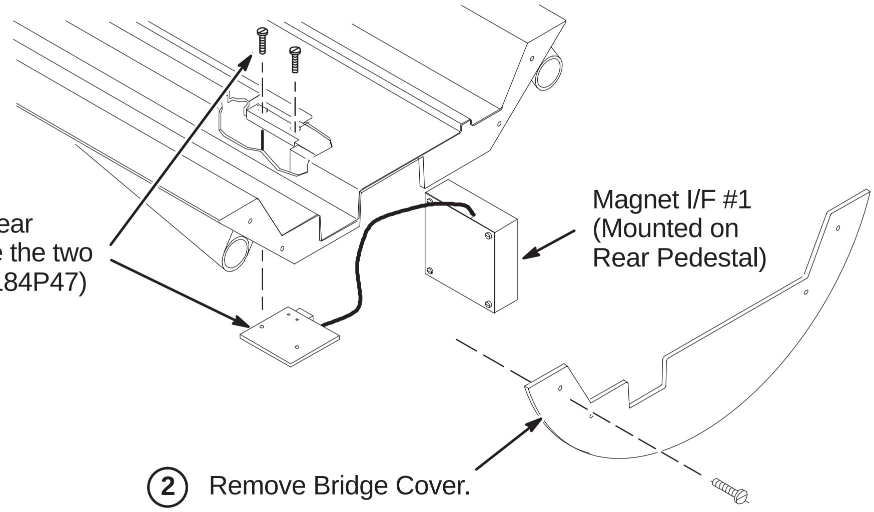
1. Unfasten the four (4) M6 x 35mm long Socket Head screws (2109867–17) and Support Bushings and save. These will be reused for installing the new Bridge.
2. Carefully lift and remove Front Split Bridge and return for recycling.

6–3–2–4 Unfasten Front of Upgraded Bridge



3. Unfasten the four (4) M6 x 35mm long Socket Head screws (2109867–17) and Support Bushings and save. These will be reused for installing the new Bridge.

6–3–2–5 Disconnect Bridge Mount Asm

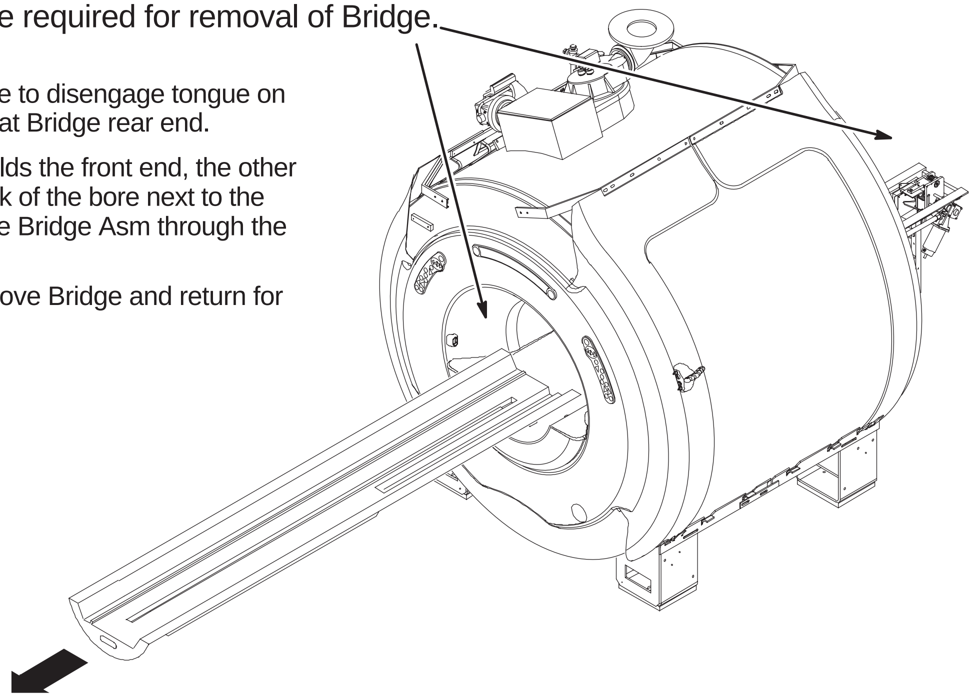


1. Unfasten the Bridge, Tongue and Rear Bridge Mount Asm (2132034). Save the two (2) 10–32 x 1–3/8 Screws (46–220184P47) for refastening later.
2. Remove Bridge Cover.

6–3–2–6 Remove Upgrade Long Bridge

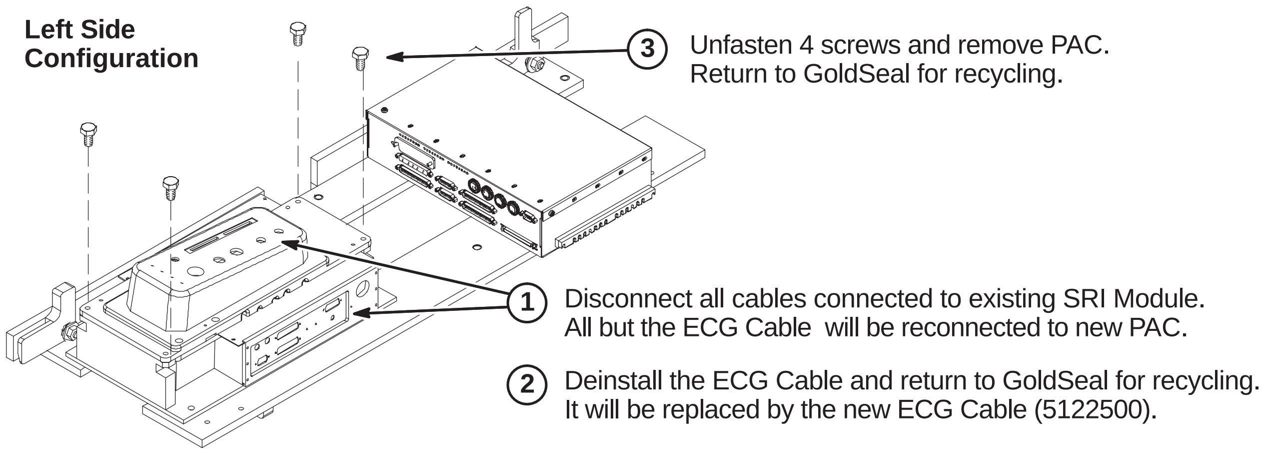
NOTE: Two people are required for removal of Bridge.

1. Lift front end of bridge to disengage tongue on long drive assembly at Bridge rear end.
2. While one person holds the front end, the other person stands in back of the bore next to the bridge and guides the Bridge Asm through the bore to the front end.
3. Carefully lift and remove Bridge and return for recycling.



6–3–3 DE–INSTALL PAC MODULE FOR TwinSpeed

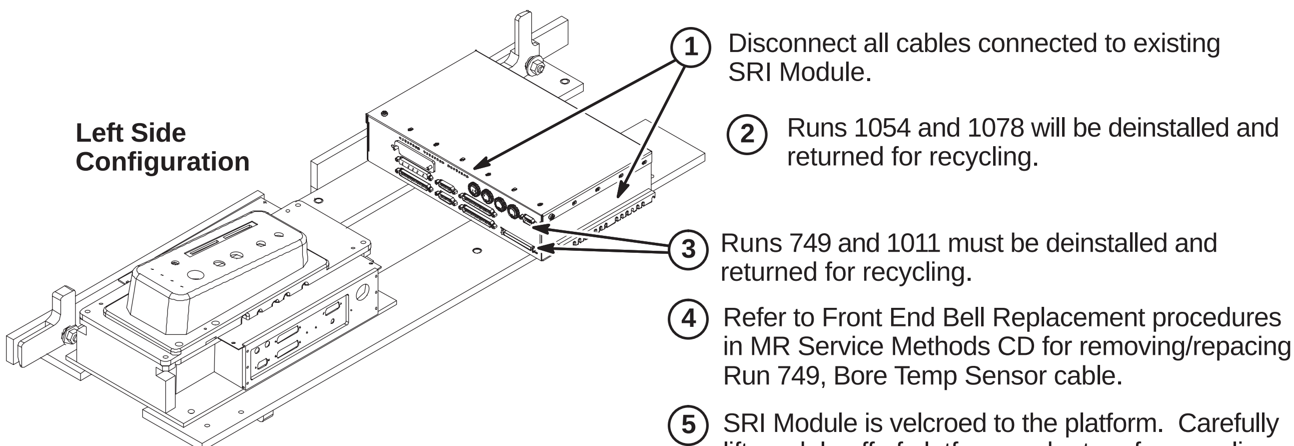
The existing PAC Module will be replaced by a new PAC Module with independent vector gating.



6–3–4 DE–INSTALL SRI MODULE FOR TwinSpeed

The existing SRI Module will be replaced by the SRI–3 Module.

NOTE: Existing Runs 749 and 1011 must be removed and replaced with new Runs 1260 and 1261. There is a functionality difference.



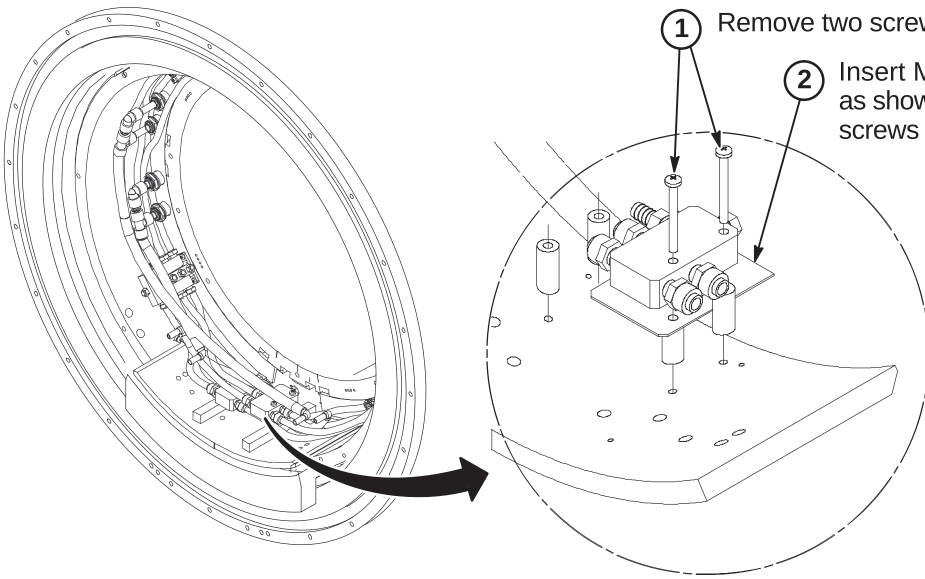
6–3–5 1.5T TwinSpeed RF BODY COIL REPLACEMENT

For replacement procedure, refer to applicable MR Service Methods CD–ROM and go to:

Replacements/1.5T Twin RF Body Coil Replacement Procedure.

When front End Bell is removed, perform Section 6–3–6 for installing the Manifold Protector plate.

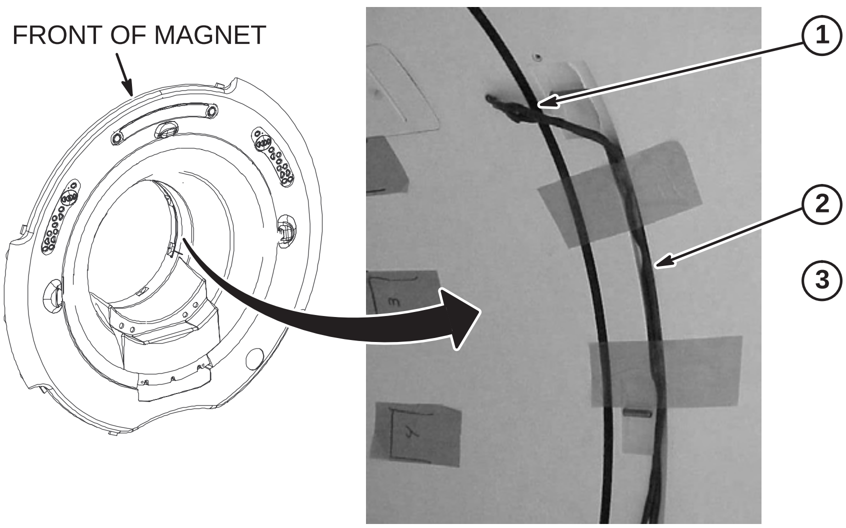
6–3–6 INSTALL MANIFOLD PROTECTOR PLATE



1. Remove two screws.
2. Insert Manifold Protector Plate (2390796) as shown and fasten in place with two screws removed in Step 1.
3. Refer back to MR Service Methods CD for instructions on re–installing end bell.

6–3–7 INSTALL NEW DUAL TEMPERATURE SENSOR CABLE, RUN 1260

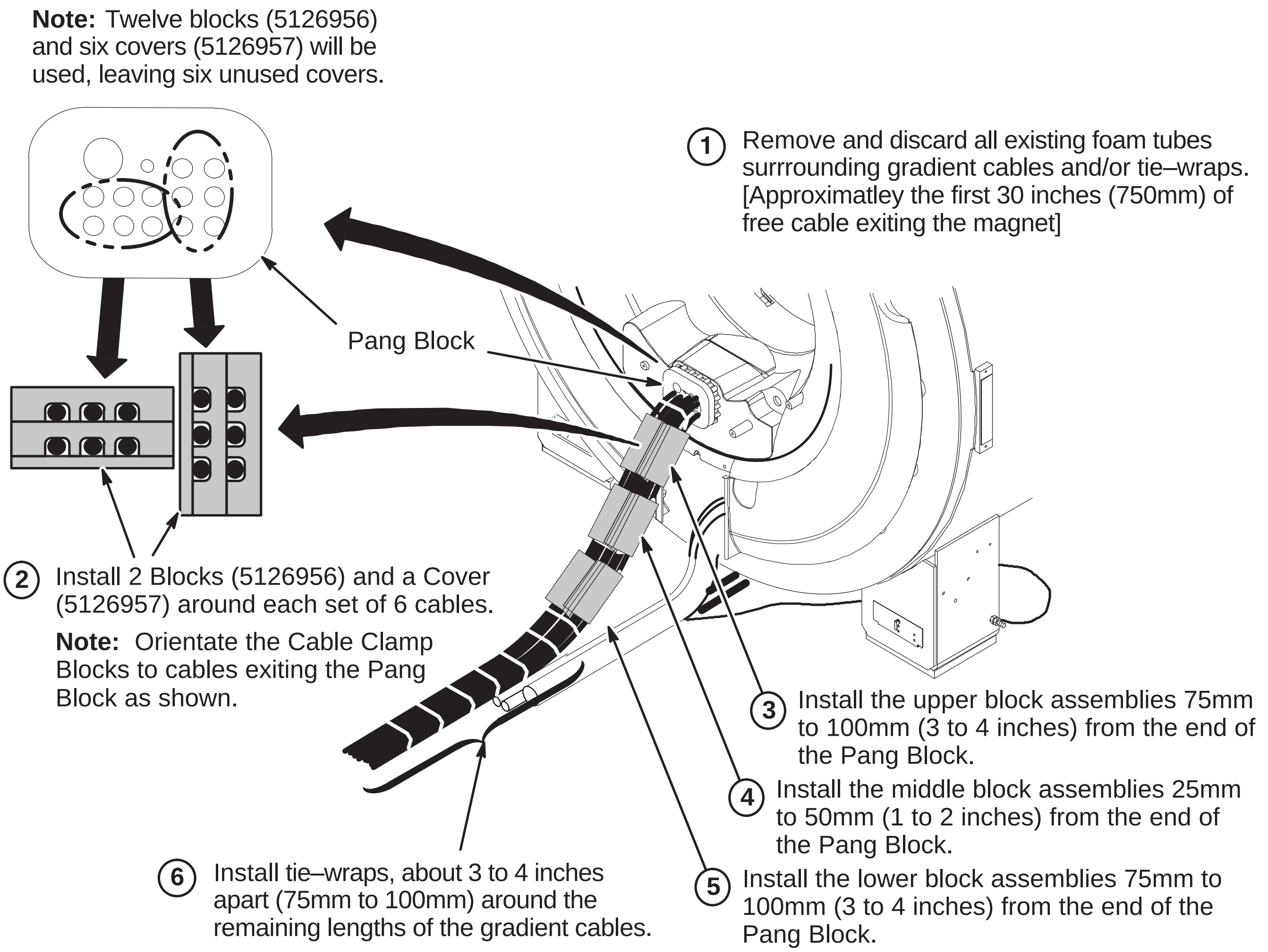
Install Run 1260 after the new RF Body Coil and front and rear end bells have been installed. Refer to the RF Body Coil Replacement procedure in Section 6–3–5 for more detailed steps.



1. Insert the new Thermal Sensor Cable, Run 1260 (5123812), in the provided holes at the top inside surface of the RF Coil and secure them to the Velcro Pads (46–136246P11 & 12).
2. Route Sensor Cable in groove.
3. Re–install Front Cover Strap over the End Bell gasket capturing the sensor cable in the groove.

6–3–8 REPLACE GRADIENT CABLE FOAM TUBES WITH CLAMP BLOCKS

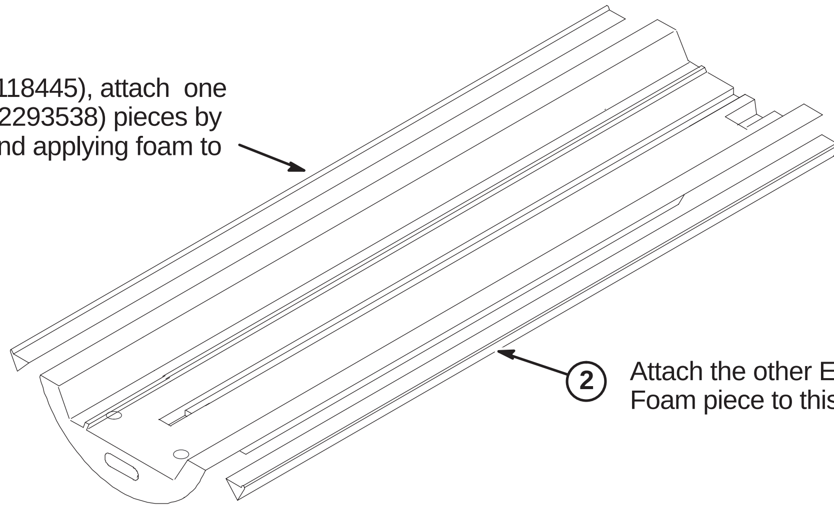
The existing foam tubes that are wrapped around the gradient cables at the rear of the magnet are to be removed and replaced with cable clamp blocks.



6–4 INSTALLATION OF NEW WideOpen AND TwinSpeed PARTS

6–4–1 ATTACH EDGE FOAM TO BRIDGE

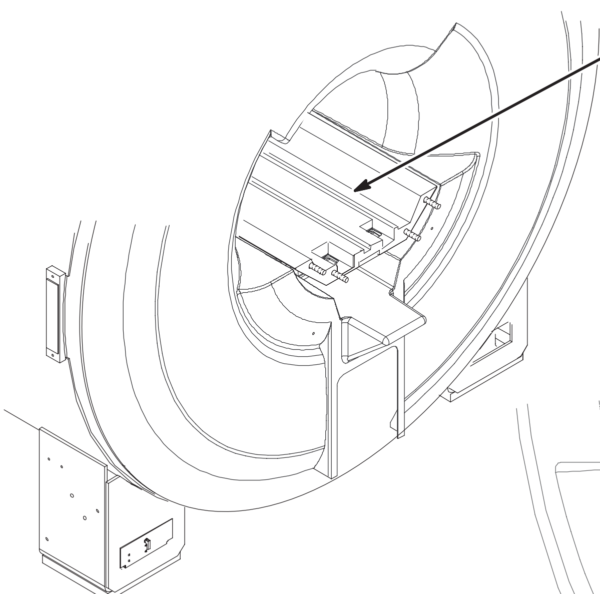
1 Starting at the front of Bridge (5118445), attach one of the Twin Bridge Edge Foam (2293538) pieces by removing the protective cover and applying foam to bottom side of Bridge surface.



2 Attach the other Edge Foam piece to this side.

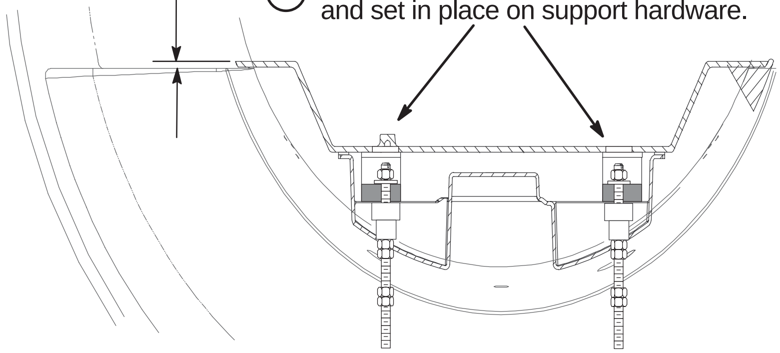
6–4–2 INSTALLATION OF FRONT BRIDGE For WideOpen

1 Install new Front Bridge Asm (5118445) into magnet.

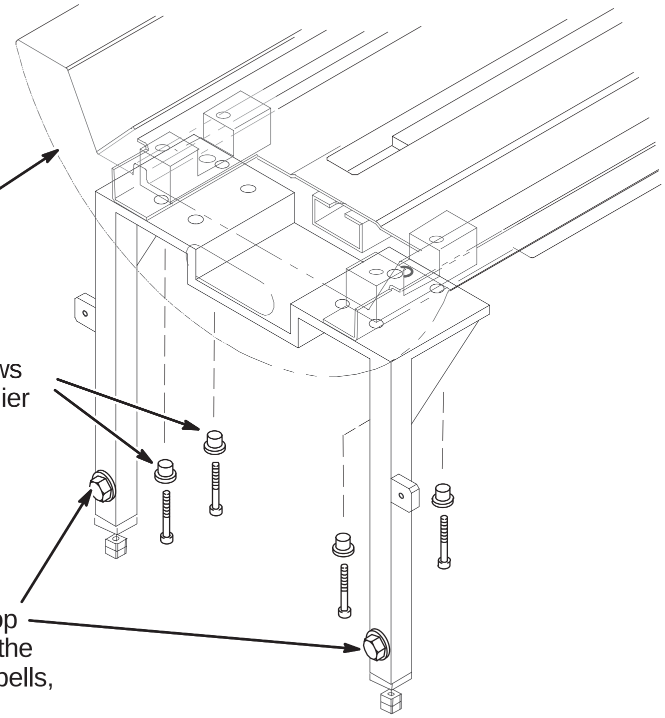


3 Adjust the support hardware until the Bridge's rop surface is 2mm to 4mm above the horizontal reference plane portion of the end bells,

2 At front of Bridge, align Bridge with studs and set in place on support hardware.



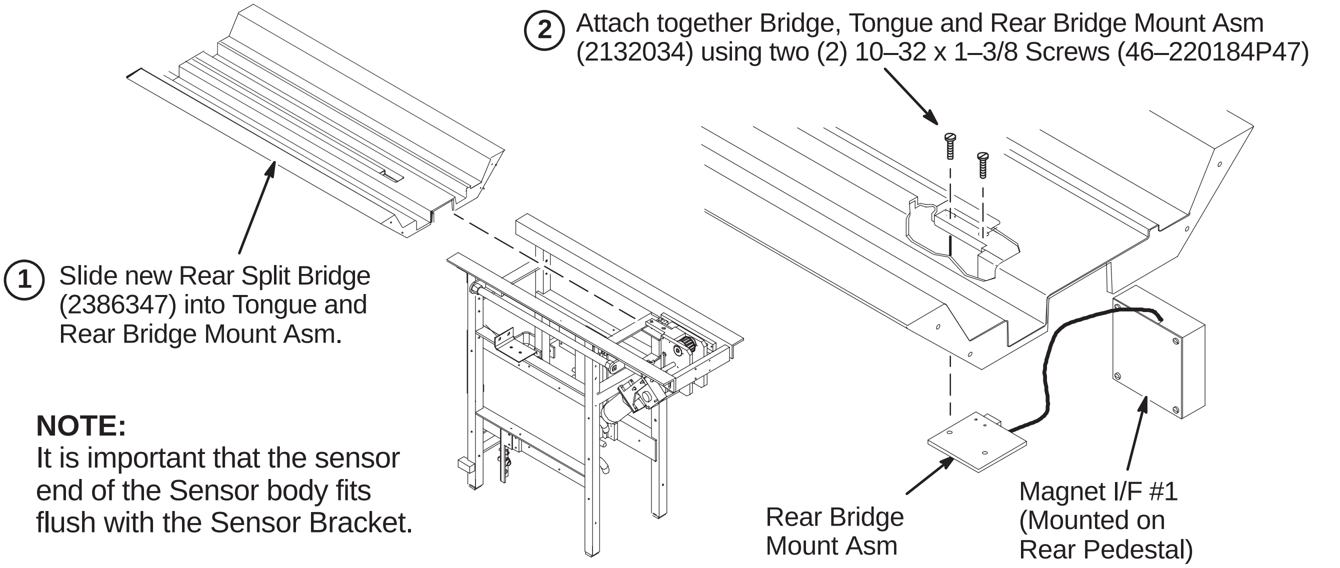
6–4–3 INSTALLATION OF FRONT BRIDGE For TwinSpeed



1. Install new Front Bridge Asm (5118445) into magnet.
2. Install the four (4) M6 x 35mm long Socket Head screws (2109867–17) and Support Bushings removed in earlier procedure.
3. Adjustment screws for leveling the Bridge’s top surface which should be 2mm to 4mm above the horizontal reference plane portion of the end bells,

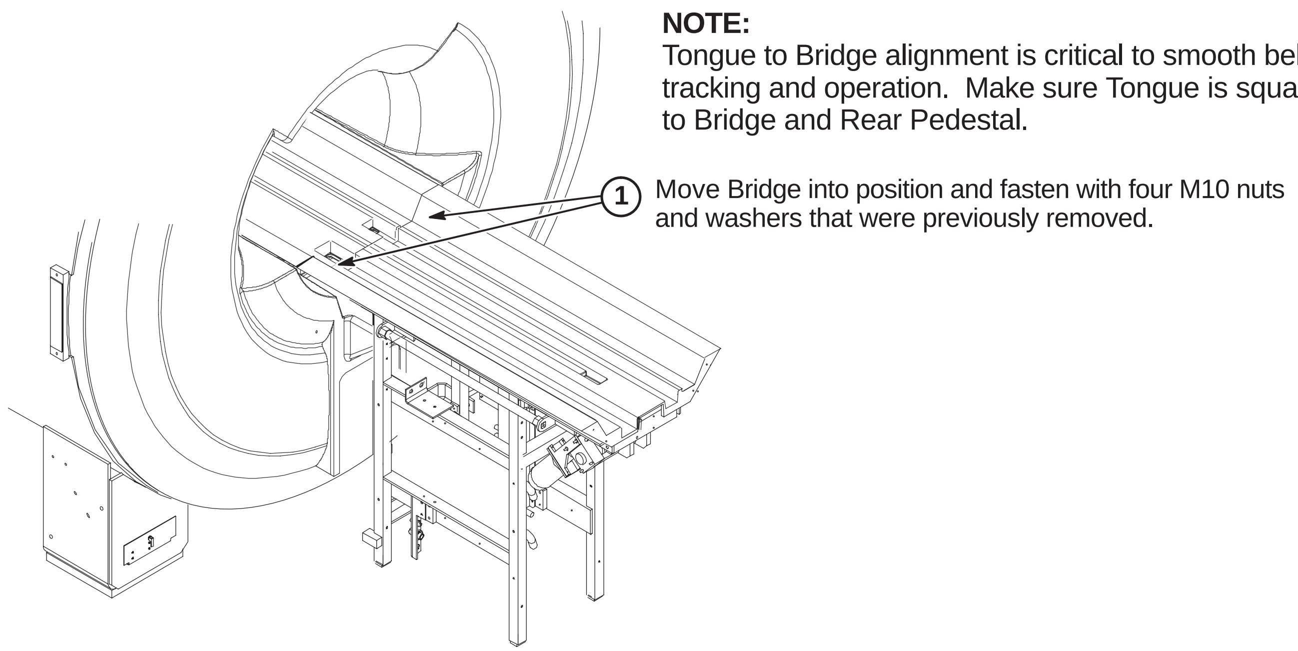
6–4–4 INSTALL REAR SPLIT BRIDGE

6–4–4–1 Install Rear Split Bridge on Rear Pedestal



6–4–4–2 Connect Rear Split Bridge to Front Split Bridge

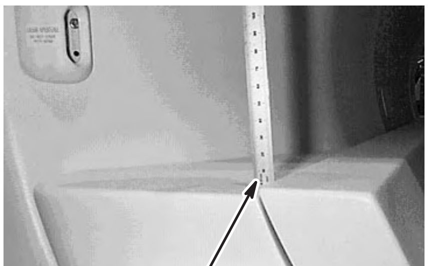
NOTE:
Tongue to Bridge alignment is critical to smooth belt tracking and operation. Make sure Tongue is square to Bridge and Rear Pedestal.



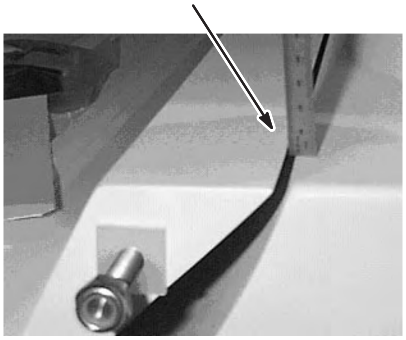
6–4–4–3 Level Bridge

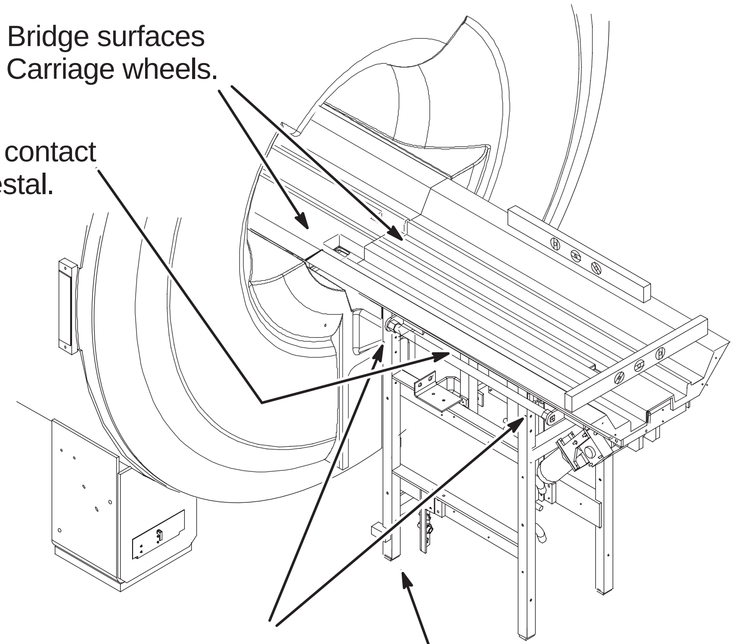
NOTE:
Make sure that surfaces between front and rear Bridge surfaces are even. This will provide smooth travel for the Carriage wheels.

NOTE:
Rear Pedestal must be adjusted to maintain full contact between bottom of Bridge and top of Rear Pedestal.



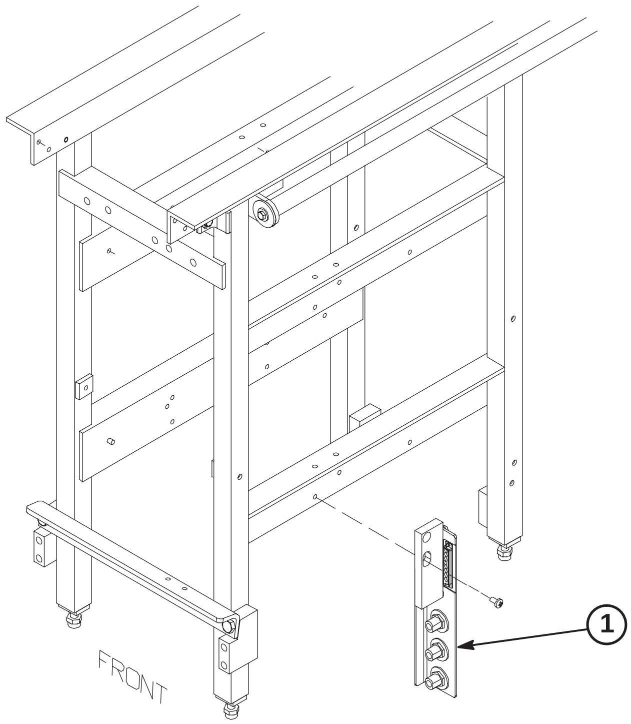
2mm to 4mm above End Bell surface.





1. Level Bridge and make sure that Bridge surface is no less than 2mm and no more than 4mm above the surface of front and rear End Bells, as indicated in photos at left. Tighten Vertical Adjustment Fasteners.
2. Adjust leg screws on both sides as needed to level Rear Pedestal.

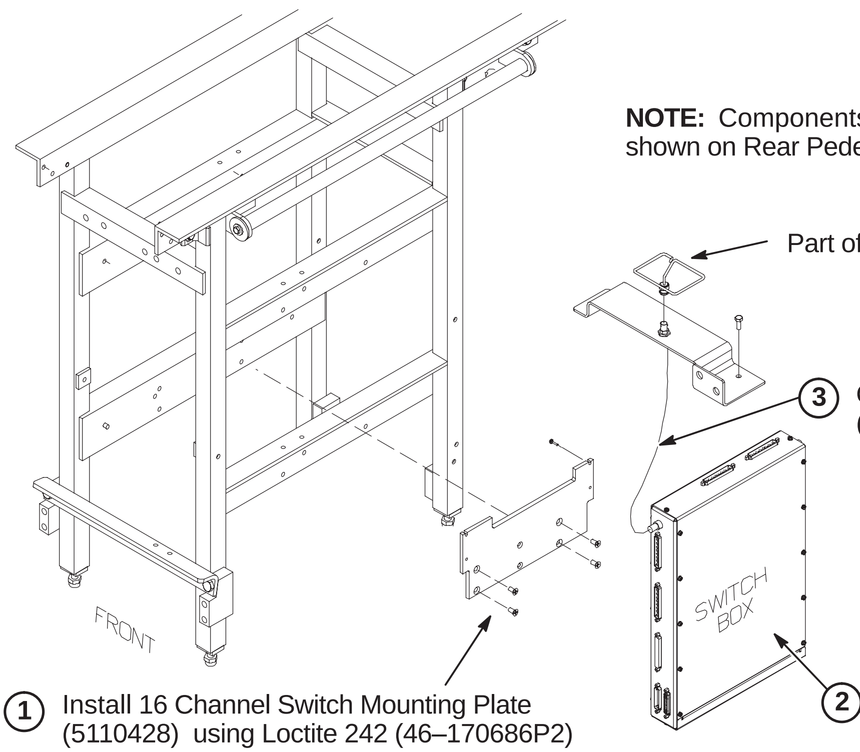
6–4–5 INSTALL BULKHEAD INTERFACE



NOTE: Components and cables not shown on Rear Pedestal for clarity.

Install new Bulkhead Interface (5113250) using Loctite 242 (46–170686P2) and two M6 x 12mm pane Head Screws (2109873–24).

6–4–6 INSTALL 16 CHANNEL SWITCH



NOTE: Components and cables not shown on Rear Pedestal for clarity.

Part of Rear Pedestal

Connect Run 1204 (5116105) as shown.

Install 16 Channel Switch Mounting Plate (5110428) using Loctite 242 (46–170686P2) and four M6 x 12mm Slotted Flat Head Screws (25120369–86) in the four larger outside holes.

Attach 16 Channel Switch (5109645) to Mounting Plate using Loctite 242 (46–170686P2) and four M3 x 12mm Pan Head screws (46–312342P24).

6–4–7 HYBRID SPLITTER UPGRADE

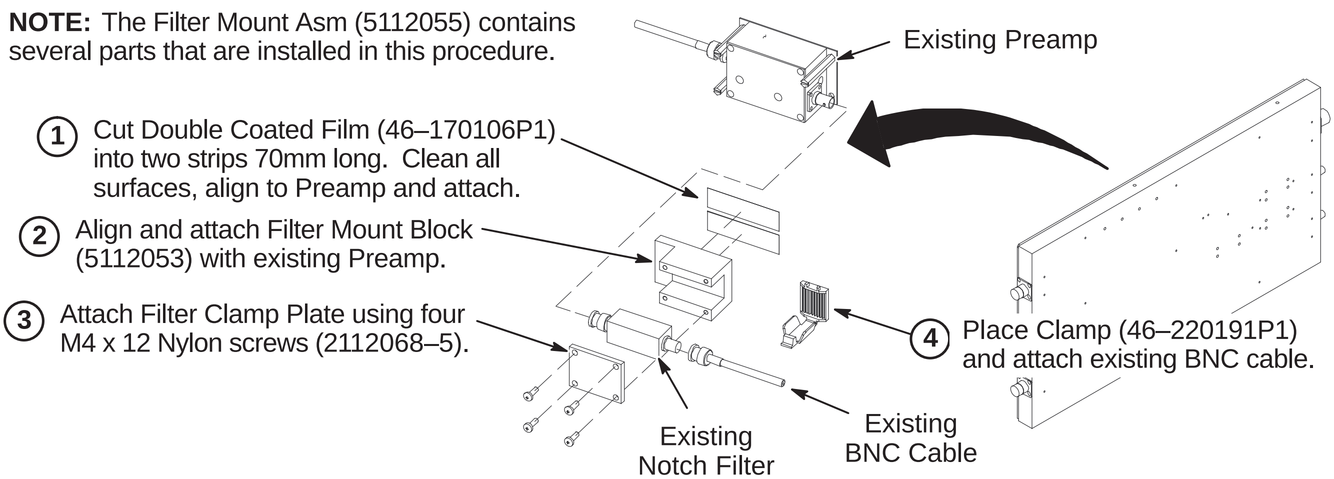
Depending on Rear Pedestal design, the Body Splitter may be in the horizontal or vertical position. Also:

- Refer to Section 6–4–7–1 for systems that do not have mount assembly for the Notch Filter
- Refer to Section 6–4–7–2 for instructions on installing Isolation Filters on systems with Netcom Hybrid Splitter
- Refer to Section 6–4–7–3 for instructions on installing Isolation Filters on systems with Teledyne Splitter

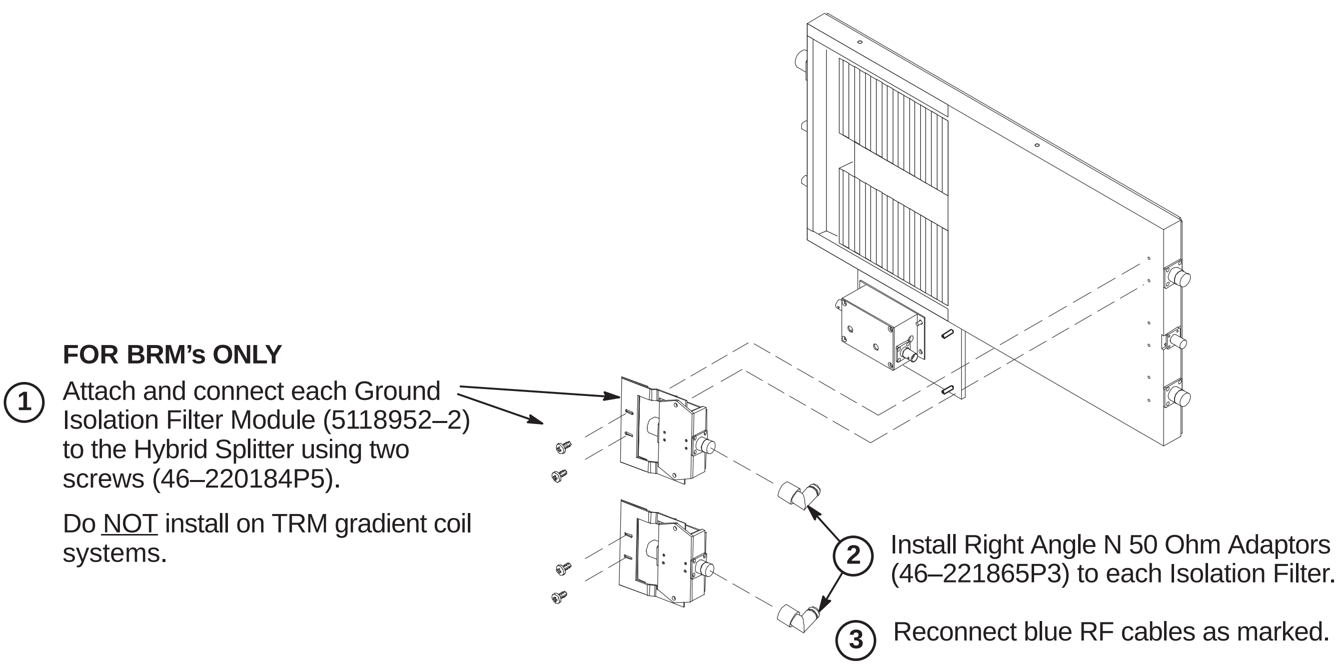
The Ground Isolation Filter Modules are only installed on systems with BRM gradient coils. Do NOT install on systems with TRM gradient coils (TwinSpeed).

6–4–7–1 Install Filter Mount Asm to Body Hybrid Splitter

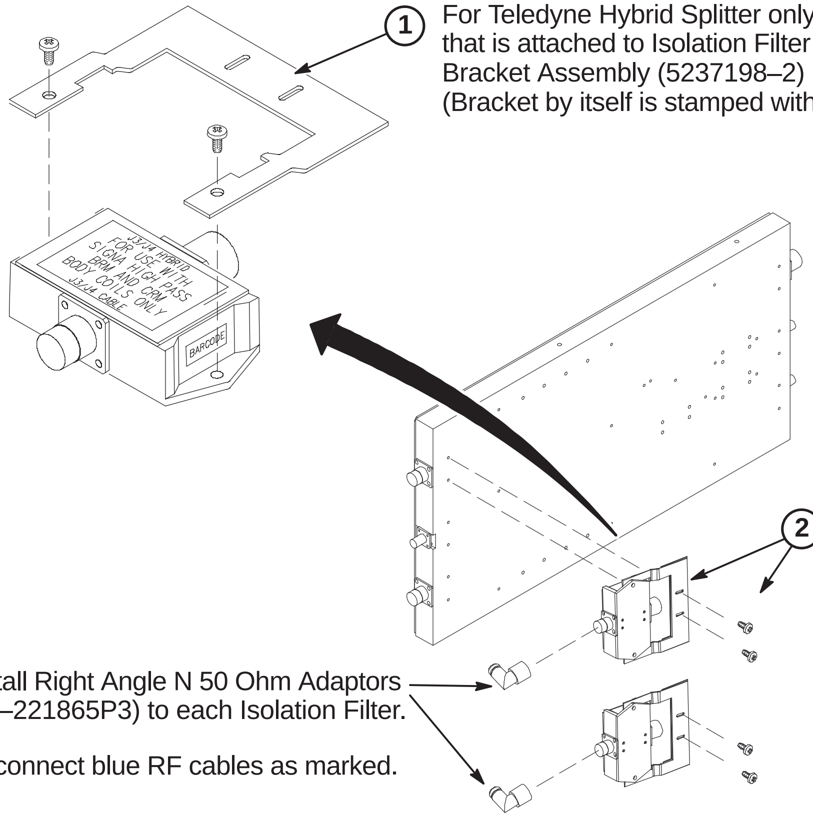
This Filter Mount assembly may not be required on newer Netcom Hybrid Spitter as it may already have a Notch Filter mounting clamp of a different style installed.



6–4–7–2 Install Isolation Filters to Netcom Body Hybrid Splitter (BRM Only)



6–4–7–3 Install Isolation Filters to Teledyne Body Hybrid Splitter (BRM Only)



1. For Teledyne Hybrid Splitter only, remove existing bracket that is attached to Isolation Filter and replace with Mounting Bracket Assembly (5237198–2) provided with upgrade kit. (Bracket by itself is stamped with 2254629)

FOR BRM's ONLY

2. Attach and connect each Ground Isolation Filter Module (5118952) to the Hybrid Splitter using two screws (46–220184P5) for each module.

   Do NOT install on TRM gradient coil systems.

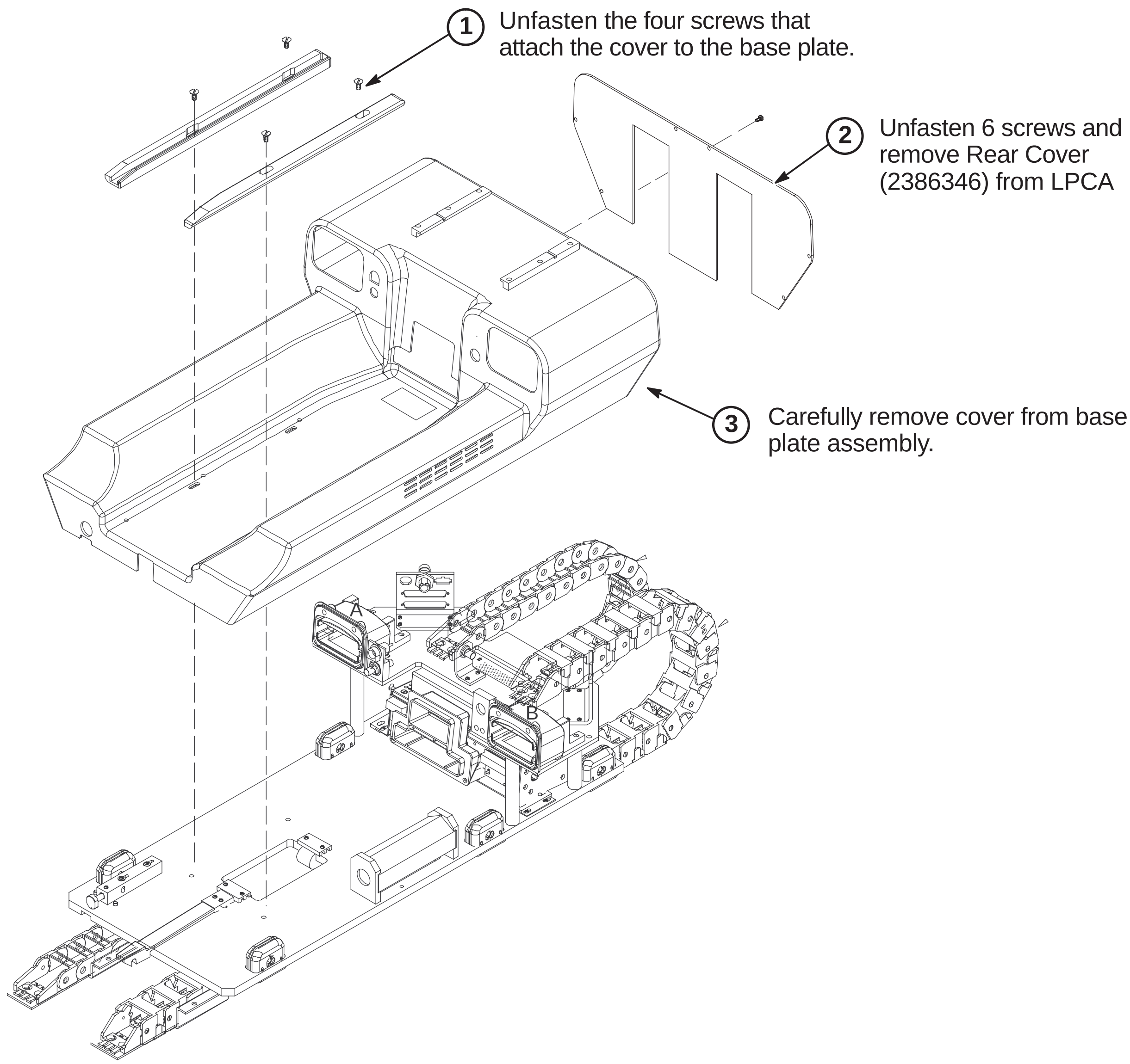
3. Install Right Angle N 50 Ohm Adaptors (46–221865P3) to each Isolation Filter.

4. Reconnect blue RF cables as marked.

6–4–8 INSTALL LPCA AND CABLE TRACK(S)

The WideOpen style enclosure will be upgraded to an LPCA with a dual cable track. The S–Series and Cx magnet enclosures will install a new LPCA with only one cable track.

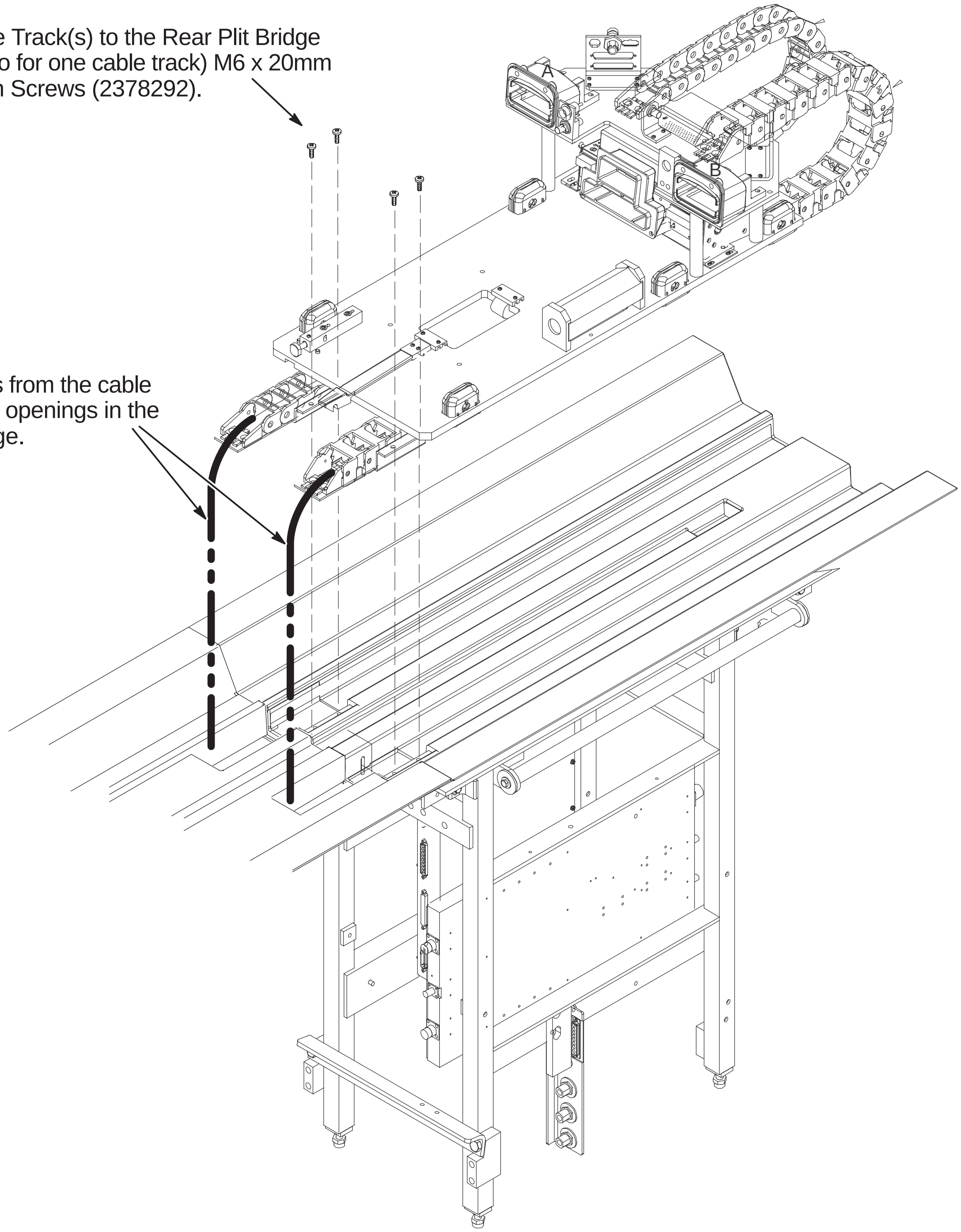
6–4–8–1 Remove LPCA Cover



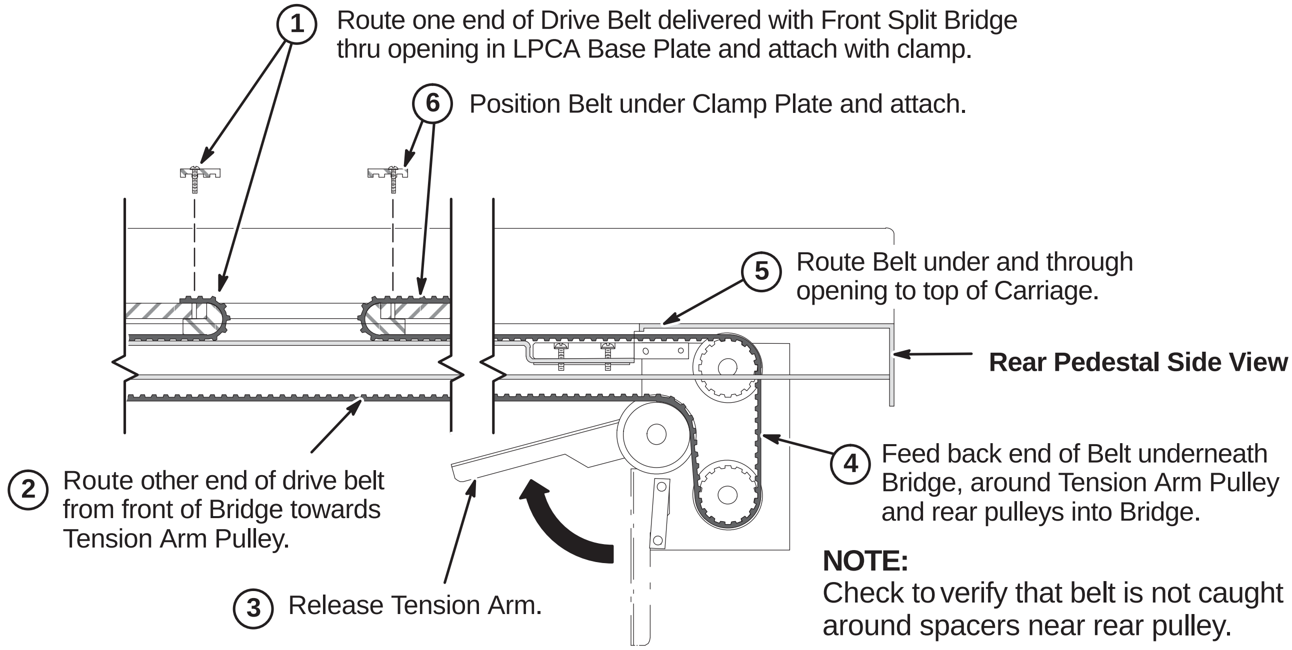
6–4–8–2 Attach Cable Track(s) to Bridge

2 Attach the Cable Track(s) to the Rear Plit Bridge using four (or two for one cable track) M6 x 20mm Pan Head Nylon Screws (2378292).

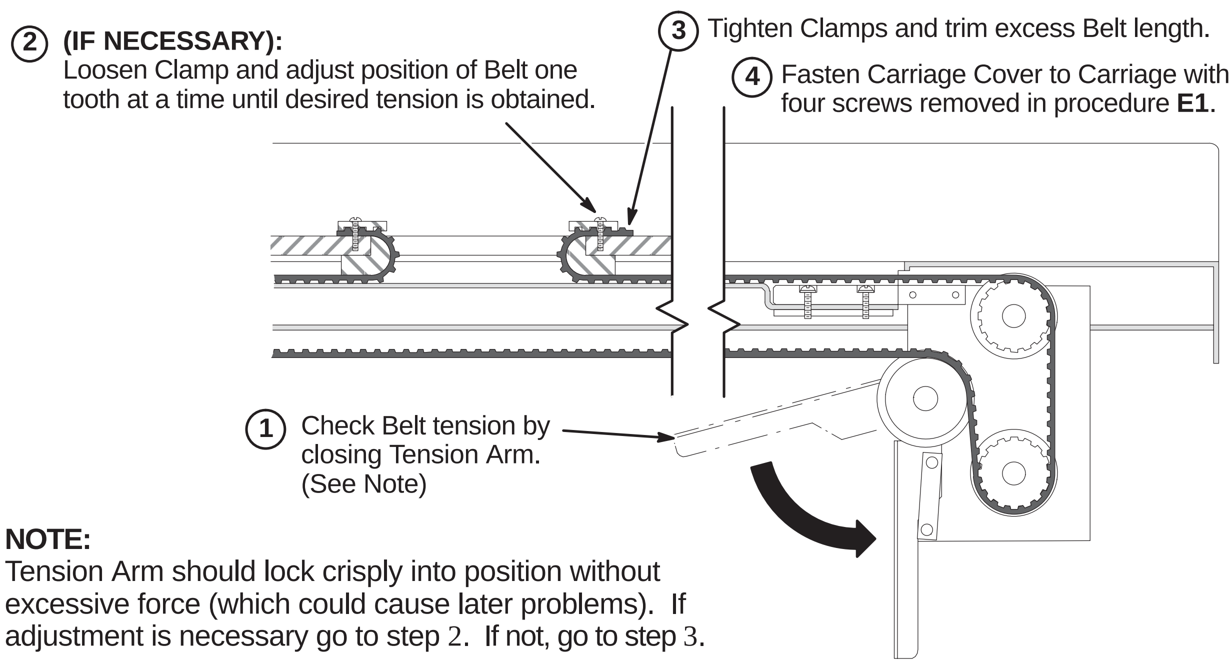
1 Feed the cables from the cable track(s) thru the openings in the Front Split Bridge.



6–4–8–3 Route and Attach Drive Belt



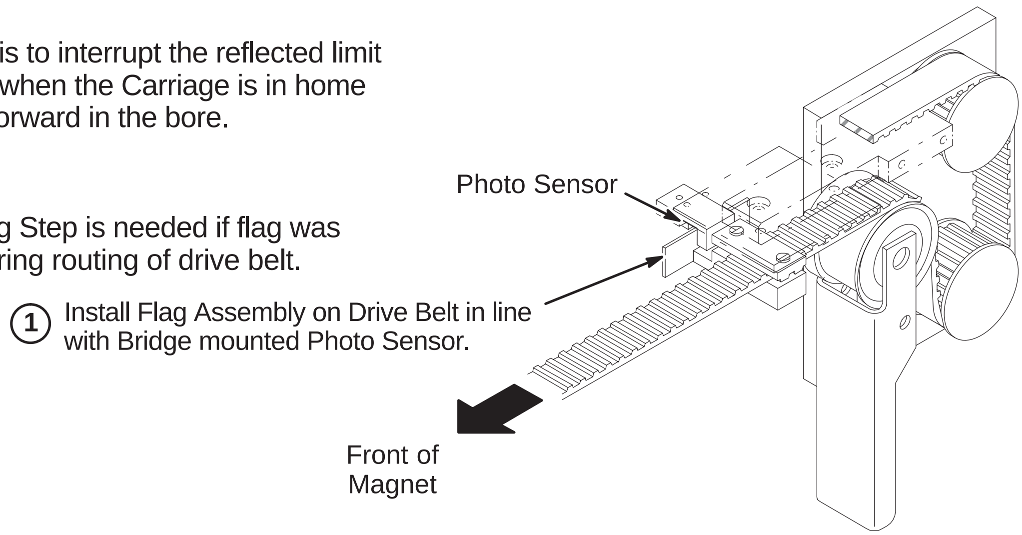
6–4–8–4 Final Adjustment of Drive Belt



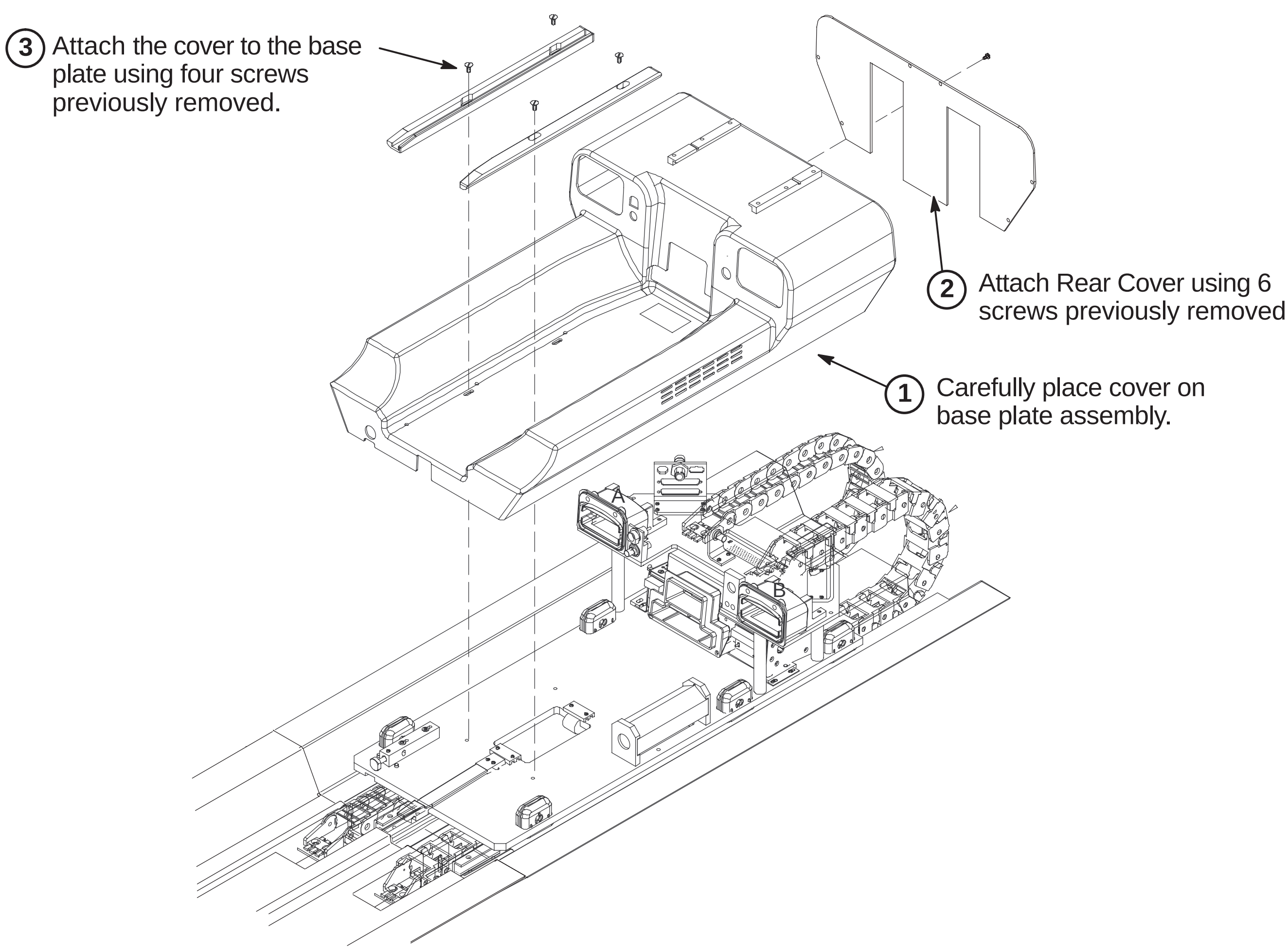
6–4–8–5 Installation of Sensor Flag

NOTE:
Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

NOTE:
The following Step is needed if flag was removed during routing of drive belt.

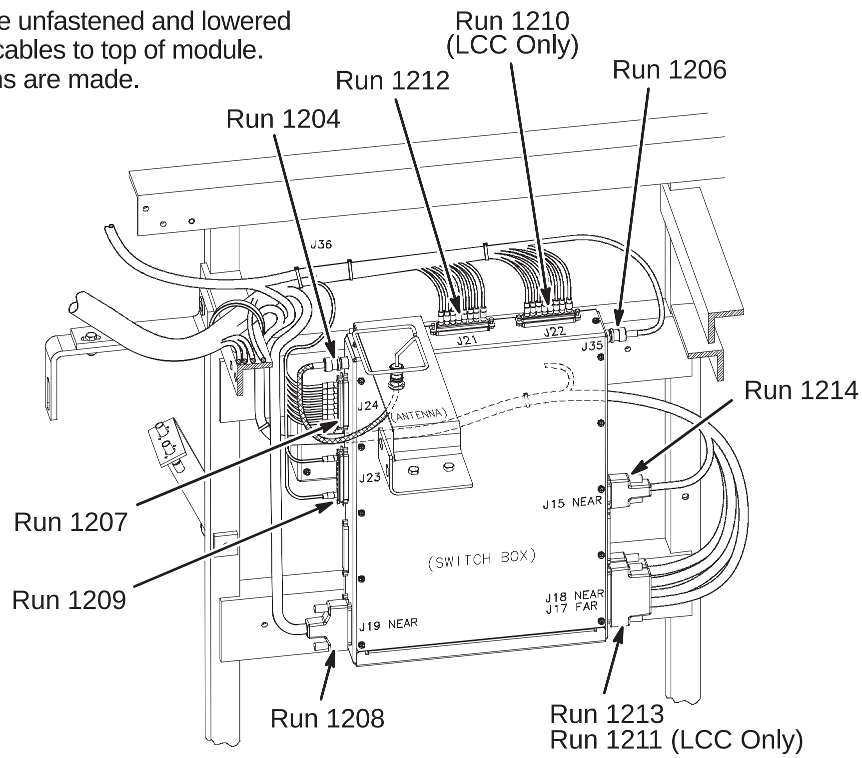


6–4–8–6 Installation of LPCA Cover



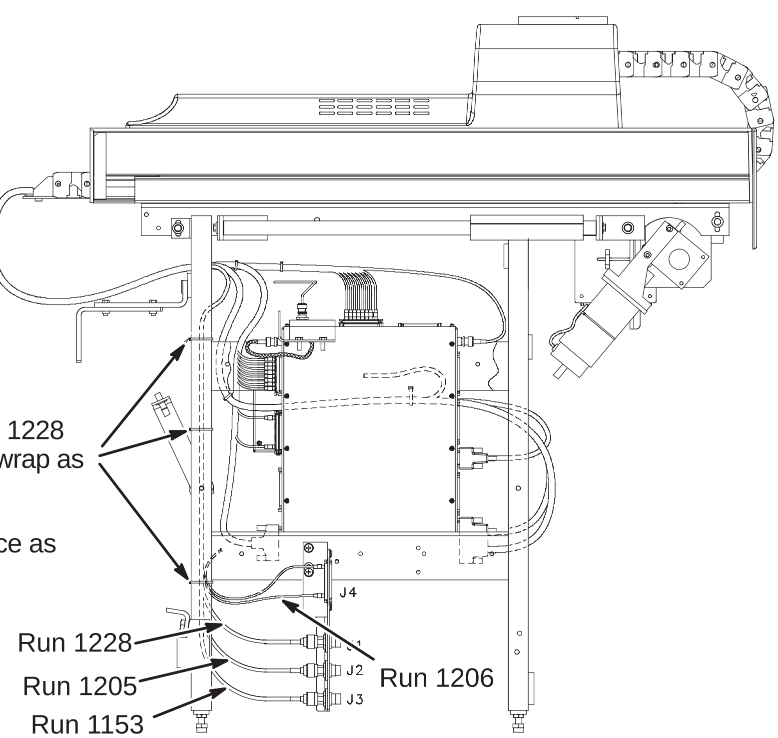
6–4–8–7 Connect Cables from Cable Track to 16 Channel Switch Module

1. 16 Channnel Switch may need to be unfastened and lowered from mounting position to connect cables to top of module. Re–mount module after connections are made.
2. Connect cable Runs as indicated.



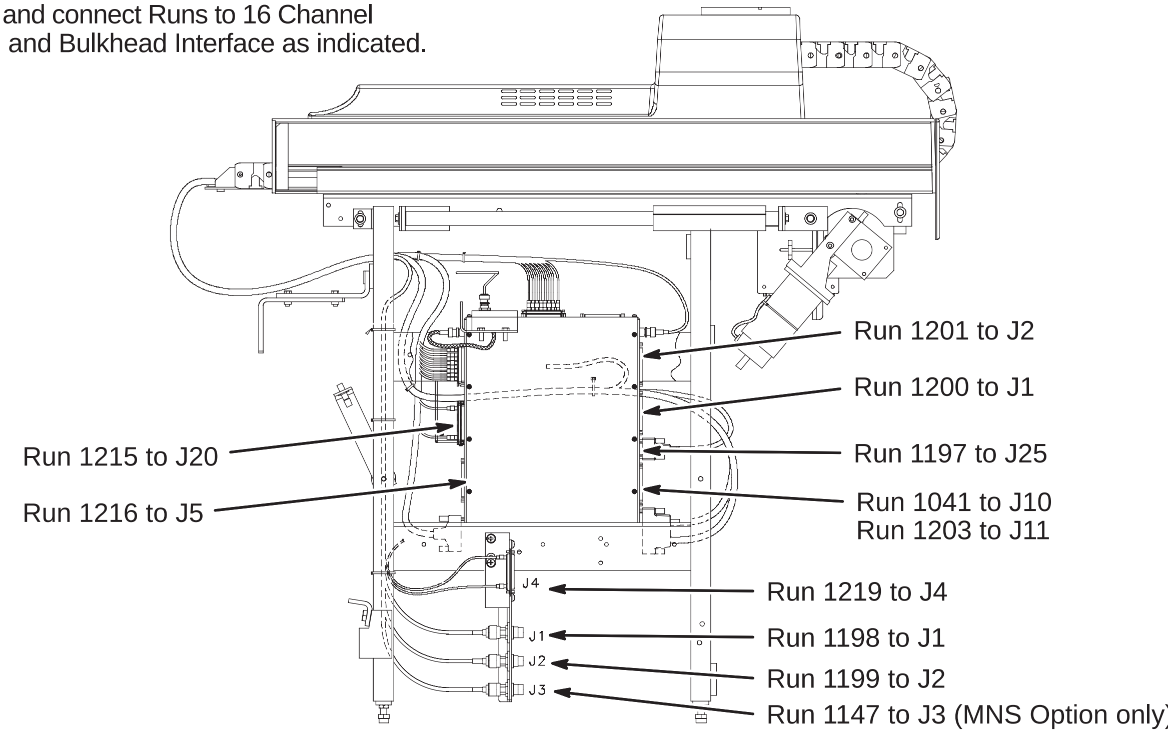
6–4–8–8 Connect Cables from Cable Track to Bulkhead Interface

1. Route Runs 1153, 1205, 1206, and 1228 along leg of Rear Pedestal and ty–wrap as indicated to leg.
2. Connect Runs to Bulkhead Interface as indicated.



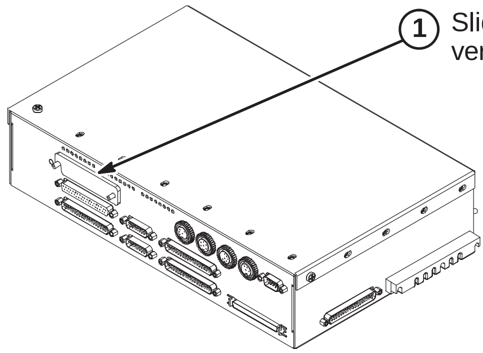
6–4–9 CONNECT REMAINING CABLES TO REAR PEDESTAL

1. Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated.



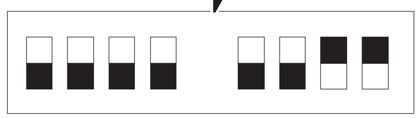
6–4–10 INSTALL SRI–3 MODULE

6–4–10–1 Verify SRI–3 Jumper Settings

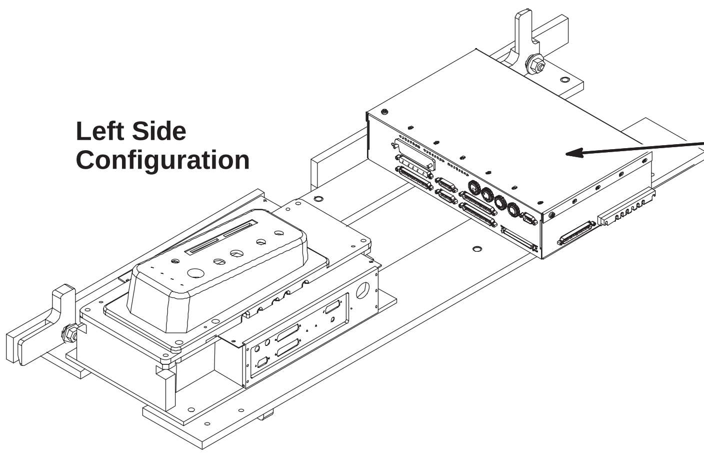


1 Slide cover plate away on new SRI–3 and verify jumper settings for 1.5T with Coil ID.

2 For the eight jumpers, the two jumpers on the right should be in the "ON" or located in the up position. All others should be "OFF" or down.



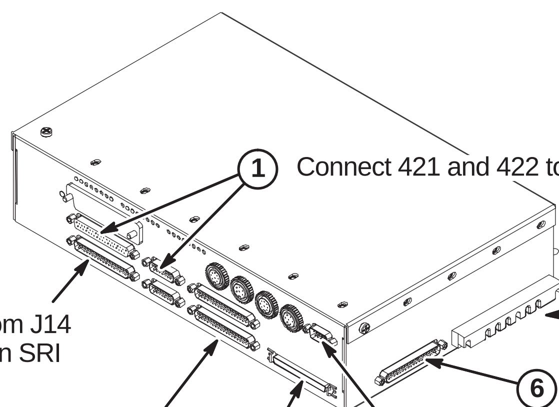
6–4–10–2 Postion Module on Platform



1 On bottom of SRI–3 Module, remove protective cover from Velcro

2 Position SRI–3 in same location as removed SRI–2 Module and press into place.

6–4–10–3 Connect Cables to SRI–3 Module



1 Connect 421 and 422 to J3 and J2.

2 Connect Run 715 from J14 on Pen Panel to J1 on SRI Module.

3 Connect Run 897 from Operator Display to J4.

4 Route Run 1261 from Operator Display and connect to J5.

5 Route Run 1260, Temperture Sensor, from Front End Bell and connect to J10.

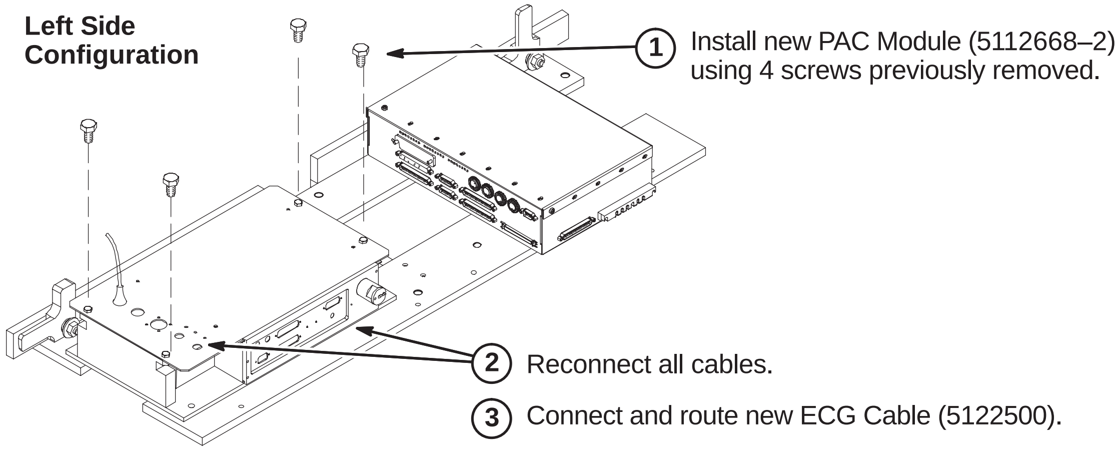
Refer to Front End Bell Replacement procedures in MR Service Methods CD for installing Run 1260, Bore Temp Sensor cable.

6 Route Run 1215 from J20 on 16 Channel Switch and connect to J20.

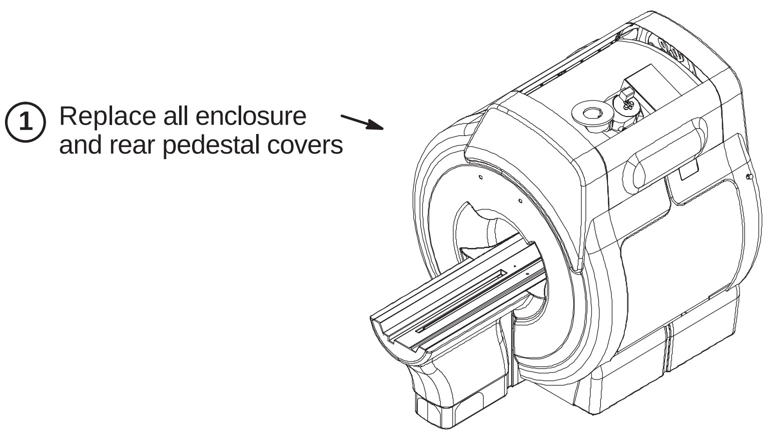
7 Connect fiber optic cable Run 711 to J11, J12, and J13.

6–4–11 INSTALL PAC MODULE AND ECG CABLE

Reverse the deinstallation steps for installation of the new PAC Module. Also install the new ECG Cable.



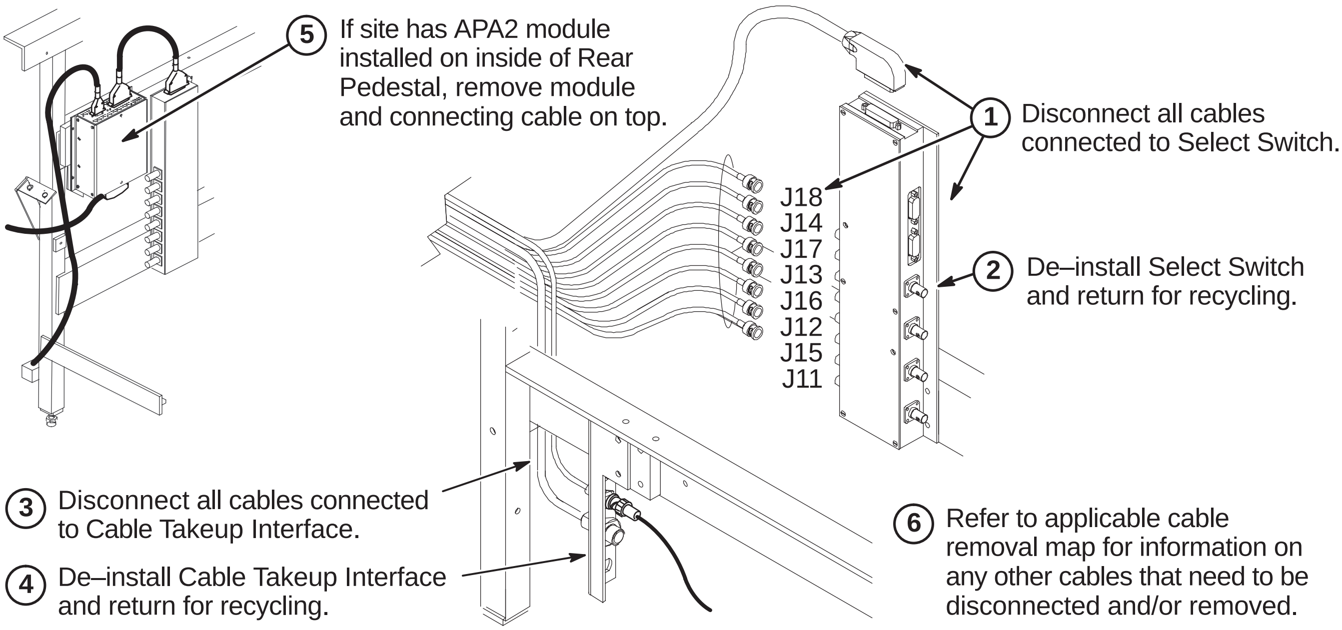
6–4–12 INSTALL COVERS



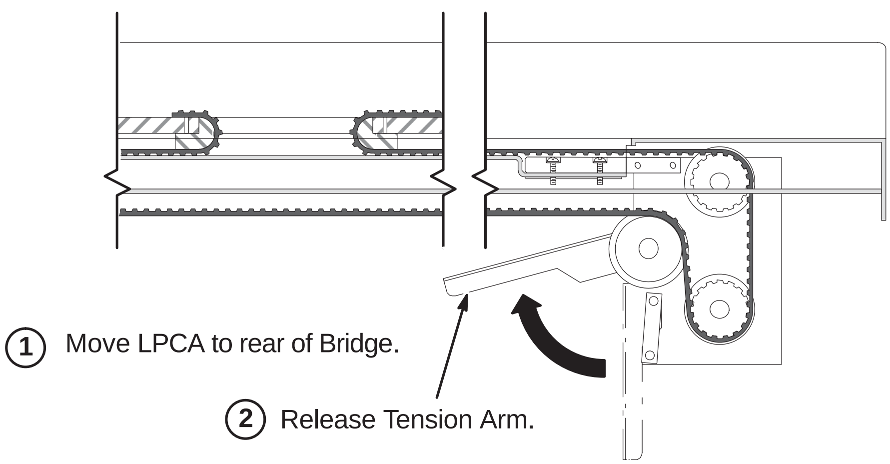
6–5 UPGRADE PROCEDURES FOR HORIZON STYLE MAGNET ENCLOSURES

6–5–1 DE–INSTALL EXISTING LPCA & CABLE TRACK

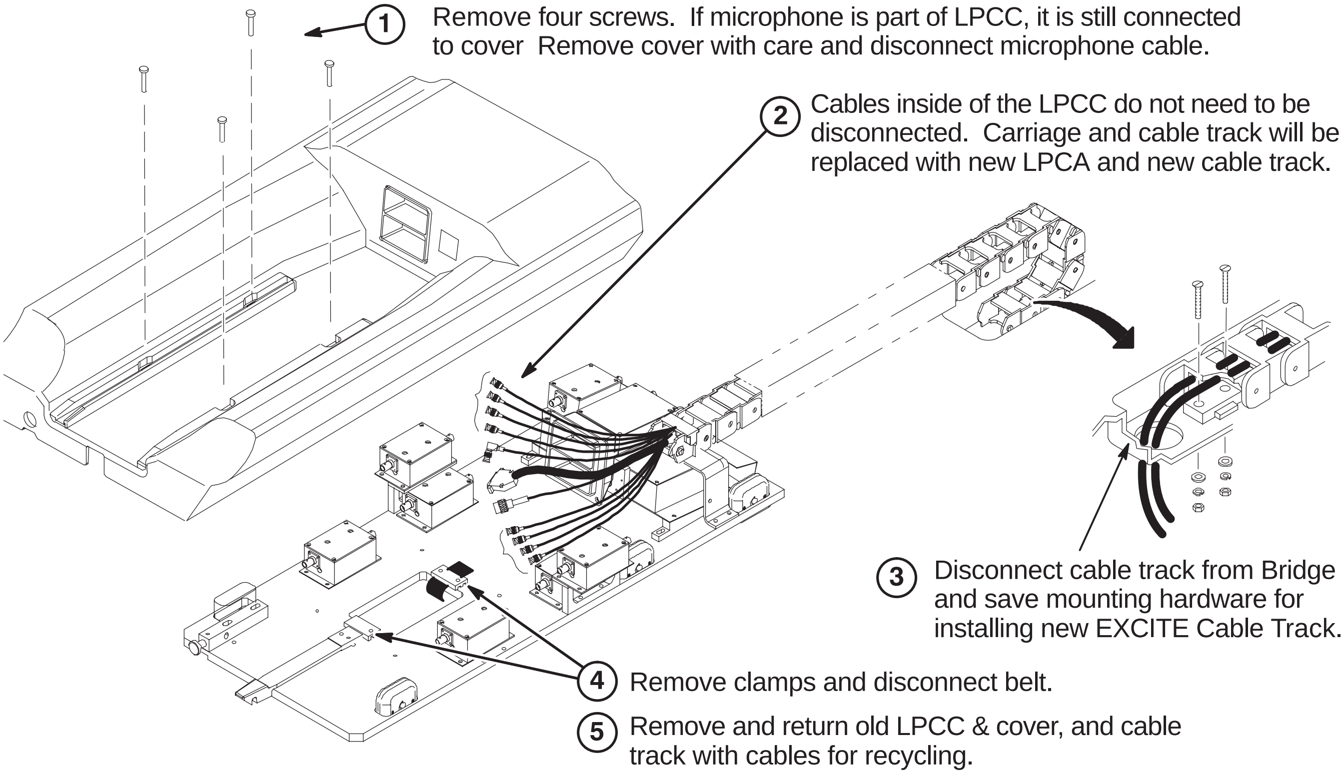
6–5–1–1 Disconnect Cables In Rear Pedestal



6–5–1–2 Relese Tension Arm Under Rear Pedestal

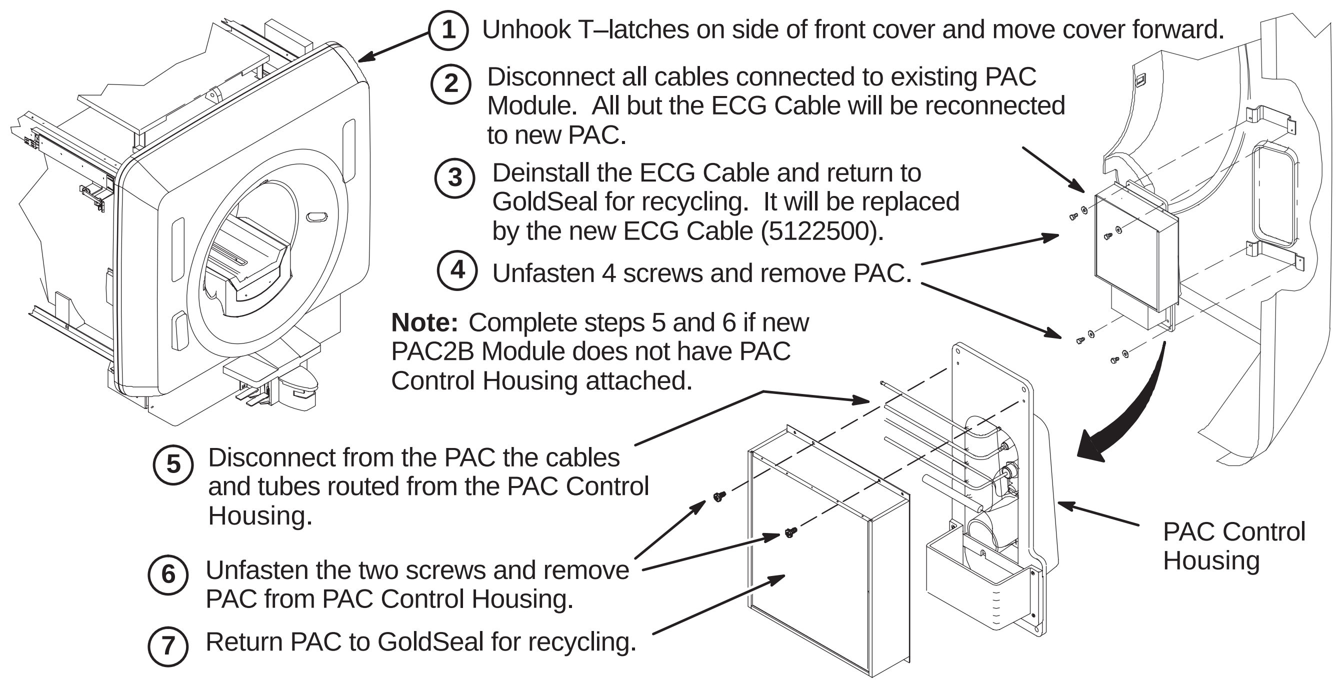


6–5–1–3 Remove Existing Low Profile Carriage Cover & Cable Track



6–5–2 DE–INSTALL PAC MODULE

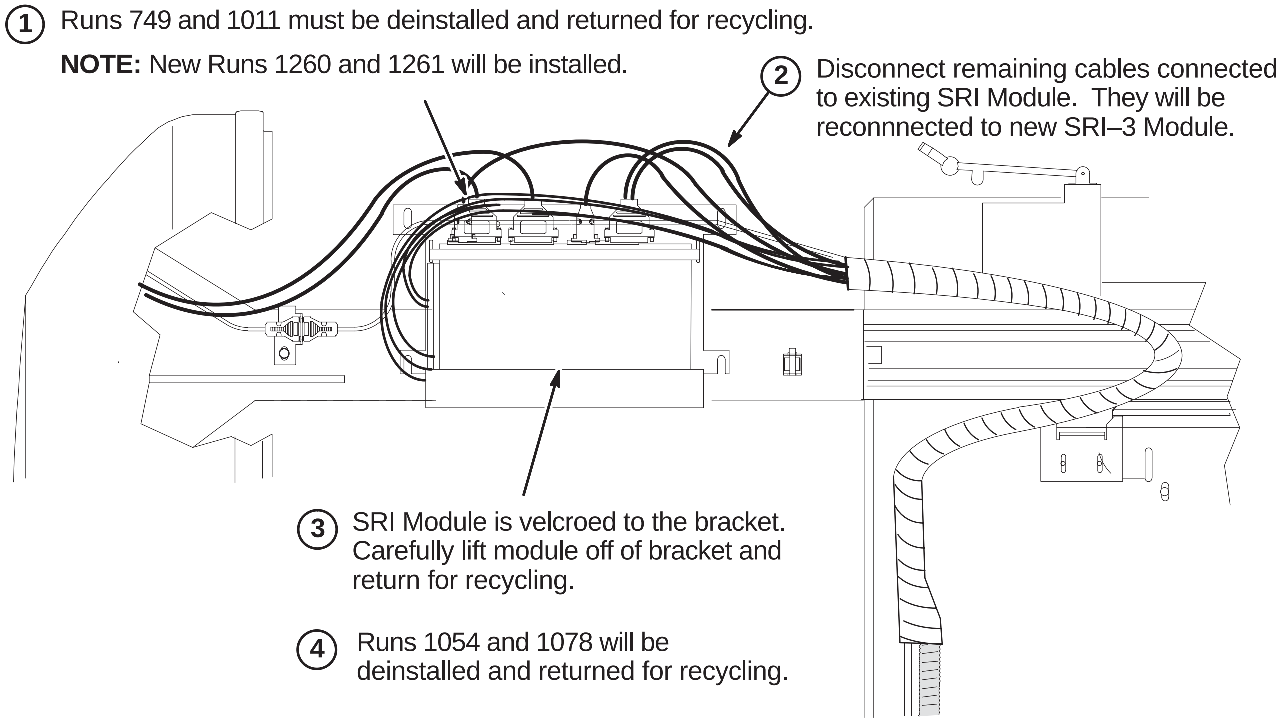
The existing PAC Module will be replaced by a new PAC Module with independent vector gating.



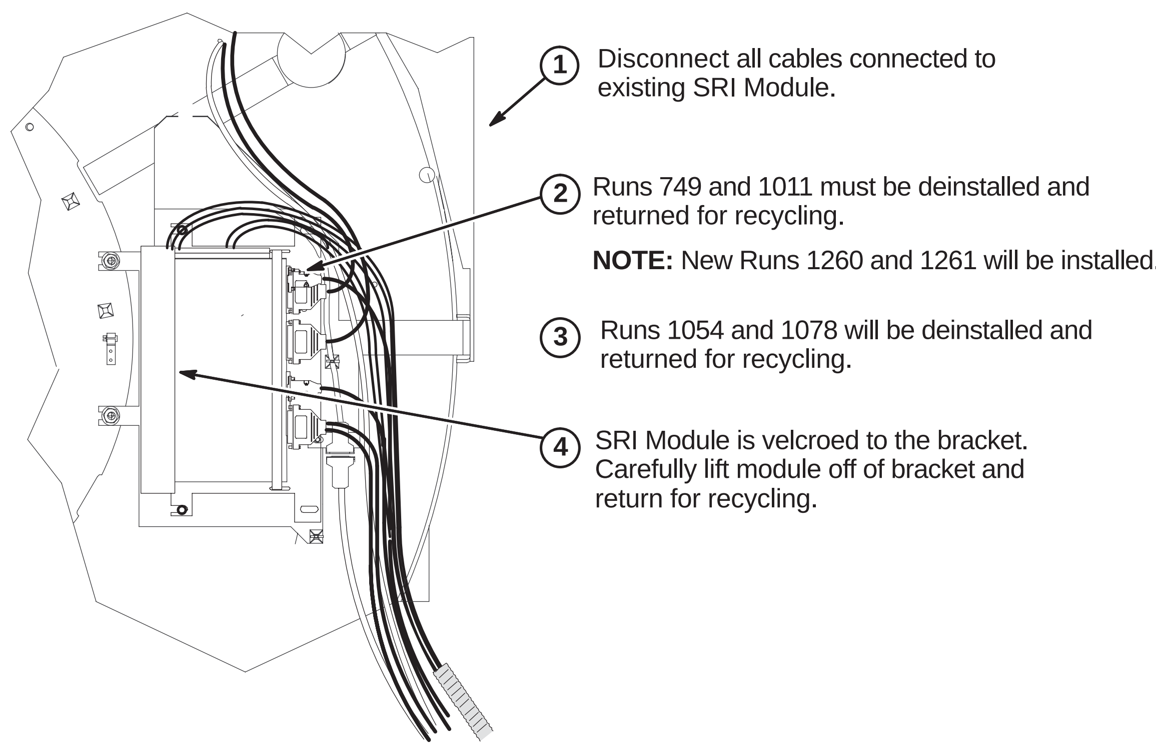
6–5–3 DE–INSTALL SRI MODULE

Section 6–5–3–1 contains instructions for removing the existing SRI module on an S–Series magnet.
Section 6–5–3–2 contains instructions for removing the existing SRI module on an Cx magnet.

6–5–3–1 Remove SRI Module for S–Series Enclosure



6–5–3–2 Remove SRI Module for Cx or LCC Horizon Enclosure



6–5–4 PREPARE END BELL FOR INSTALLATION OF NEW DUAL THERMAL SENSOR CABLE

The Bridge needs to be repositioned or removed prior to removal of Front End Bell. Review sections **6–5–4–3** and **6–5–4–4**. Determine which method would best fit the needs of site being upgraded.

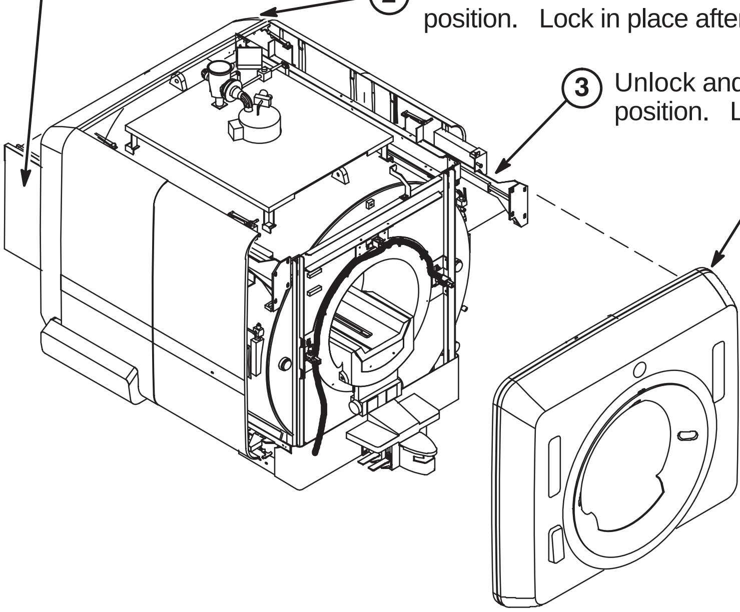
6–5–4–1 Removal of Enclosure Covers

1 Remove rear pedestal covers.

2 Unlock and extend rear cover to service position. Lock in place after extension.

3 Unlock and extend front cover to service position. Lock in place after extension.

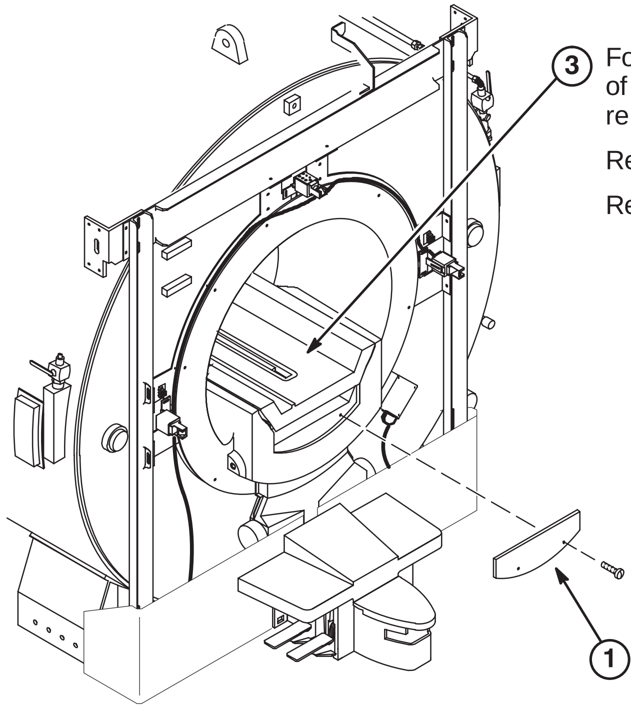
4 For ease of removal of Front End Bell, front cover may need to be removed. If so, make sure to follow WARNING below.



WARNING!

PERSONAL INJURY AND/OR EQUIPMENT DAMAGE POSSIBILITY! FRONT ENCLOSURE COVER IS HEAVY AND CUMBERSOME. AT LEAST TWO PEOPLE ARE REQUIRED TO LIFT AND MOVE IT; THREE IS PREFERABLE.

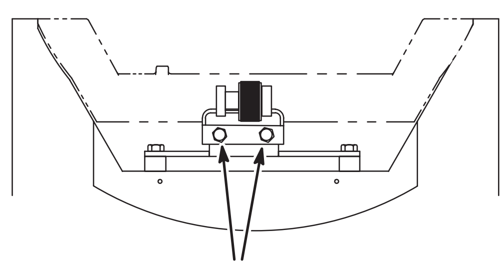
6–5–4–2 Removal of Front End Bell Panel



3 For removal of Front End Bell, Bridge must be moved out of the way either by lifting the front end or complete removal.

Refer to Section **6–5–4–3** for procedure to lift the Bridge.

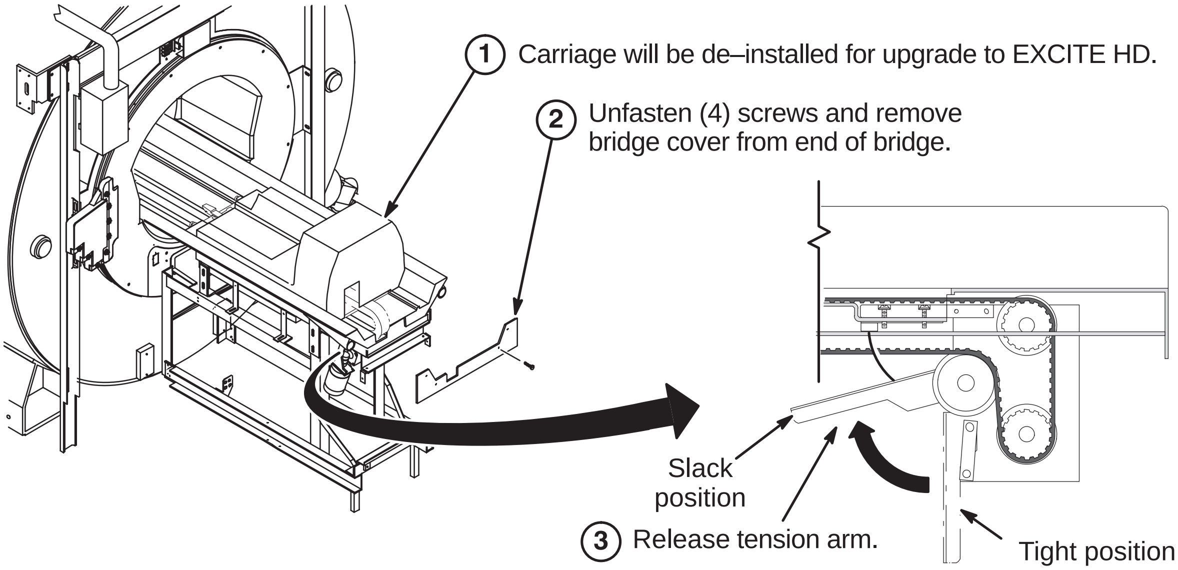
Refer to Section **6–5–4–4** for removal of Bridge.

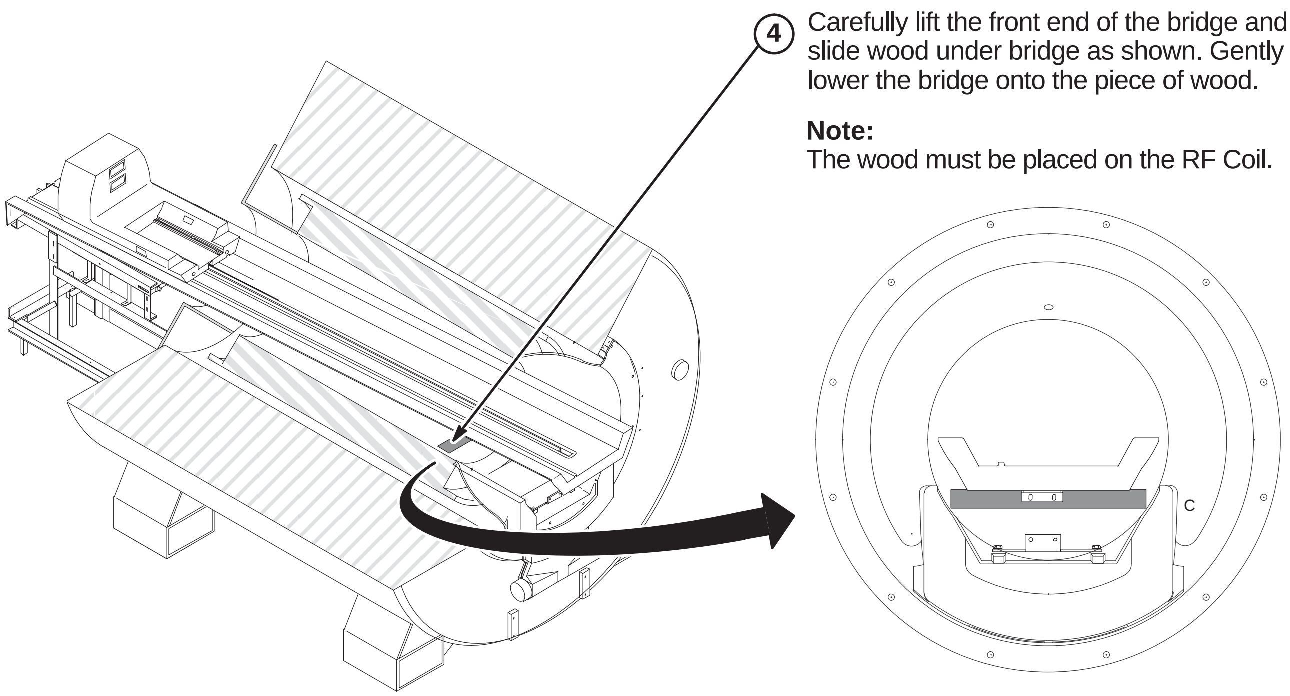


2 Unfasten M10 flat washers, lock washers, and screws attaching front end of bridge asm to front end bell.

1 Unfasten (2) M6 x 25 pan screws and remove end bell panel.

6–5–4–3 Lifting Front End Of Bridge Method

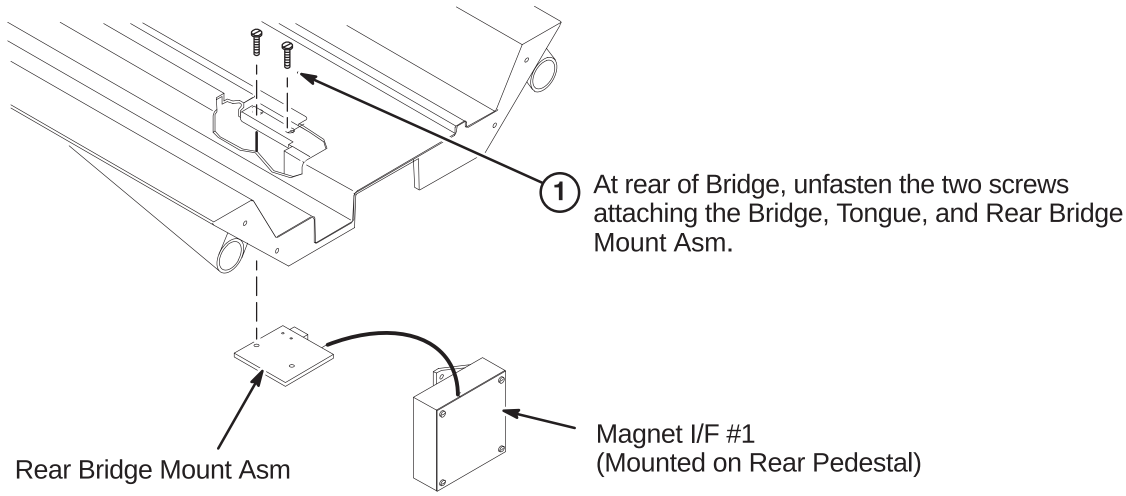


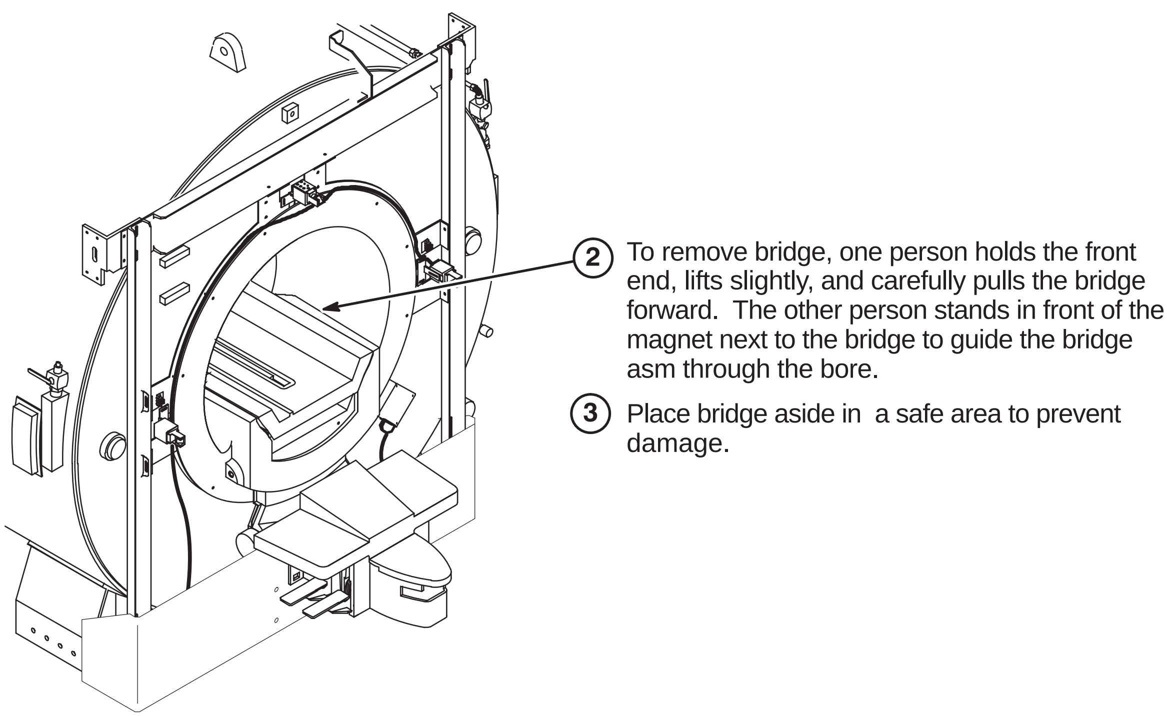


6–5–4–4 Removal of Bridge Method

CAUTION

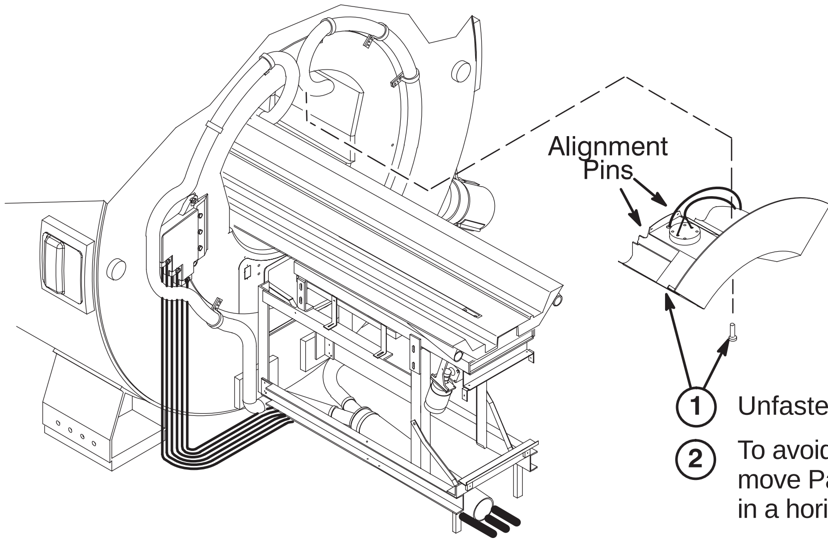
Possible equipment damage! Be careful not to damage the home flag on the drive belt when removing or replacing the bridge. The home flag is fragile and can easily be damaged.





L

6–5–4–5 Remove Patient Comfort Module



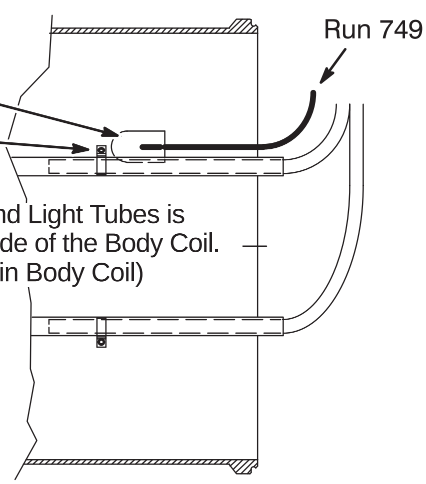
Note:
Hold Patient Comfort Module in place until mounting screws in Step 1 are removed. Alignment Pins can break off during removal if over–stressed.

1. Unfasten two M6 x 25 SST Pan Head Screws.
2. To avoid breaking alignment pins during removal, move Patient Comfort Module away from End Bell in a horizontal path.

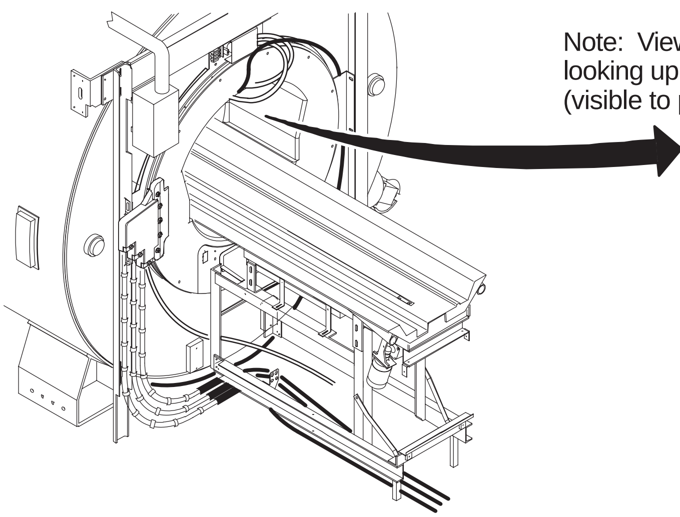
The site can have the existing Temperature Sensor Cable routed to the Rear end bell (earlier location) or Front end bell (later location). Either way, the cable needs to be removed and replaced with new cable, Run 1260. One of the two following procedures will be applicable to your site.

6–5–4–6 Remove Existing Single Temperature Sensor Cable at Rear End Bell Location

1. Unfasten existing Temperature Sensor Cable (Run 749) from top inside of Body Coil. It may be attached with:
   adhesive fastener (as shown)
   OR
   with the Light and Cable clamp



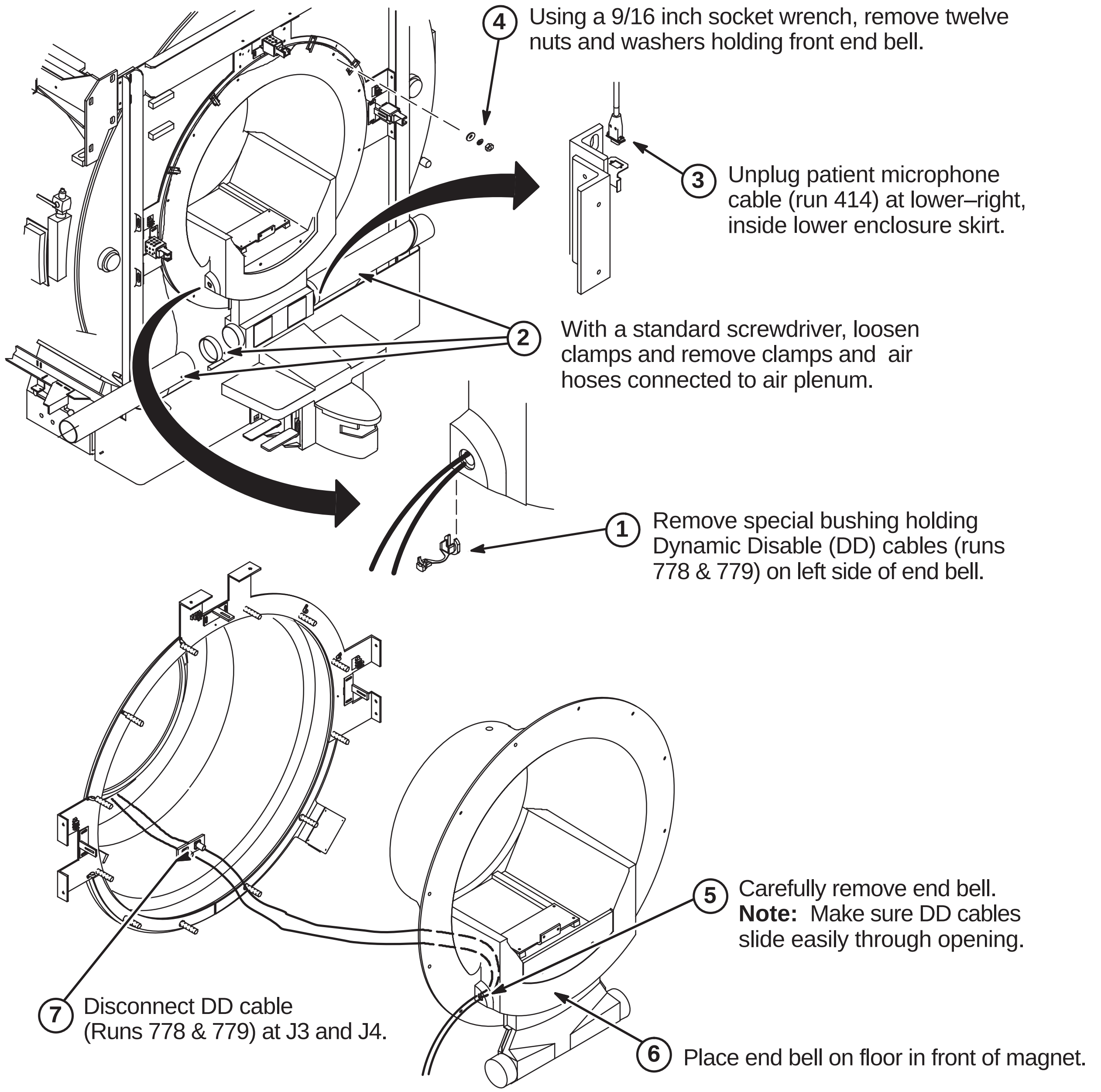
Note: View of Sensor and Light Tubes is looking up at the top inside of the Body Coil. (visible to patient laying in Body Coil)



2. Disconnect other end of Run 749 from J10 on SRI module.
3. Remove and discard cable.

6–5–4–7 Unfasten Front End Bell

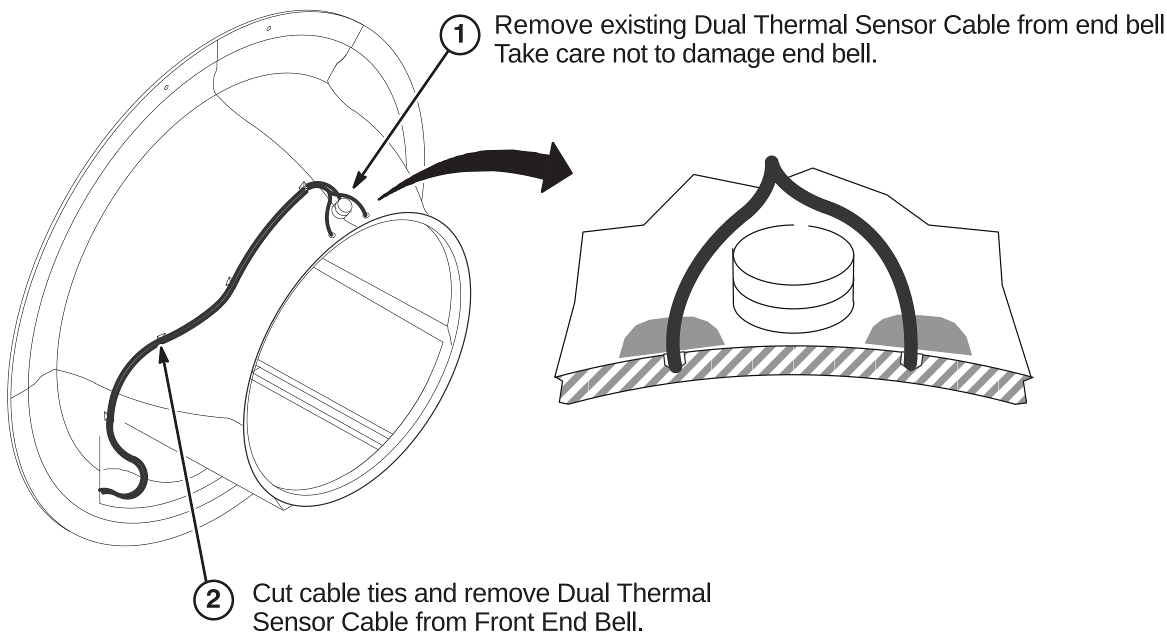
The site may have upgraded to a Dual Thermal Sensor Run 749 recently, but this cable has to be removed and replaced with new Dual Thermal Sensor cable, Run 1260 (5123812). EXCITE HD requires Run 1260 to be installed.



6–5–4–8 Existing Dual Temperature Sensor Cable Removal from Front End Bell (If applicable)

WARNING!

FERROUS MATERIAL HAZARD! DO NOT DRILL, HAMMER, CHISEL, ETC. OR USE ANY FERROUS TOOLS IN MAGNET ROOM UNLESS MAGNET IS RAMPED DOWN!

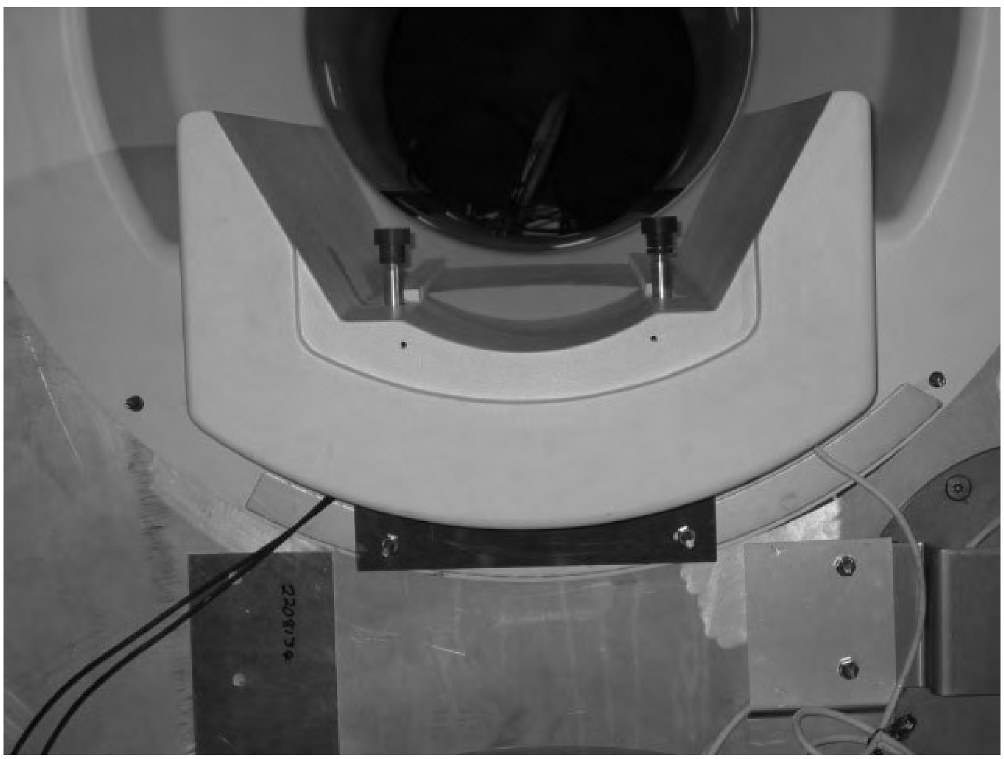


6–5–5 INSTALL IIQ KIT AND RF GROUND STRAP (If Required)

Prior to this upgrade, the site should have been audited to determine if the IIQ Kit (M3090NE) and the RF Ground Strap (M3090NF) have been installed.

The IIQ Kit includes parts and instructions for the installation of the Gradient Radial Support and the Bridge Support. It is required for all magnets with Horizon style enclosures. If this upgrade has not been performed on the existing enclosure, then M3090NE should have been ordered and must be installed at this point of the EXCITE HD ugprade.

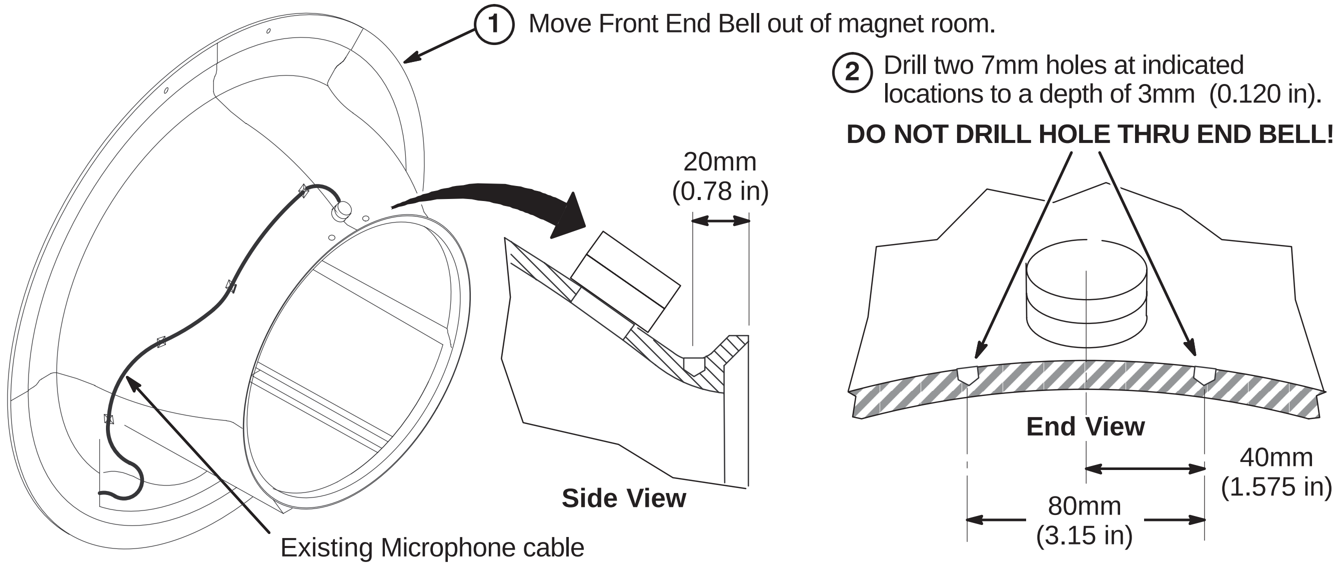
Refer to steering Direction 2401983, included in M3090NE, for guidance. Directions 2401984, *S–Series Magnet Bridge Support* (shown below) and Direction 2319693, *Signa Gradient Radial Support Upgrade*, along with associated parts, are also included in M3090NE.



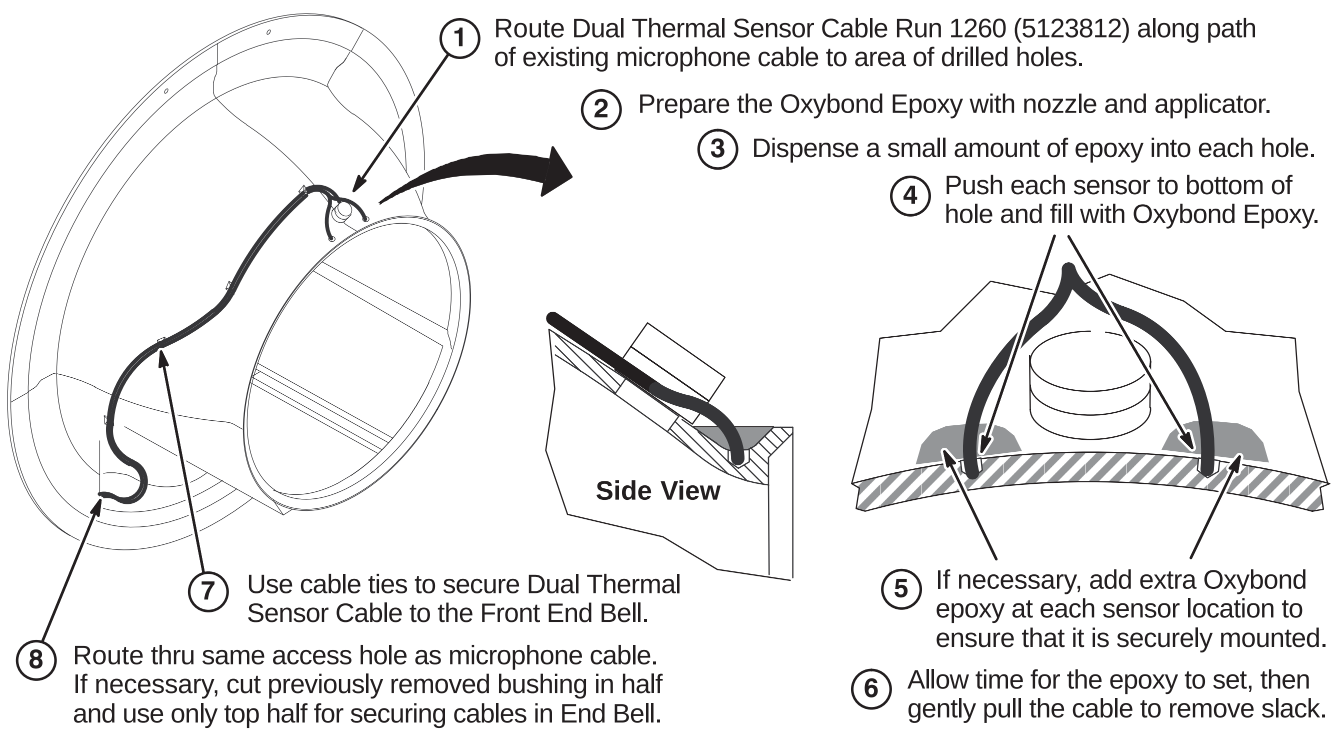
Also, the RF Ground Strap (M3090NF) must be installed if this upgrade hasn't been performed previously. This kit includes parts and instructions for installation of the RF Ground Strap. Refer to Direction 2319685, *BRM of CRM RF Shield Grounding Upgrade,* for instructions on installing this kit.

6–5–6 INSTALL NEW DUAL TEMPERATURE SENSOR CABLE, RUN 1260

6–5–6–1 Preparation of Front End Bell



6–5–6–2 Attach Dual Thermal Sensor Cable



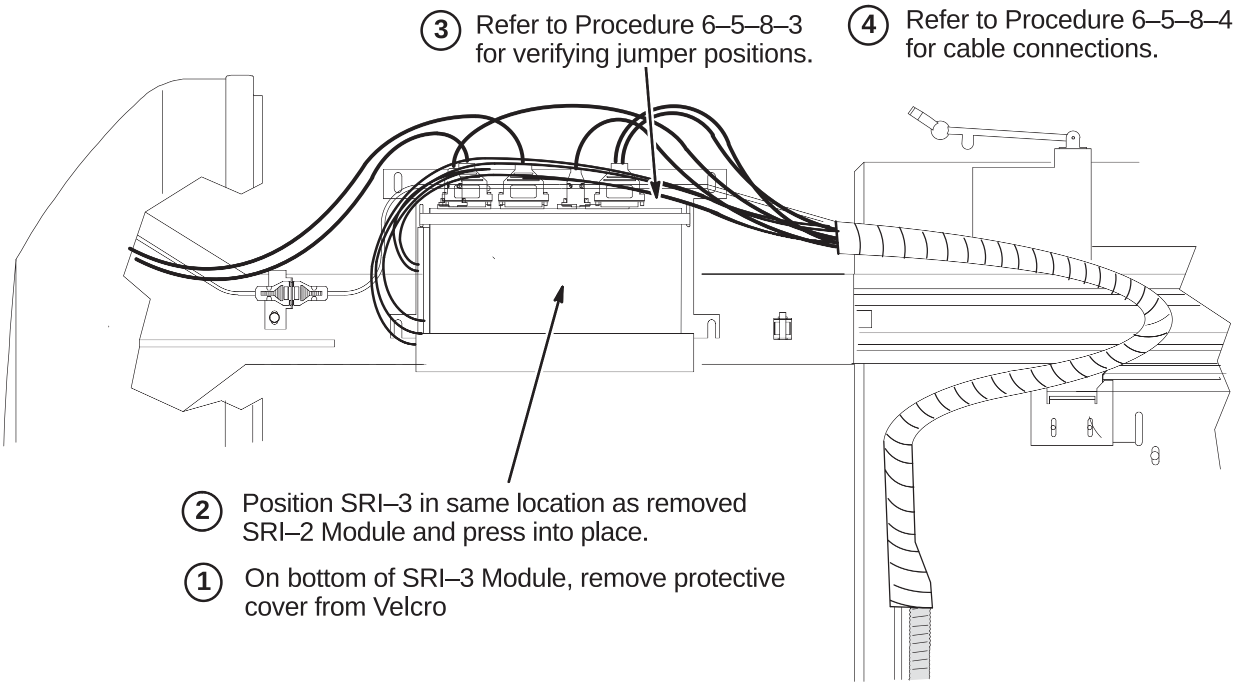
6–5–7 RE–INSTALL FRONT END BELL, COMFORT MODULE, AND BRIDGE

Reverse the applicable Front End Bell Removal procedures (Section 6–5–4) to reinstall the end bell, comfort module, and bridge.

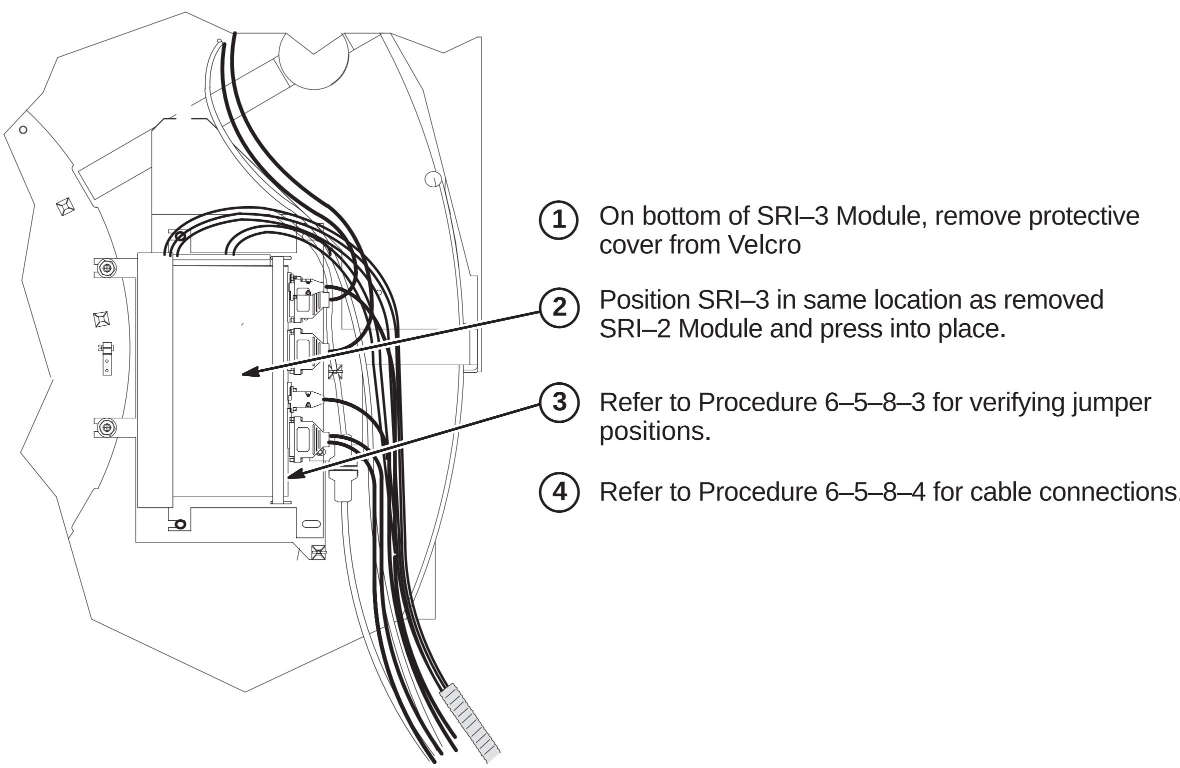
6–5–8 INSTALL SRI–3 MODULE

For site with S–series magnet, complete procedure in Section 6–5–8–1.
For site with Cx magnet, complete procedure in Section 6–5–8–2.

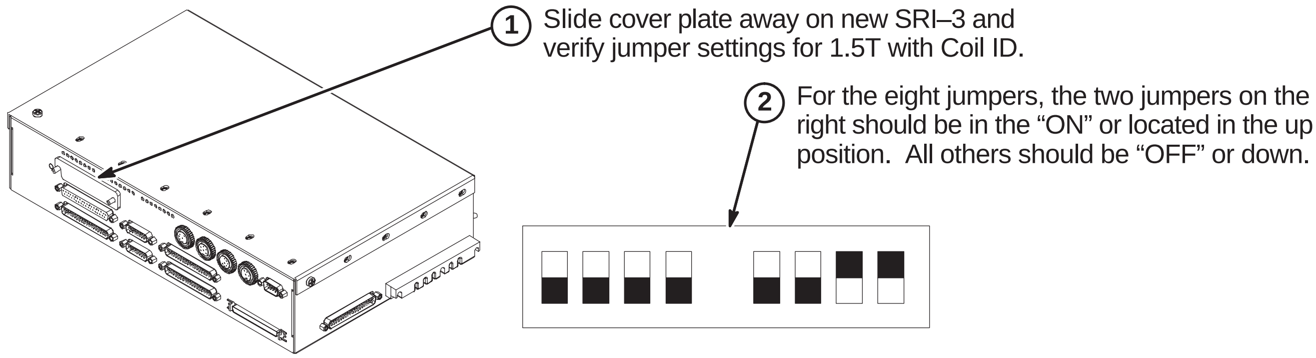
6–5–8–1 Install SRI–3 Module for S–Series Enclosure



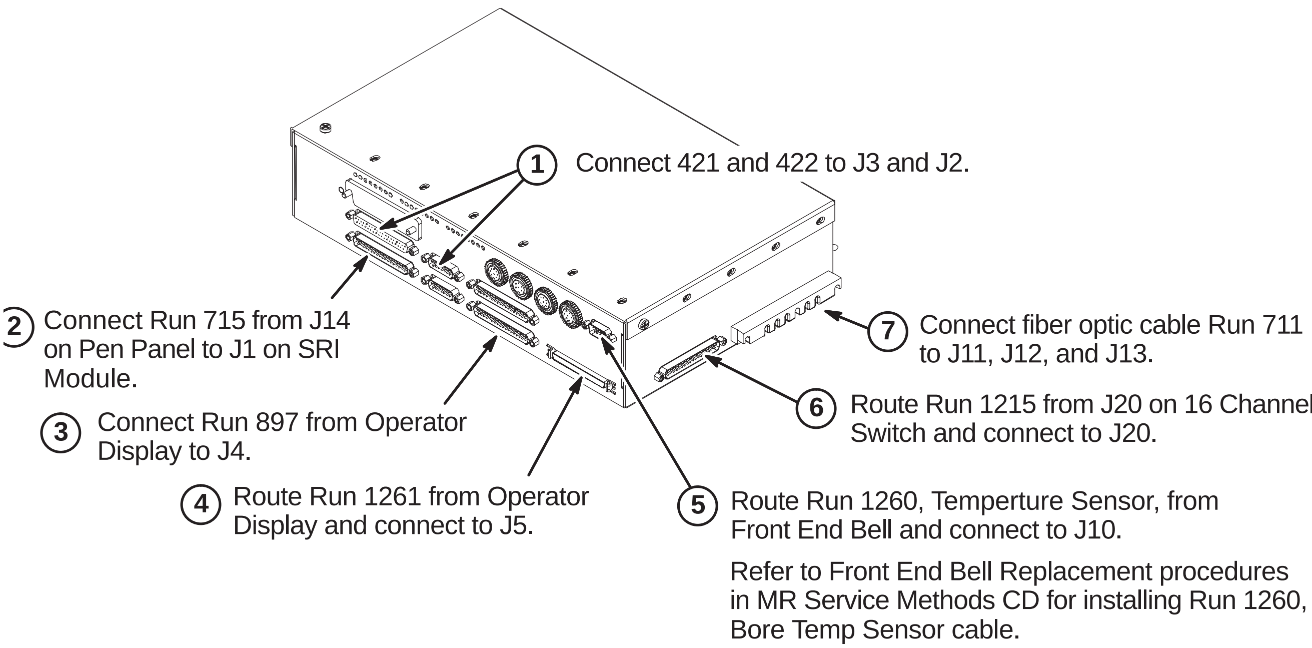
6–5–8–2 Install SRI–3 Module for Cx or LCC Horizon Enclosure



6–5–8–3 Verify SRI–3 Jumper Settings

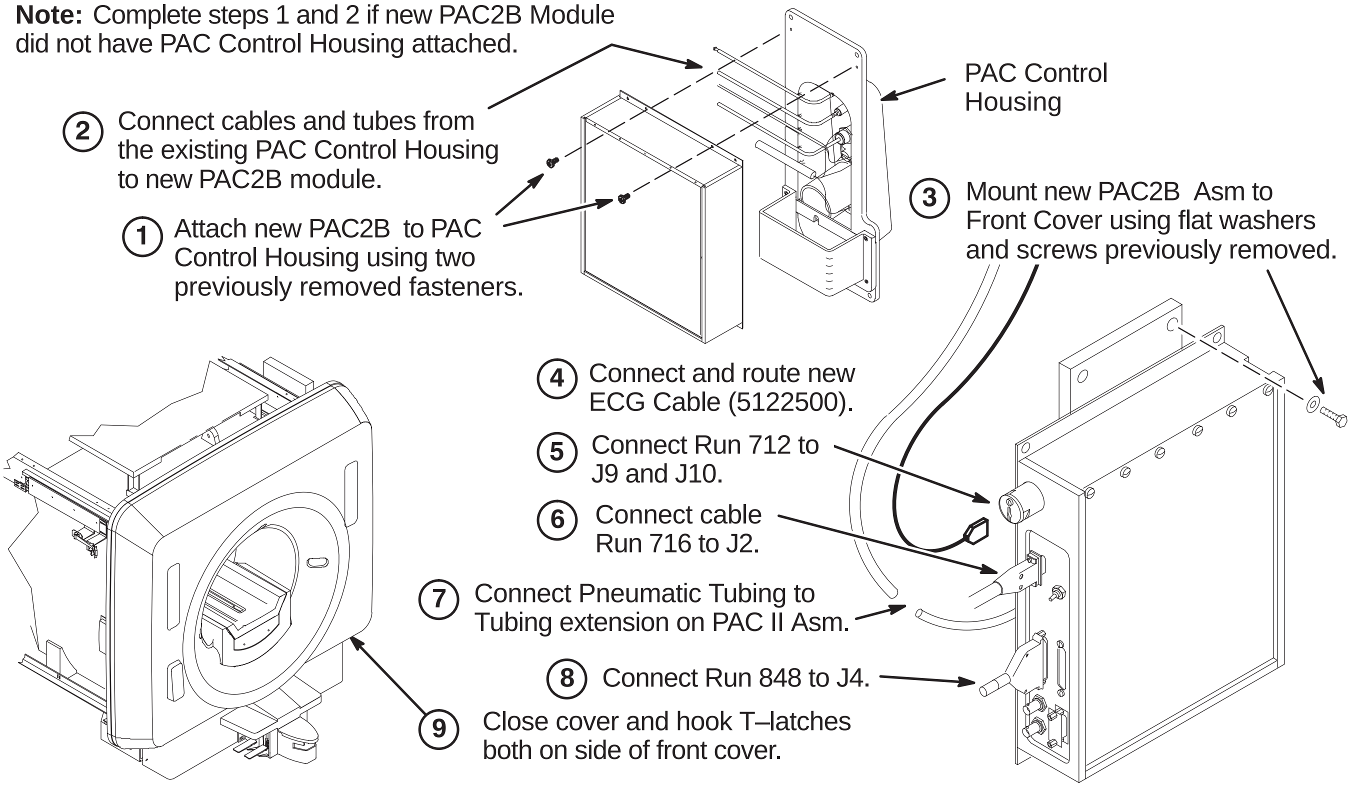


6–5–8–4 Connect Cables to SRI–3 Module



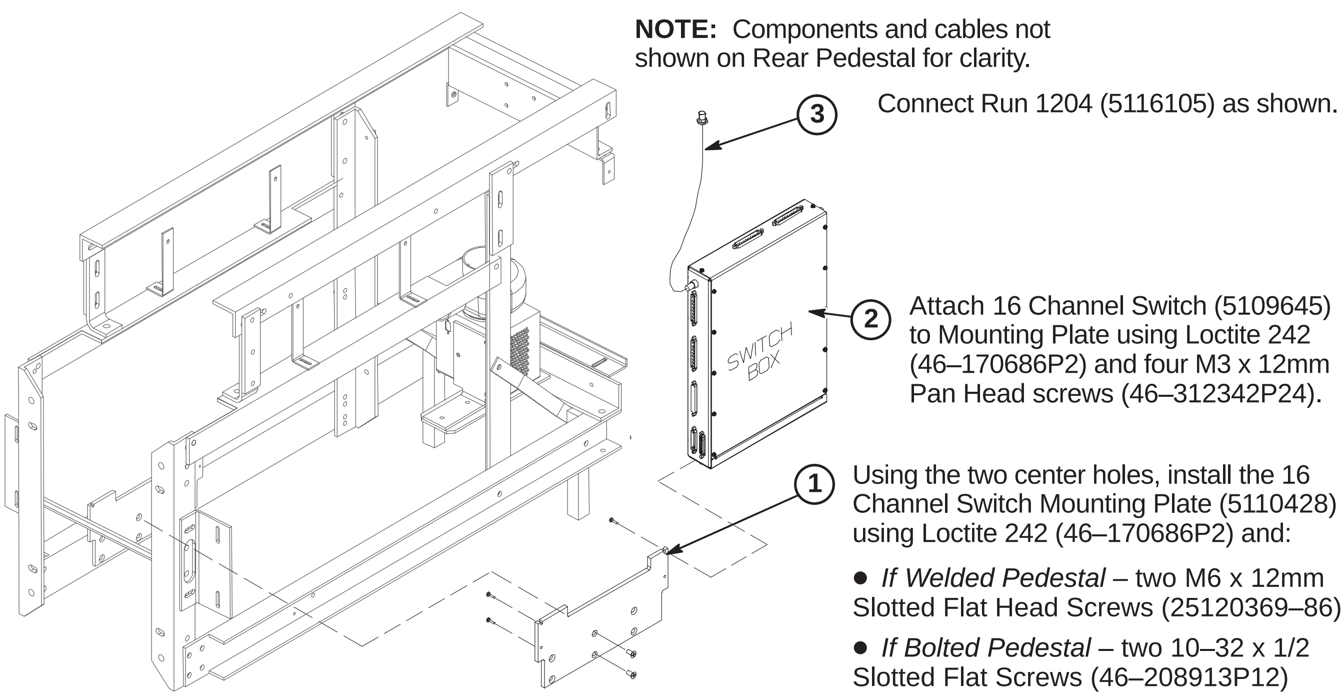
6–5–9 INSTALL NEW PAC MODULE AND ECG CABLE

Reverse the deinstallation steps for installation of the new PAC Module. Also install the new ECG Cable.

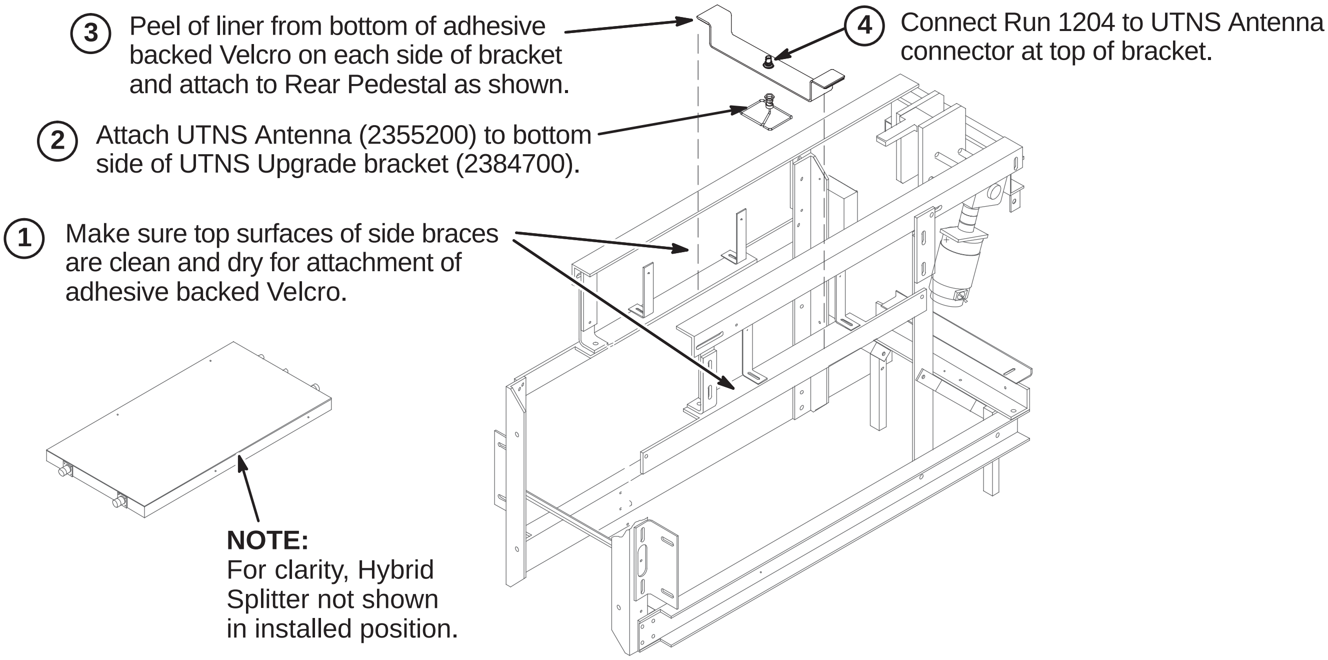


6–5–10 INSTALL NEW COMPONENTS ON REAR PEDESTAL

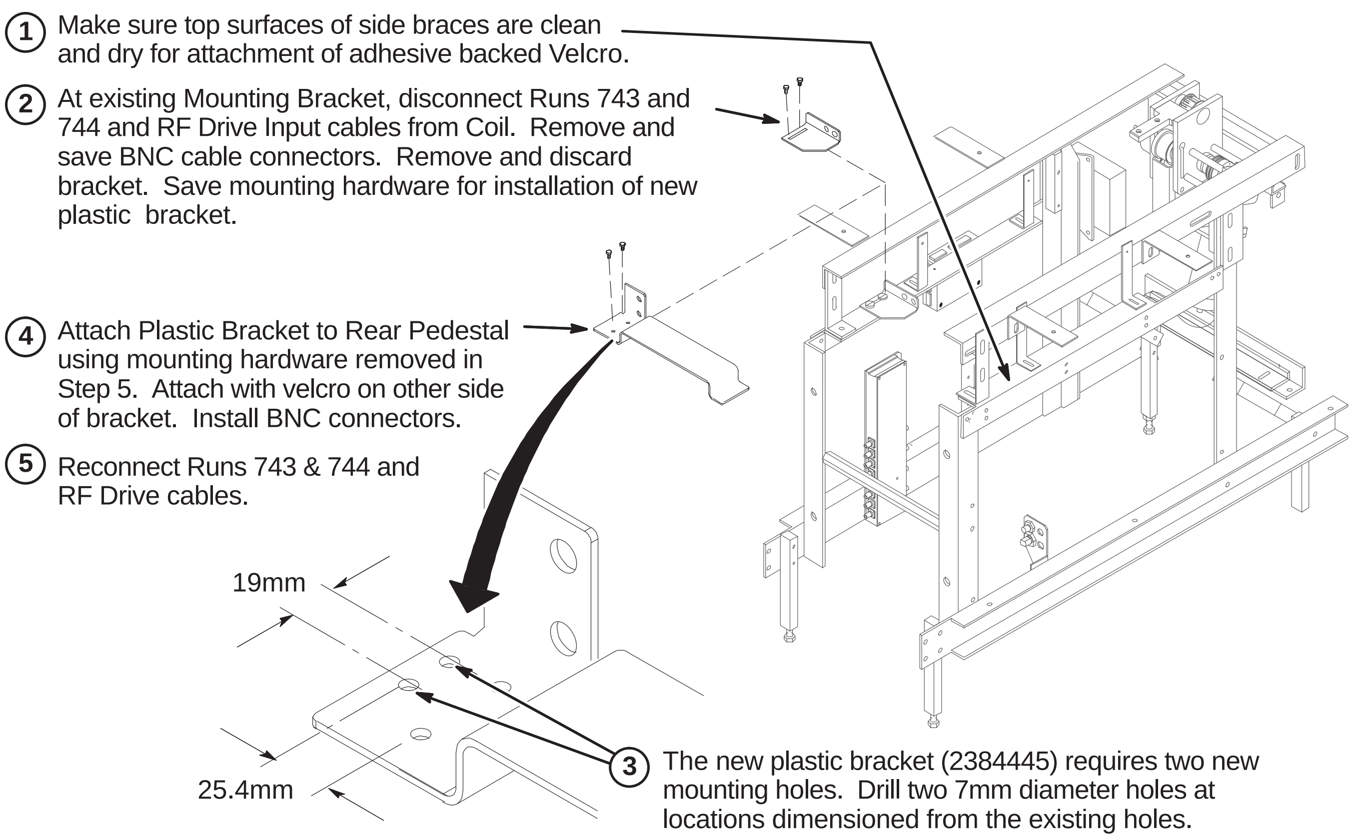
6–5–10–1 Install 16 Channel Switch



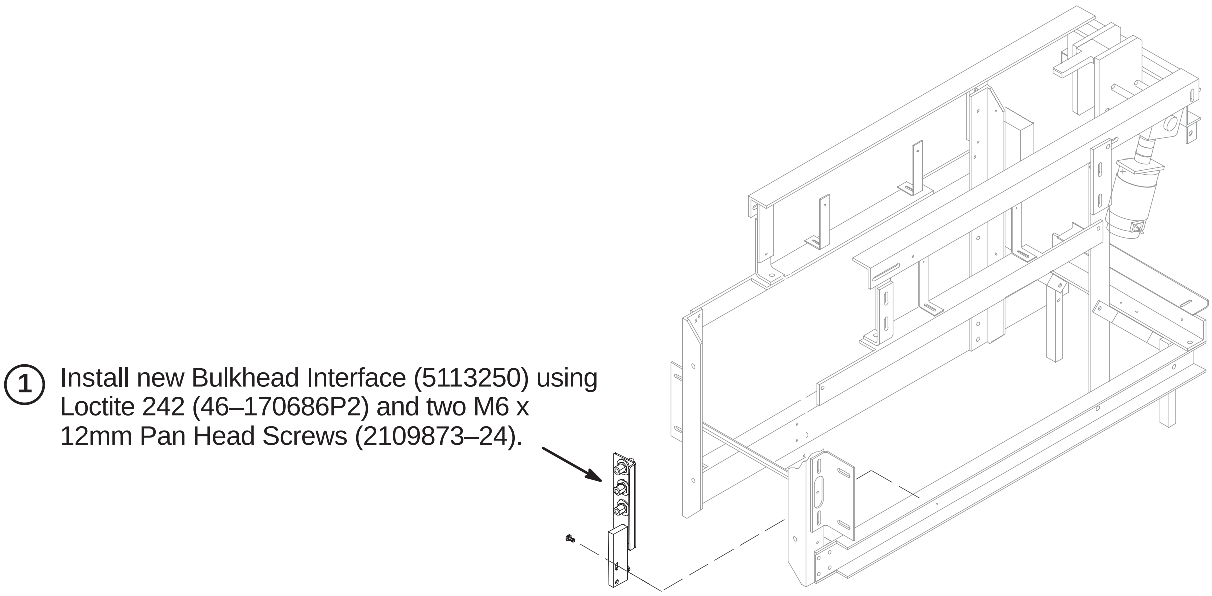
6–5–10–2 Install UTNS Antenna



6–5–10–3 Install RF Drive Cable Mounting Bracket for LCC and Cx Magnets Only

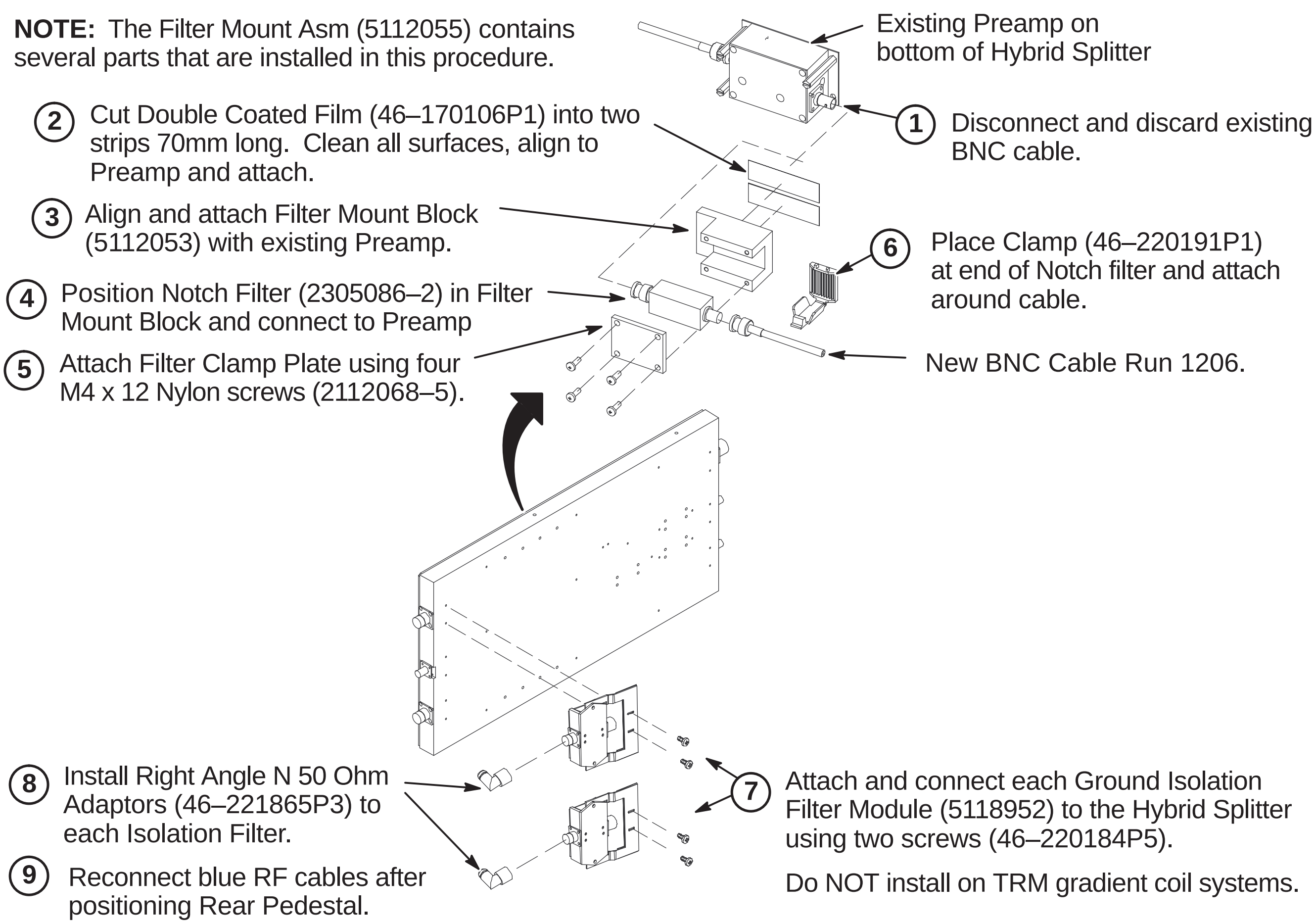


6–5–10–4 Install Bulkhead Interface

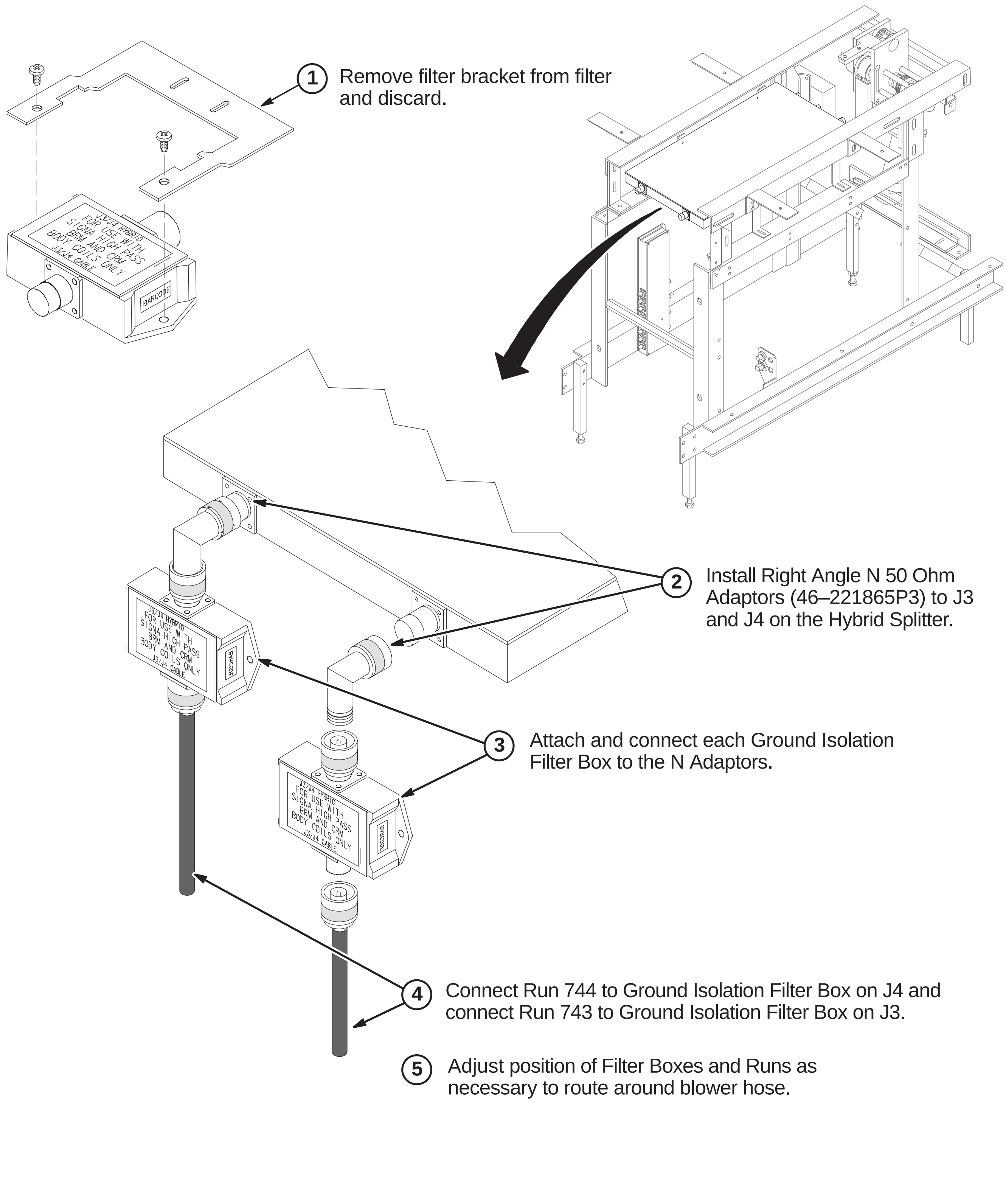


6–5–10–5 Install Filter Mount Asm and Isolation Filters to Body Hybrid Splitter

NOTE: The Filter Mount Asm (5112055) contains several parts that are installed in this procedure.

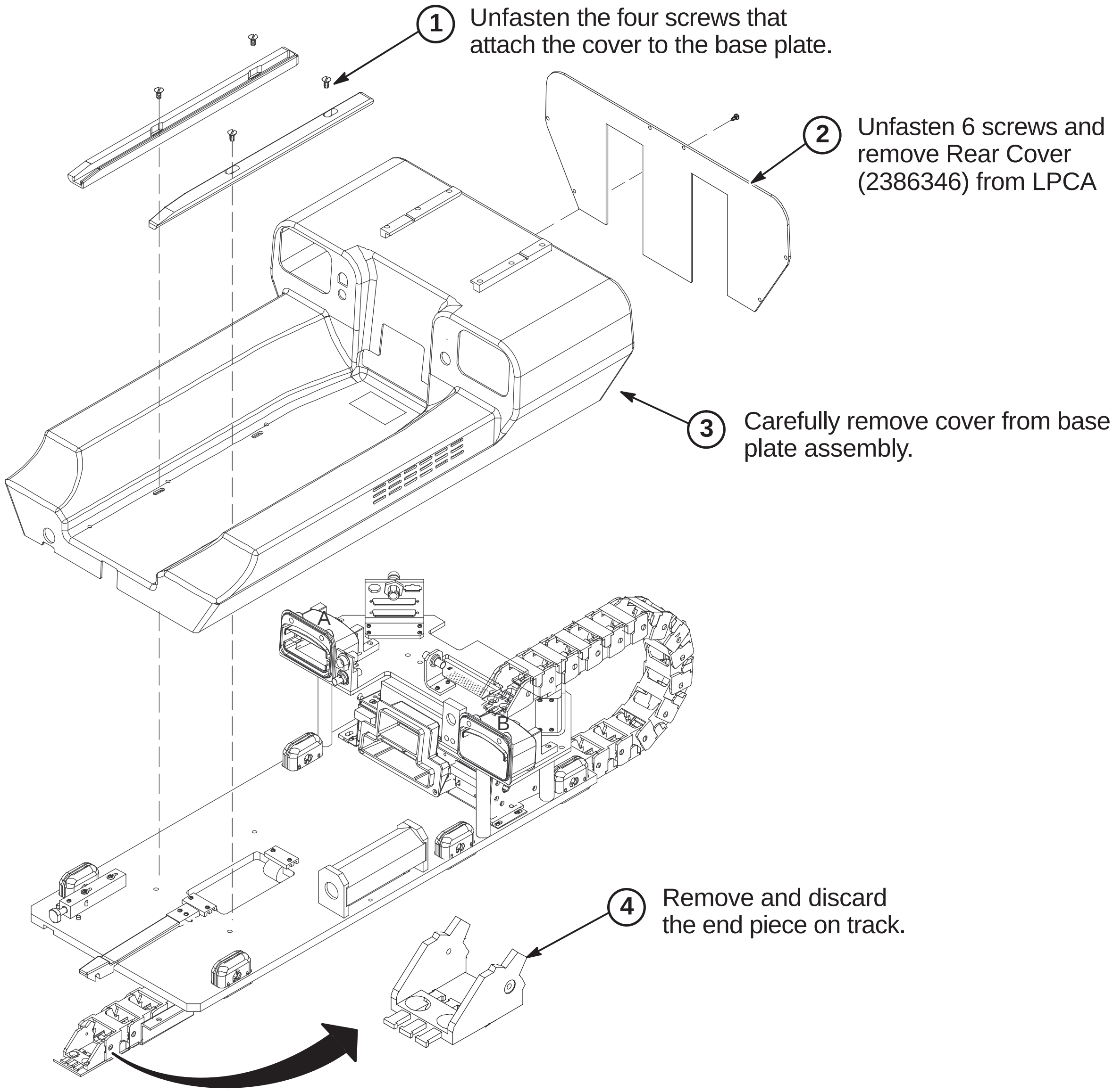


6–5–10–6 Alternative Ground Isolation Filter Installation for LCC Magnet with Horizon Enclosure

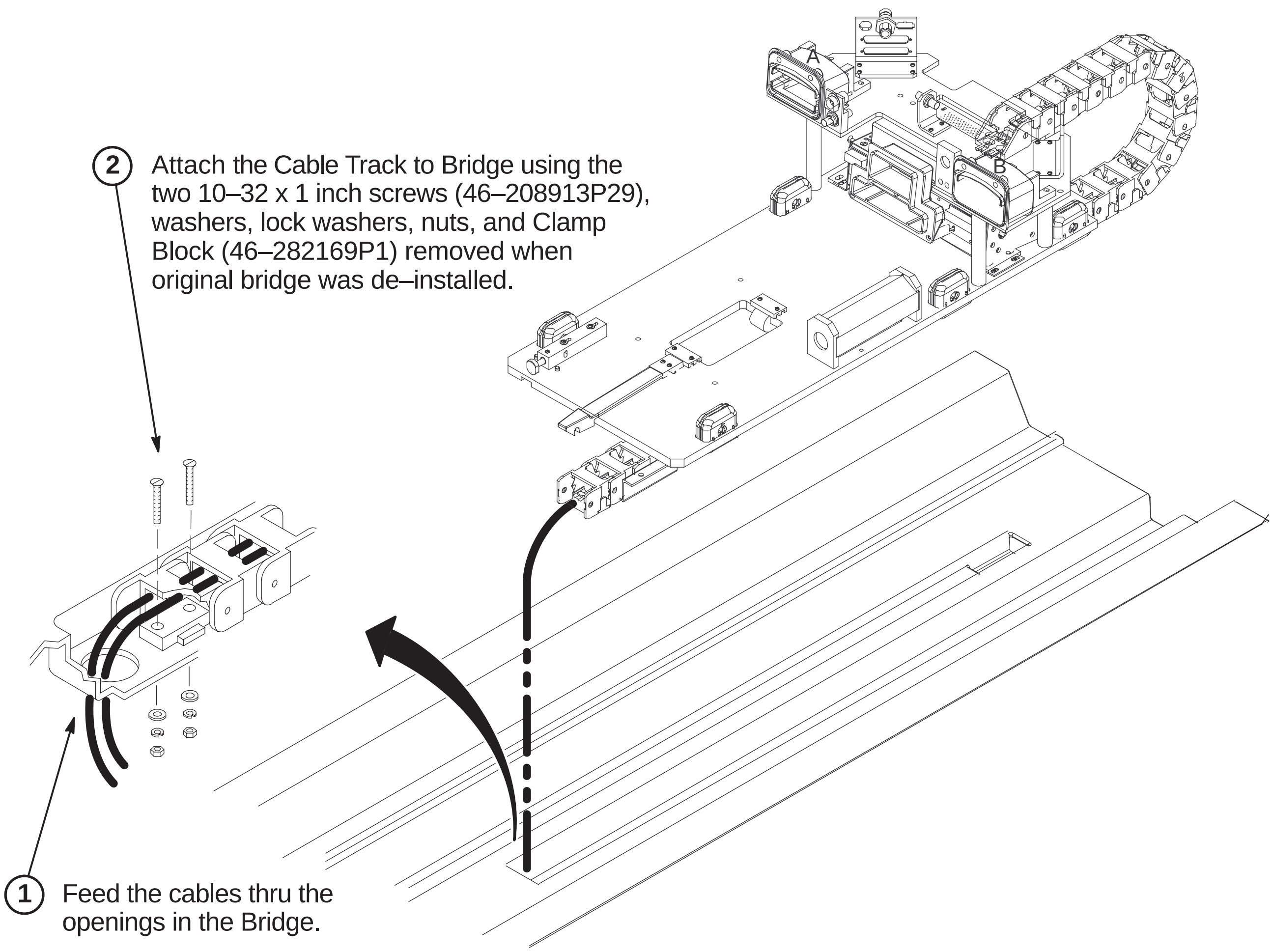


6–5–11 INSTALL LPCA AND CABLE TRACK

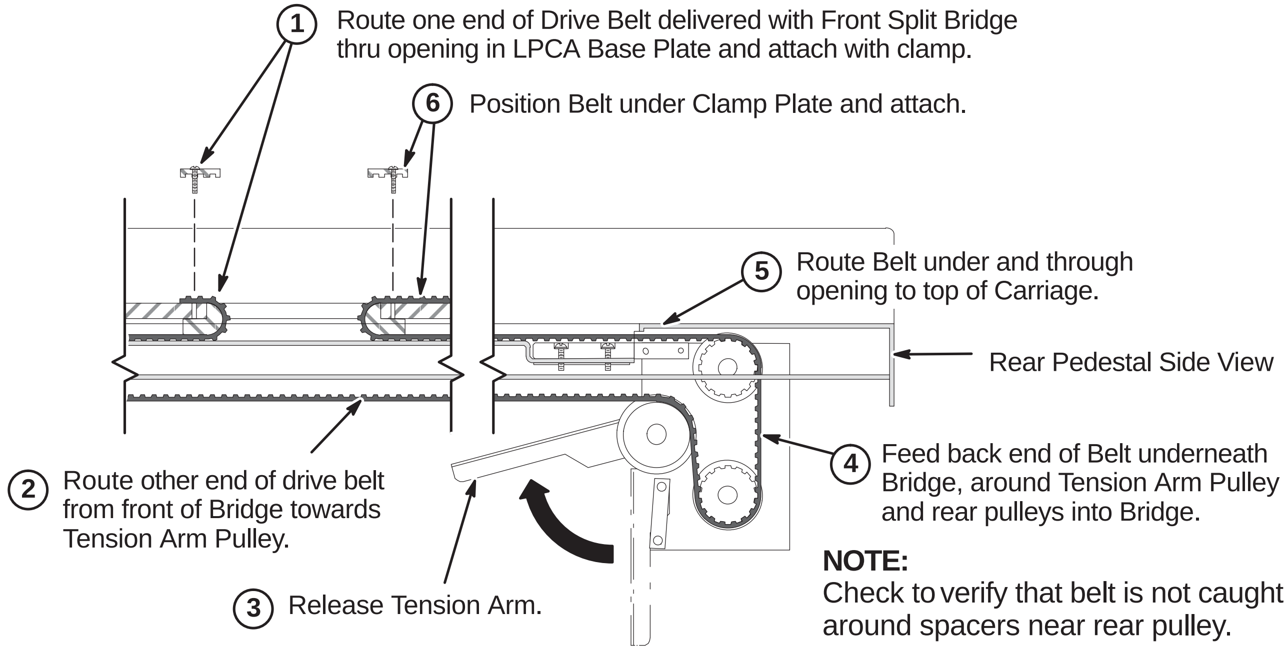
6–5–11–1 Remove LPCA Cover



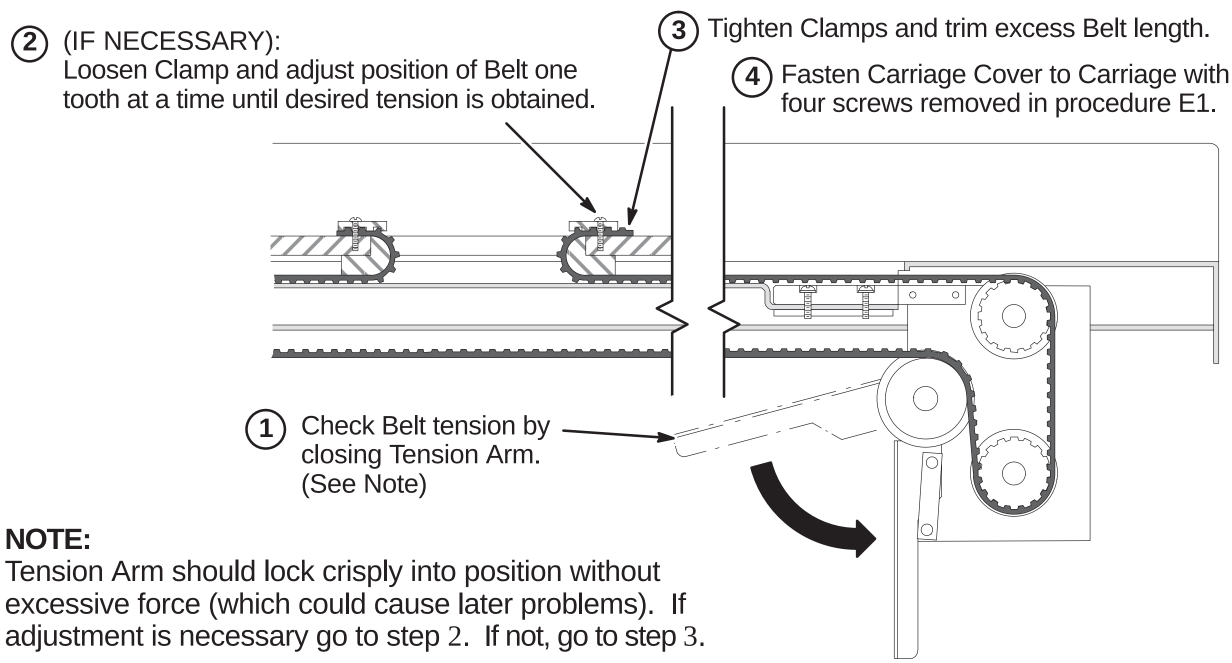
6–5–11–2 Attach Cable Track to Bridge



6–5–11–3 Route and Attach Drive Belt



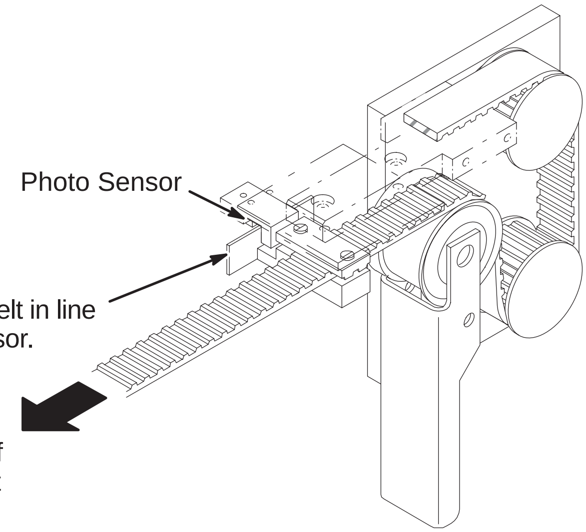
6–5–11–4 Final Adjustment of Drive Belt



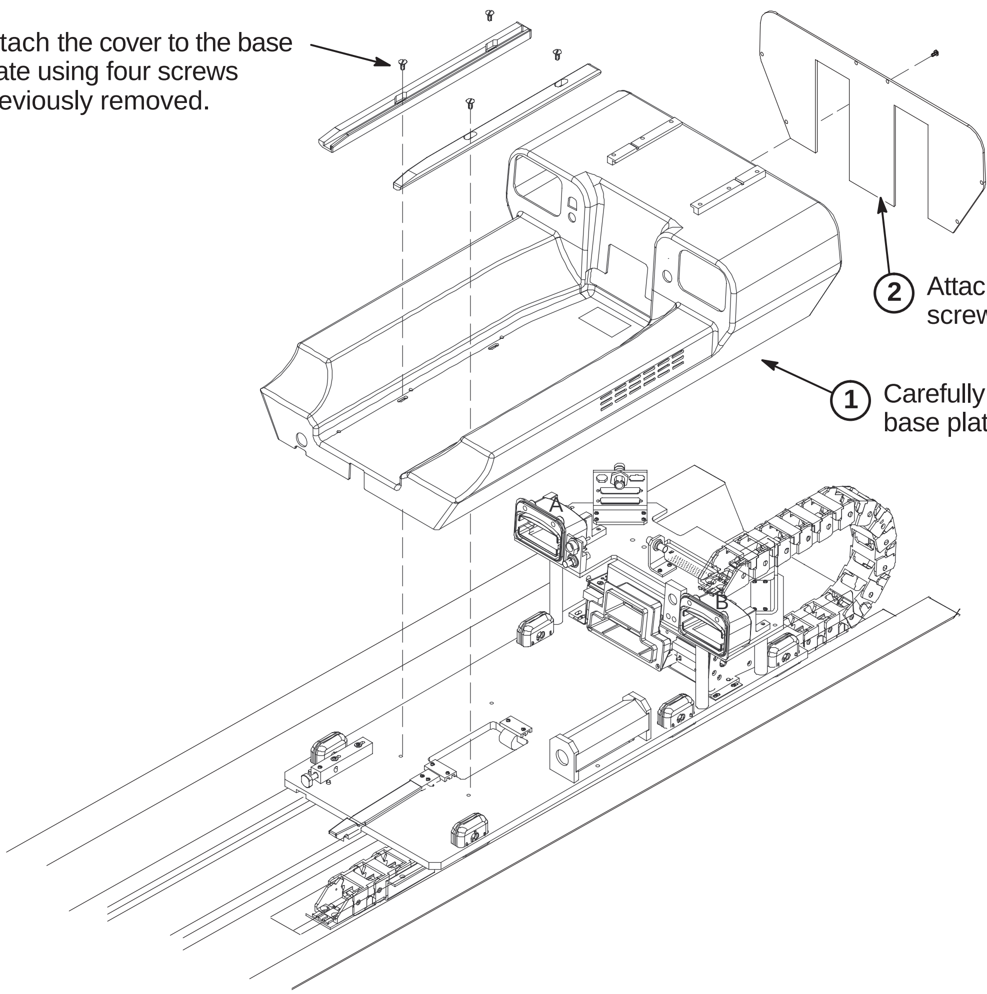
6–5–11–5 Installation of Sensor Flag

NOTE:
Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

NOTE:
The following Step is needed if flag was removed during routing of drive belt.

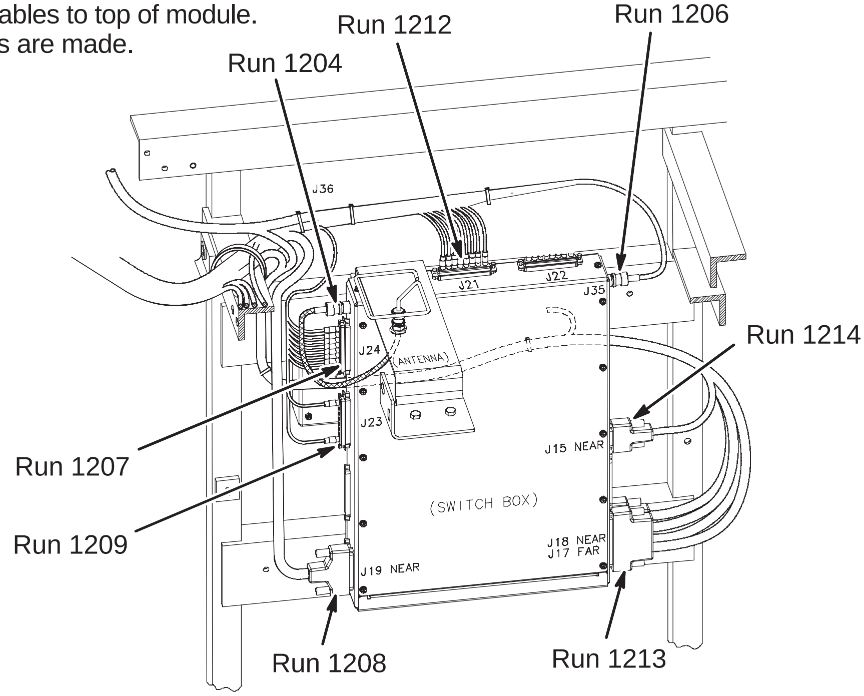


6–5–11–6 Installation of LPCA Cover



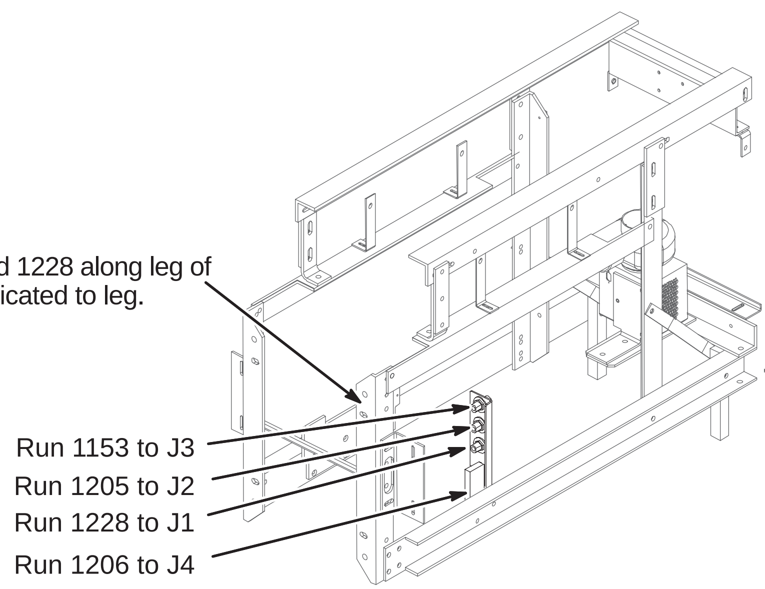
6–5–11–7 Connect Cables from Cable Track to 16 Channel Switch Module

1. 16 Channnel Switch may need to be unfastened and lowered from mounting position to connect cables to top of module. Re–mount module after connections are made.
2. Connect cable Runs as indicated.



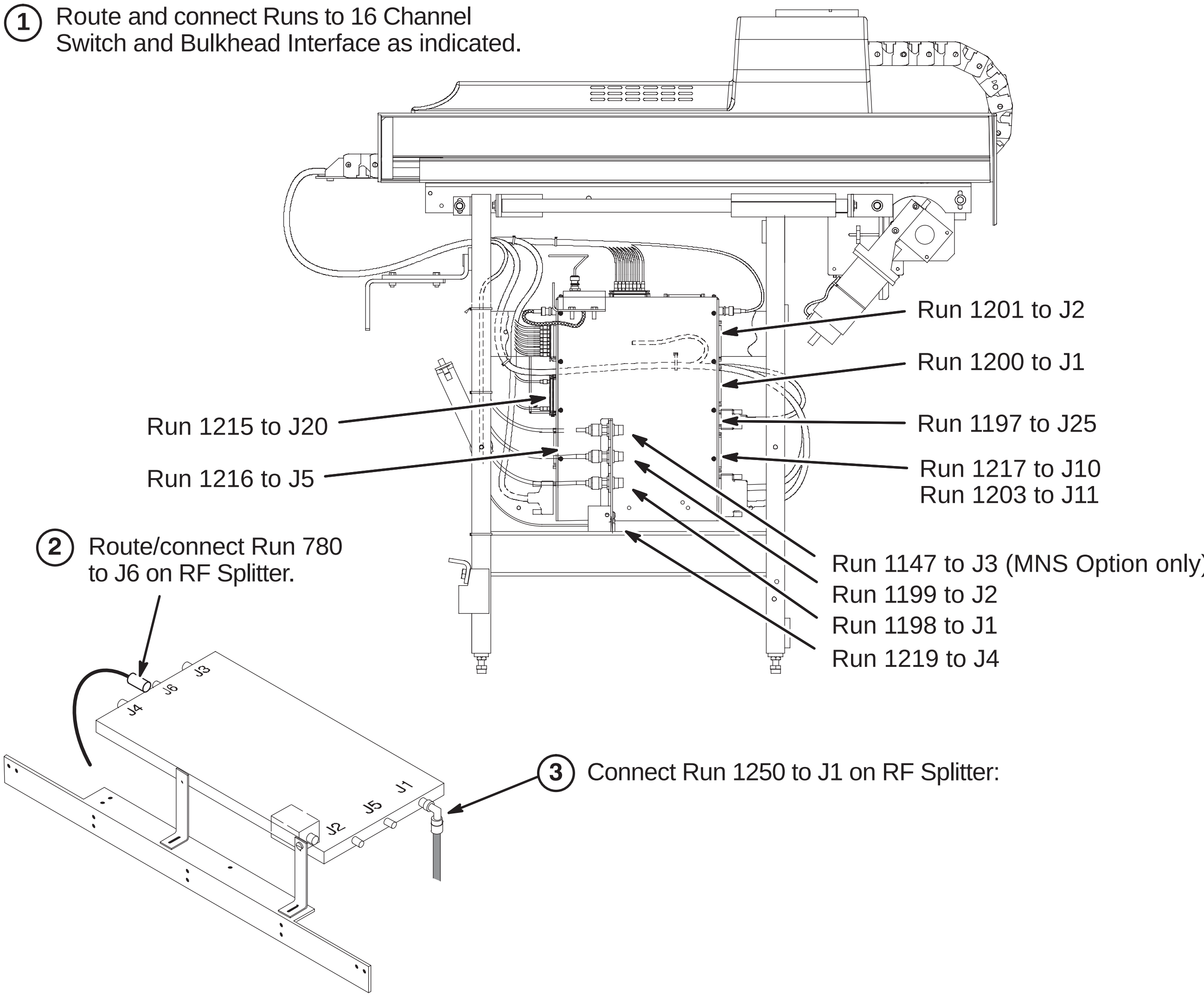
6–5–11–8 Connect Cables from Cable Track to Bulkhead Interface

1. Route Runs 1153, 1205, 1206, and 1228 along leg of Rear Pedestal and ty–wrap as indicated to leg.
2. Connect Runs to Bulkhead Interface as indicated.



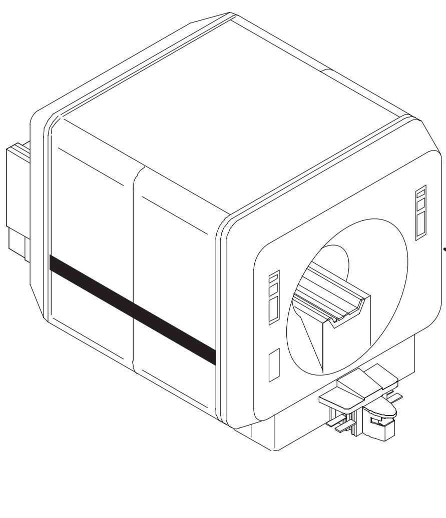
6–5–12 CONNECT REMAINING CABLES TO REAR PEDESTAL

1. Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated.



2. Route/connect Run 780 to J6 on RF Splitter.
3. Connect Run 1250 to J1 on RF Splitter:

6–5–13 INSTALL COVERS



1. Replace all covers on enclosure and rear pedestal.

6–6 TIMES MICROWAVE RF CABLES INSTALLATION

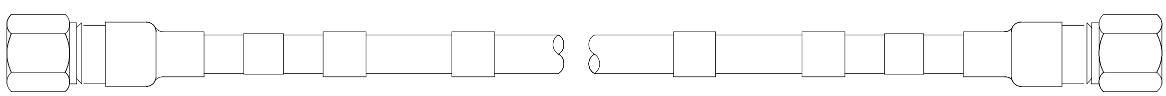
6–6–1 CABLE INFORMATION

Two cables are to be created and other associated parts to be installed are provided for this procedure. Also provided is a Times Microwave stripping tool for the LMR 600 (2352193), and a Times Microwave cutting tool (5111565).

CAUTION

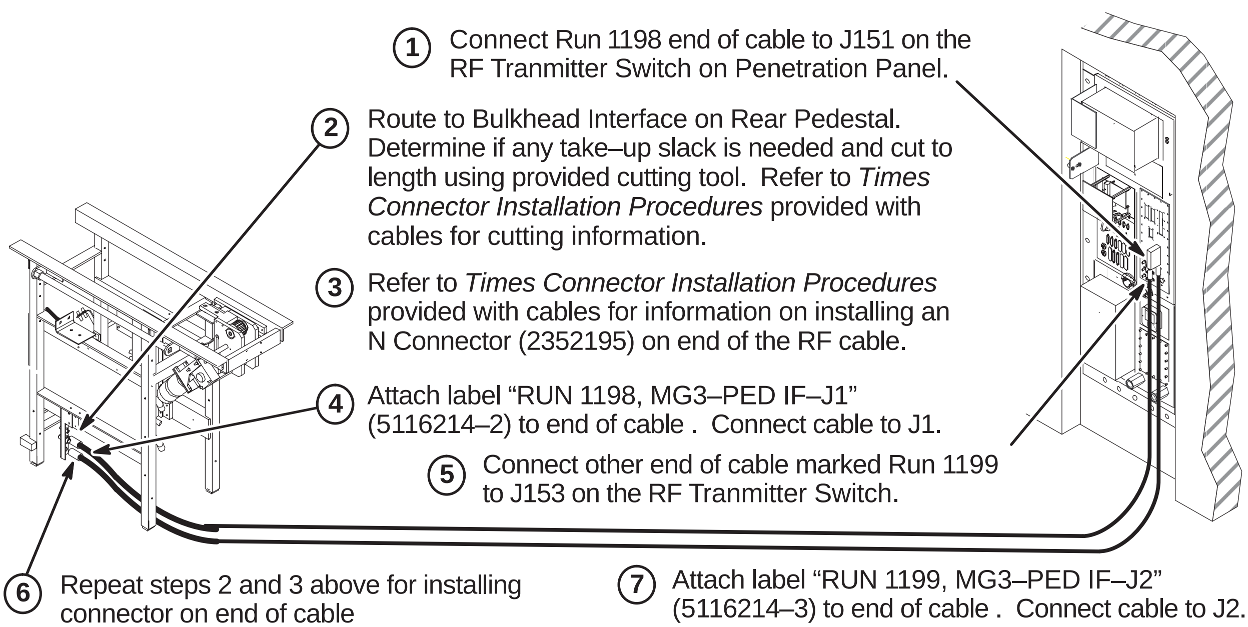
Do not subject LMR–600FR cable to a bend radius less than 3 inches (75mm). The cables will be permanently damaged if it is kinked by a bend radius less than the specified minimum.

Cable 5115915, will be cut to length to create two cables, Runs 1198 and 1199, and terminated for routing in the scan room. The remaining connectors, fasteners, and labels will be installed as directed in the following steps.



Run 1198 and Run 1199

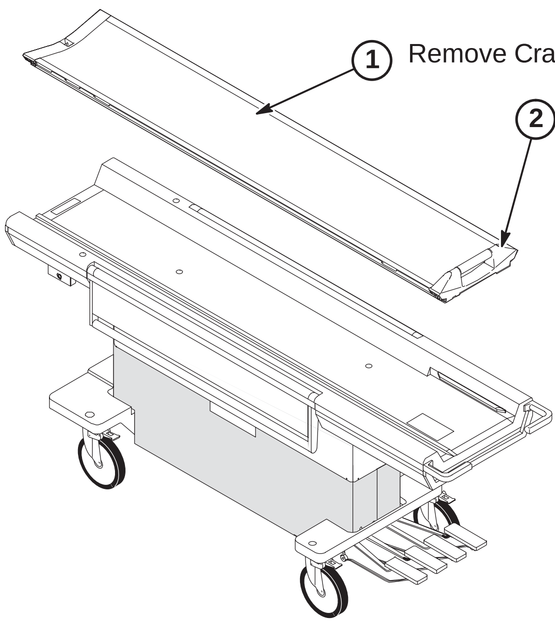
6–6–2 ROUTE, TERMINATE, AND CONNECT RF CABLES



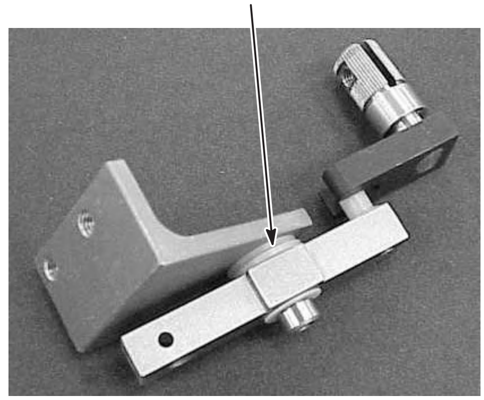
6–7 PATIENT TABLE CRADLE RELEASE UPGRADE

This upgrade is performed to improve the release mechanism and improve image quality.

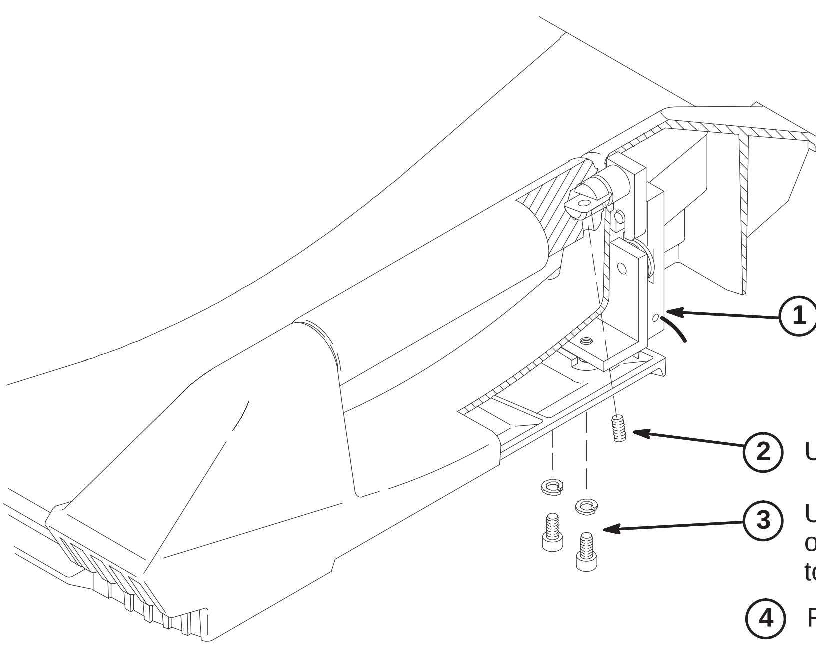
6–7–1 PREPARE CRADLE FOR UPGRADE



1. Remove Cradle from patient table.
2. Position Cradle to access release mechanism located under this corner of cradle.
3. The new mechnism is designed with two washers and two bushings inserted into the lever. The lever has also been widened in the center portion.

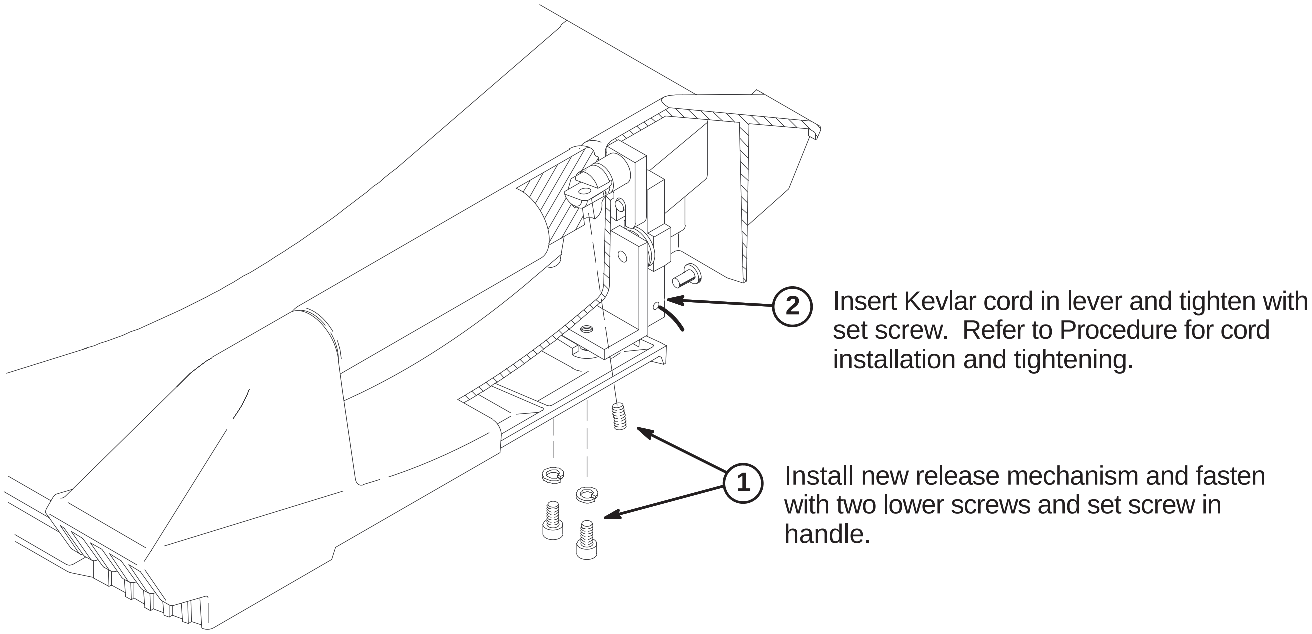


6–7–2 REMOVE EXISTING RELEASE MECHANISM

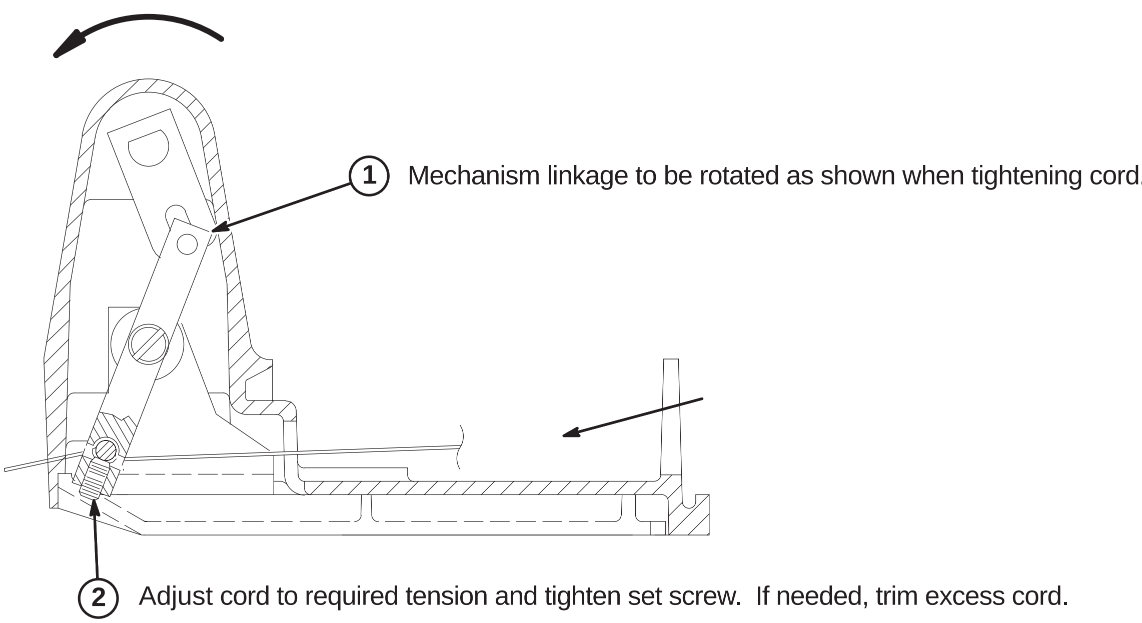


1. Unfasten set screw at bottom of lever and remove Kevlar cord from lever. Retain set screw and flat head rivet for transfer to new mechanism if required.
2. Unfasten set screw at bottom of cradle handle.
3. Unfasten two screws and washers at bottom of cradle that attach the existing mechanism to cradle.
4. Remove old release mechanism.

6–7–3 INSTALL NEW RELEASE MECHANISM



6–7–4 ATTACH CORD



SECTION 7 – OPERATOR WORKSPACE

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7–1 INTRODUCTION

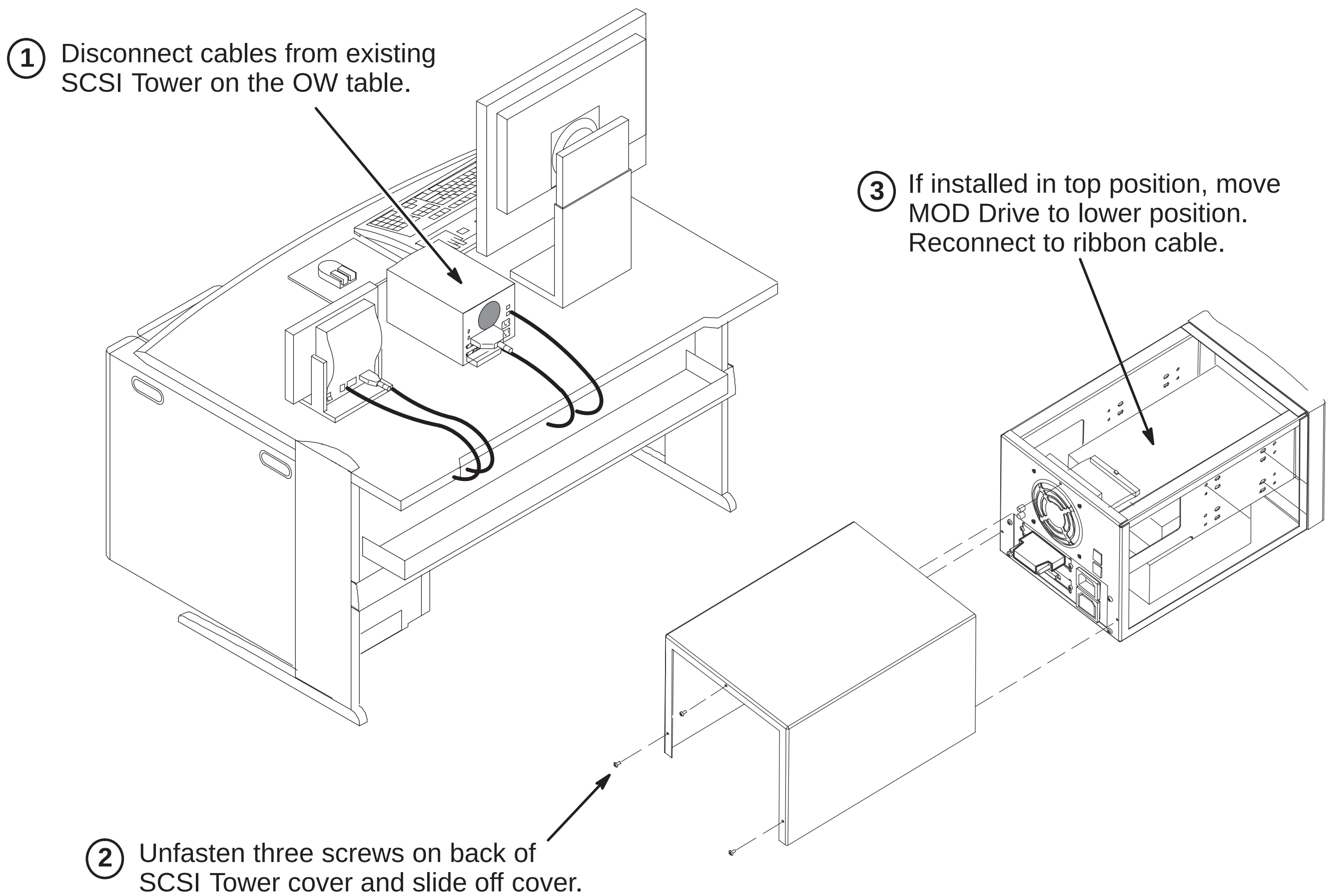
A new DVD–RW will be installed in the existing SCSI Tower on the Operator Workspace Table.

CAUTION

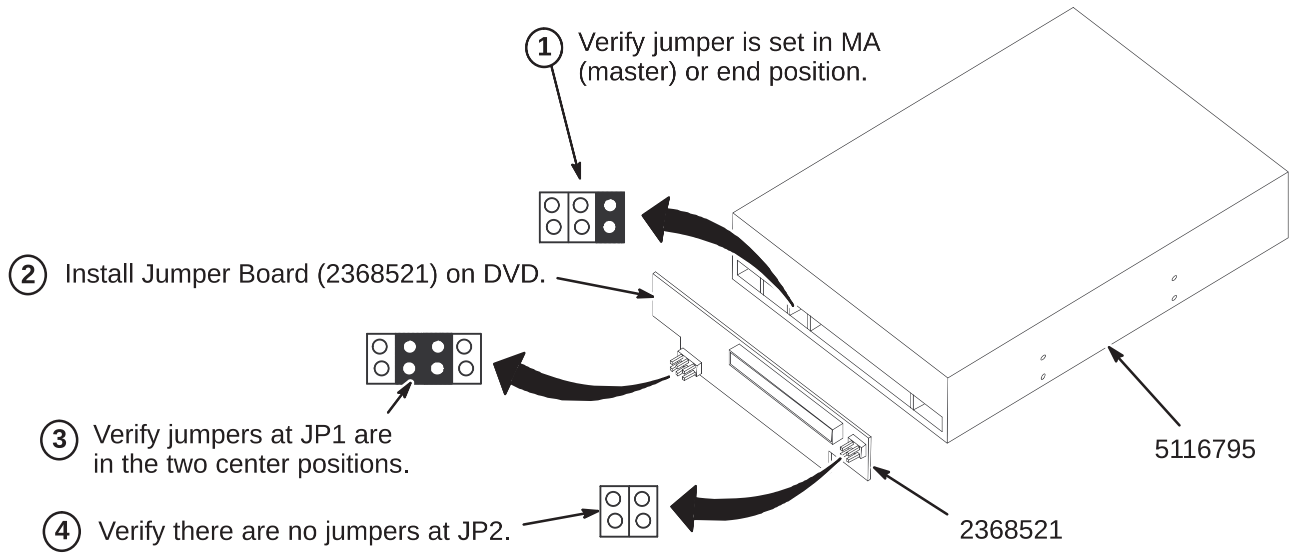
POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

7–2 INSTALLATION OF DVD IN SCSI TOWER

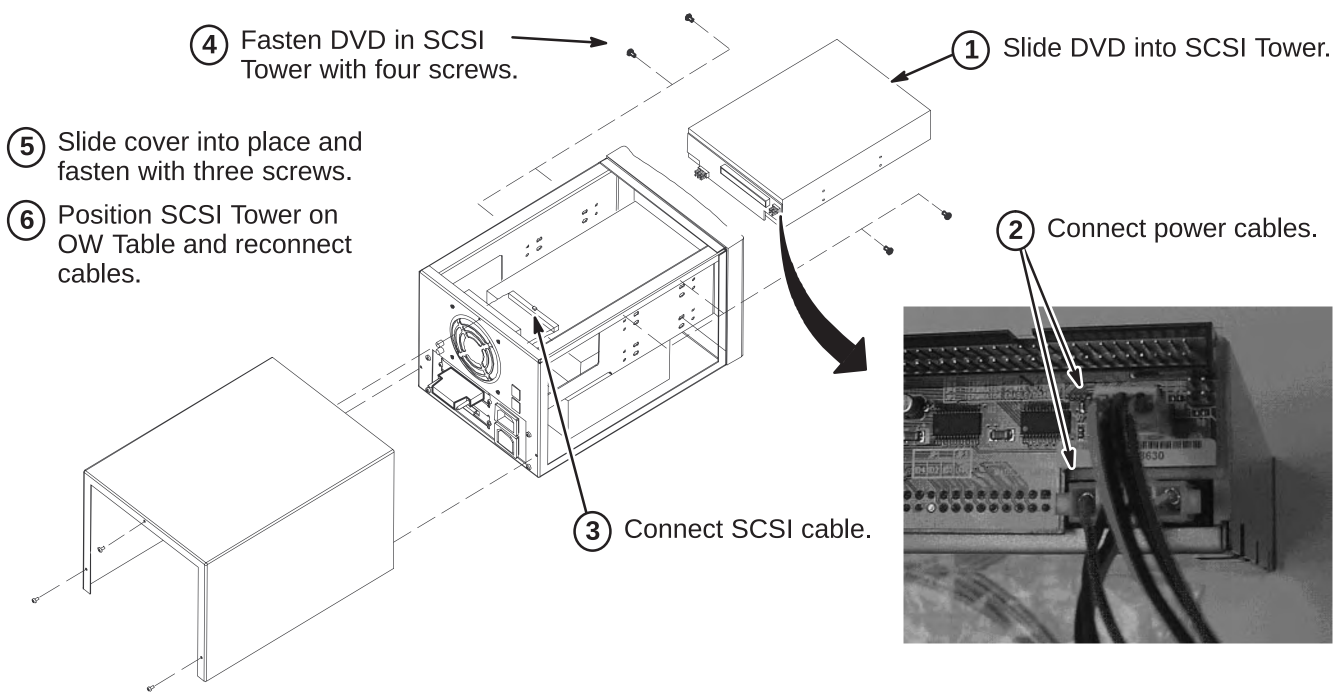
7–2–1 PREPARE SCSI TOWER FOR UPGRADE



7–2–2 VERIFY JUMPER POSITIONS ON DVD



7–2–3 INSTALL DVD INTO SCSI TOWER

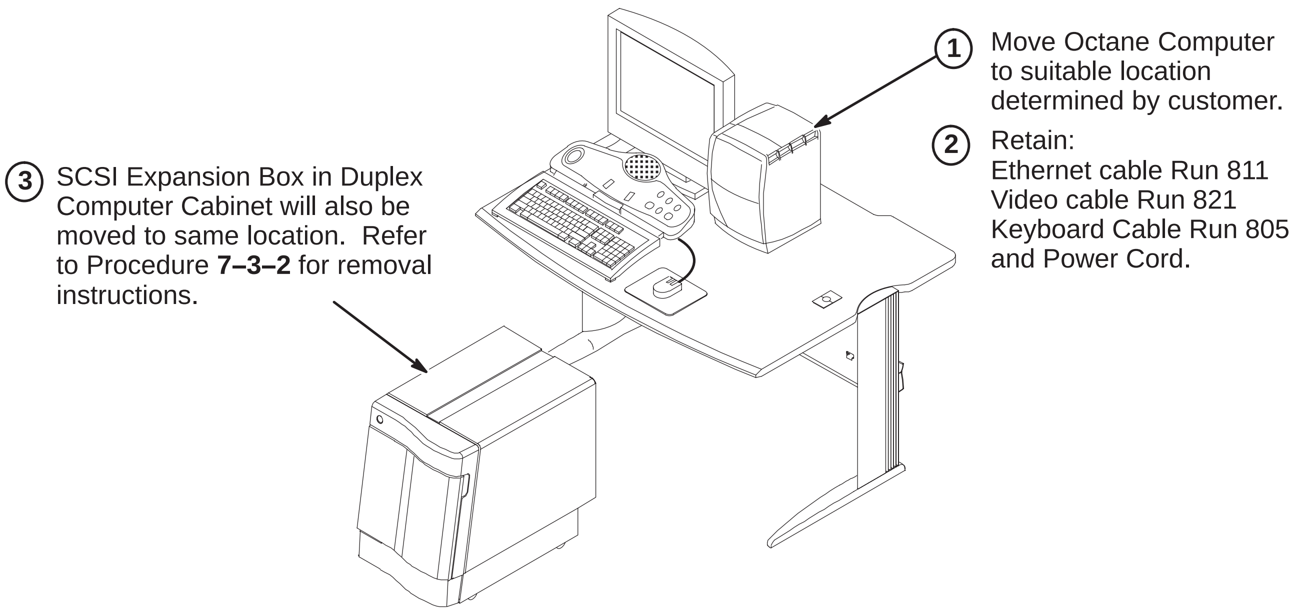


7–3 LEGACY ARCHIVE MEDIA SUPPORT

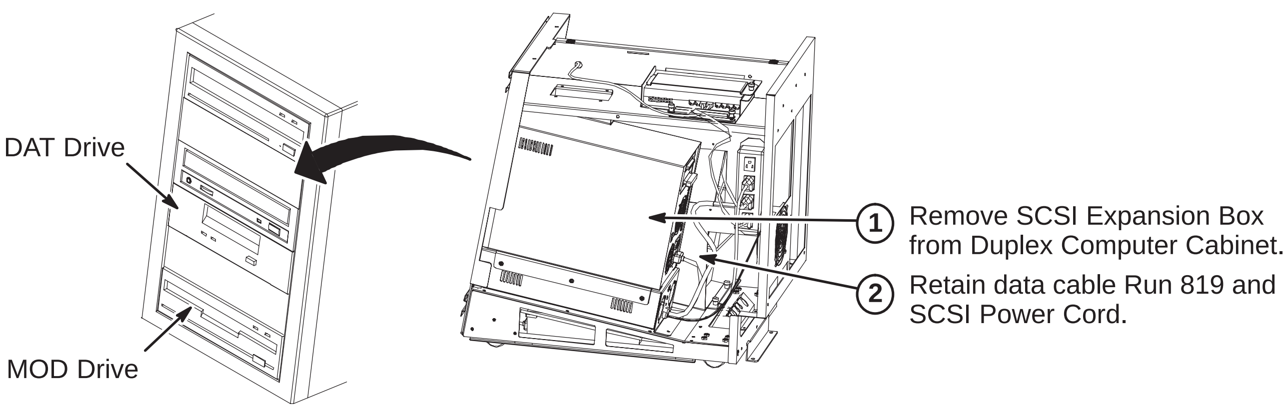
Excite 11.x Systems with Linux PC did NOT support legacy Archive Media, such as DAT and Pioneer MOD Archive Devices. This problem occurred when upgrading a 10.x systems with Octane Computers to 11.x or higher Linux computers.

To resolve this situation, sites will retain their Octane computer and legacy Media Devices. In addition, this upgrade would require a Monitor (M1000NT) and Keyboard (M1000MN).

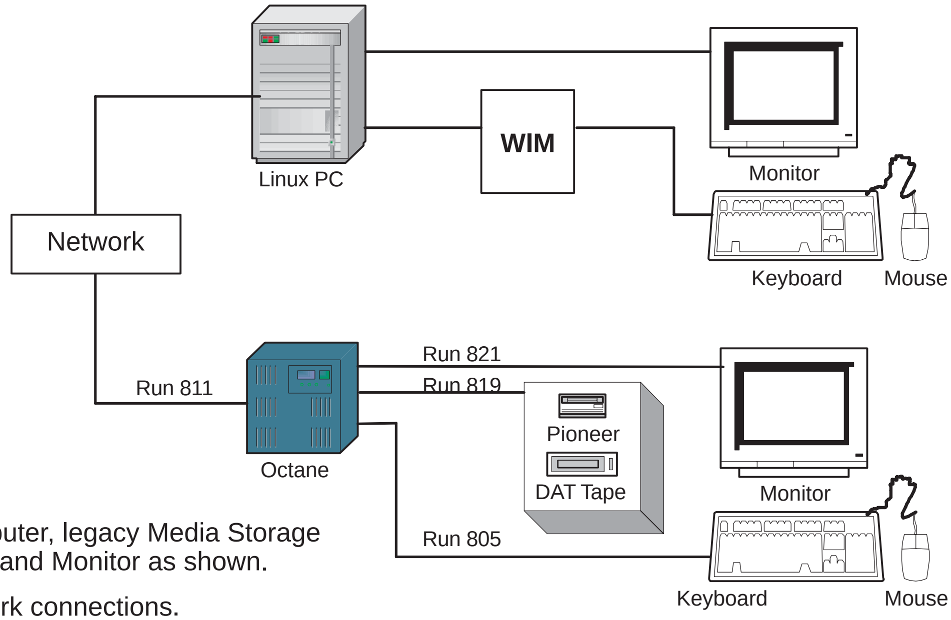
7–3–1 DEINSTALL OCTANE COMPUTER



7–3–2 DEINSTALL SCSI EXPANSION BOX AND CABLES



7–3–3 HARDWARE CONFIGURATION INTERCONNECT



1 Connect the Octane Computer, legacy Media Storage Device, Keyboard, Mouse and Monitor as shown.

2 Connect power and Network connections.

3 After Octane computer is powered up and switched ON:

1. Login as ’root’
2. cd /export/home
3. touch **.SHOW_MODE**
4. Logout
5. Login as ’signa’

4 The ’Octane’ system will now boot up in **SHOW_MODE** which will enable to further setup the DICOM Networknig to the different hosts to transfer images.

SECTION 8 – SYSTEM CABINET

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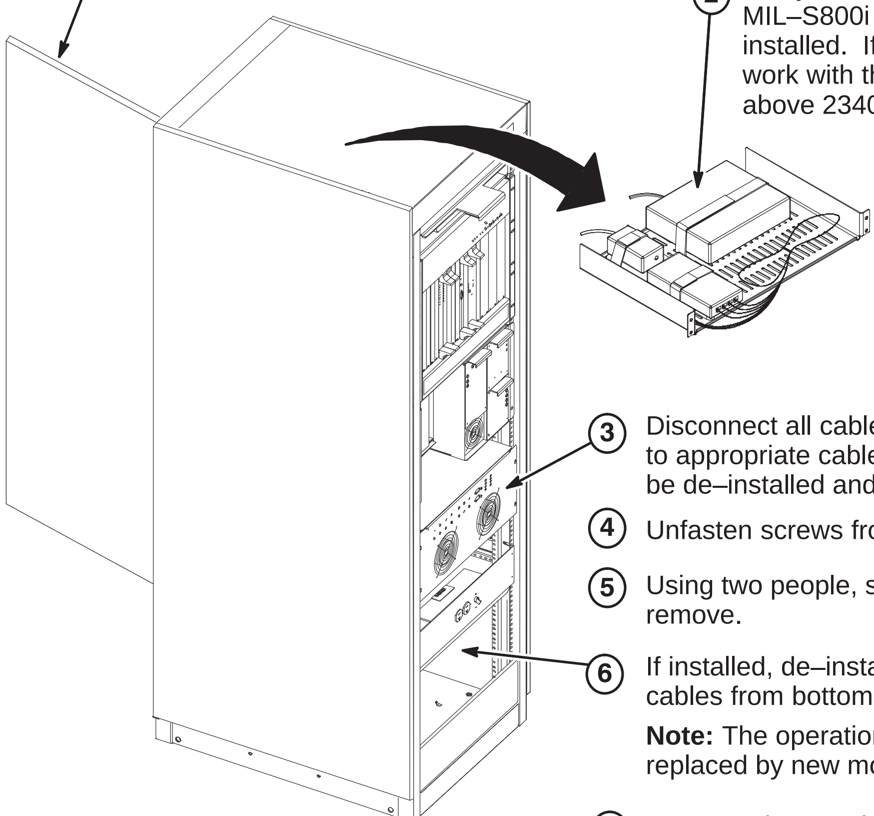
CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

8–1 INTRODUCTION

The System cabinet will be modified for the upgrade to EXCITE Release 12.x. Refer to cable maps in Section 4 for disconnection and removal of existing cables and installation of new cables. Refer to the following steps for the upgrade procedures that pertain to your site.

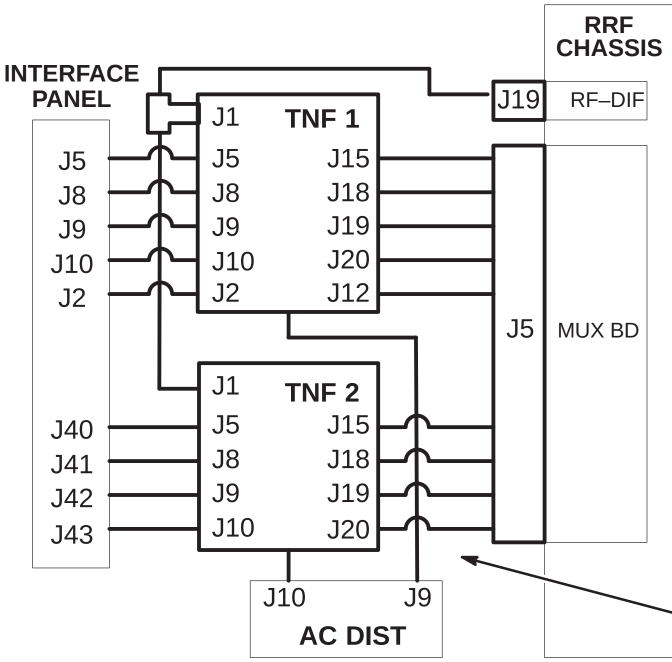
8–2 PREPARE SYSTEM CABINET FOR UPGRADE

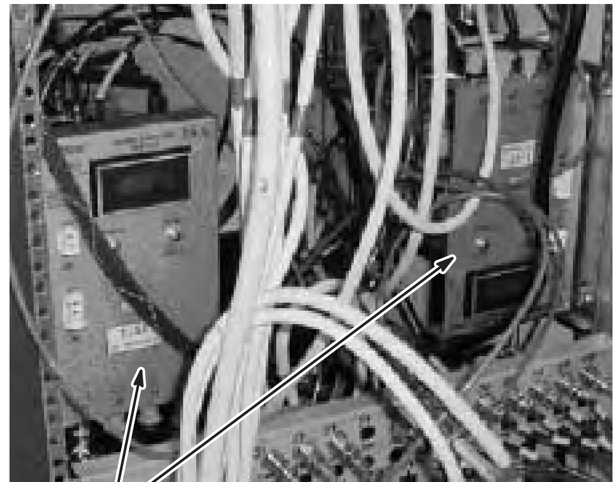


1. Remove existing rear door. New shorter door will be installed at end of cabinet upgrade procedure.
2. Verify that the proper network Switch, the Milan MIL–S800i 8 port 10/100Mbps (2340663), is installed. If an Intel hub is installed, which will not work with the EXCITE HD upgrade, order the above 2340663.
3. Disconnect all cables from rear of Driver Module. Refer to appropriate cable map to determine cables that are to be de–installed and returned for recycling.
4. Unfasten screws from front of module.
5. Using two people, slide Driver Module forward and remove.
6. If installed, de–install all TNF modules and associated cables from bottom of cabinet. Refer to Section 8–3.

   Note: The operation of the TNF modules will be replaced by new modules installed in Section 8–9–4
7. Return Driver Module, TNF Module(s) and related cables for recycling.

8–3 DE–INSTALL TNF MODULES AND CABLES

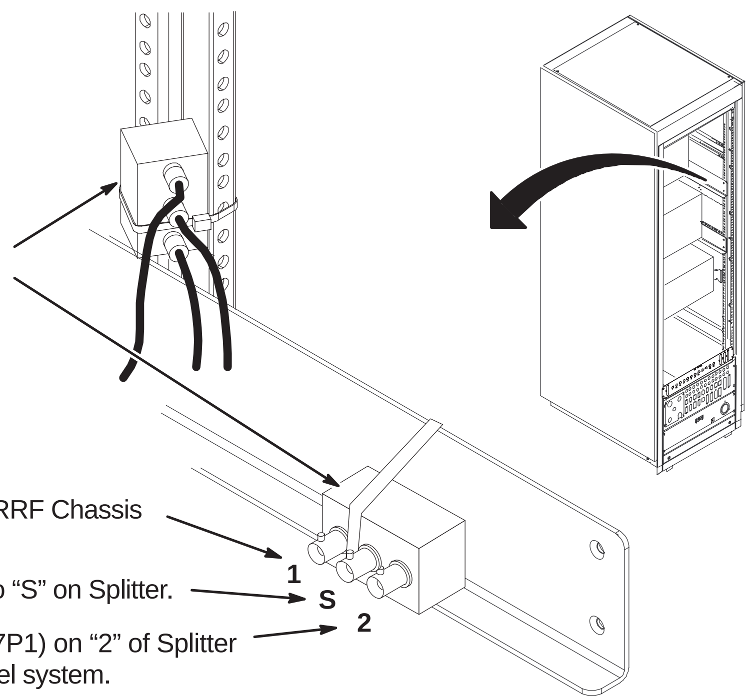




1 TNF 1 and, if installed, TNF 2 modules and associated cables will be removed and returned for recycling.

2 —— = Cables and modules to be de–installed.

8–4 INSTALL COAX POWER SPLITTER



1 Install BNC Coax Power Splitter (5115474) on vertical rail **OR** upper "L" bracket. Fasten with ty–wrap.

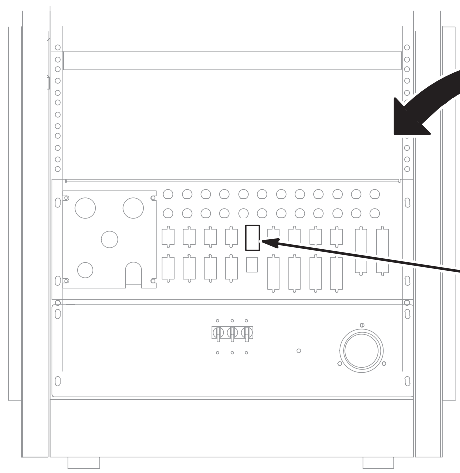
2 Connect cable from slot 12R J31 at rear of RRF Chassis to "1" on Splitter.

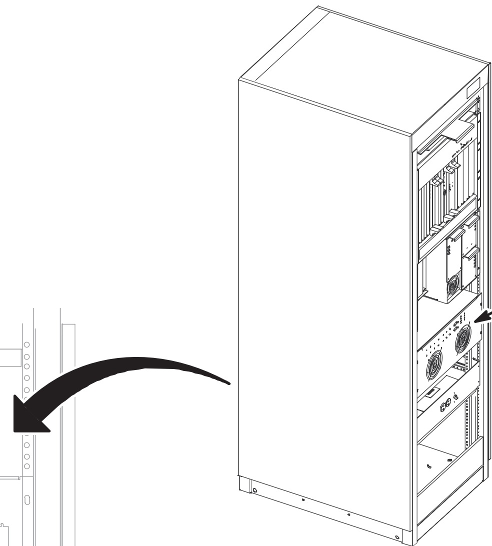
3 Connect cable from Interface Panel J202 to "S" on Splitter.

4 Install 50 Ohm BNC Terminator (46–296817P1) on "2" of Splitter if site is NOT being upgraded to a 16 channel system.

If 16 Channel is installed, connect cable from slot 3R J31 at rear of RRF Chassis to "2" on Splitter.

8–5 INSTALL NEW DRIVER MODULE

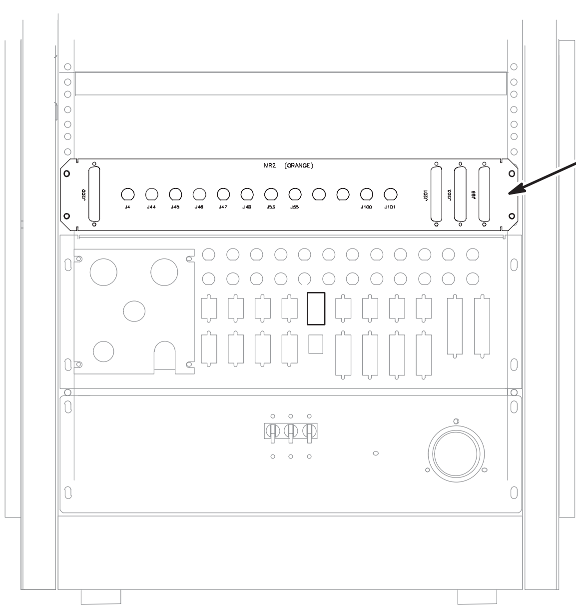




1. Using two people, slide new Driver Module into position.
2. Secure in place with previously removed screws.
3. Refer to Section 8–7 for information on connection of cables to Driver Module.
4. Make sure CAN Terminator (2303317) is connected to J56.

   Note: If ACGD Cabinet is upgraded to HFD–S, and Run 1180 is installed and connected to J56, move CAN Terminator to J8 on HFD–S Cabinet.
5. Refer to applicable cable map for connection of cables System Cabinet Interface Panel.

8–6 INSTALL NEW INTERFACE PANEL



1. At rear of cabinet, install 1.5T upgrade Interface Panel (5119238) above existing interface panel and fasten with four (4) M5 x 16mm Pan head screw (2279742–2).
2. Use table in Section 8–7 for connecting the remaining cables in the System Cabinet. All the cables connected to the new I/O Panel are pre–installed on the I/O Panel.

8–7 CONNECT CABLES FROM INTERFACE PANEL AND DRIVER MODULE

The following table provides information for connecting existing and new cables within the System Cabinet.

CABLE PART #	ONE END	OTHER END
2363363	Driver Module J1	System Cabinet Interface Panel J56
2323878–3	Driver Module J2	RF Slot 10F J23
5117295	Driver Module J4	MGD Slot 11R J21
2374675	Driver Module J8	I/O Panel J4
2382908	Driver Module J21	I/O Panel J44
2382908–2	Driver Module J22	I/O Panel J45
2382908–3	Driver Module J23	I/O Panel J46
2382908–4	Driver Module J24	I/O Panel J47
2382908–5	Driver Module J25	I/O Panel J48
2374675–2	Driver Module J7	I/O Panel J53
2374675–3	Driver Module J6	I/O Panel J55
5117963	Driver Module J10	I/O Panel J66
2394870–3	RF Slot 14R J5	I/O Panel J200
2394870–4	RF Slot 5R J5	I/O Panel J201
5115416	Splitter J5 and RF Slot 14R J13	I/O Panel J202

8–8 PROCEDURES FOR UPGRADING TO 8 CHANNEL

After upgrade, all systems will have one DRF2 board and IRF2 board.

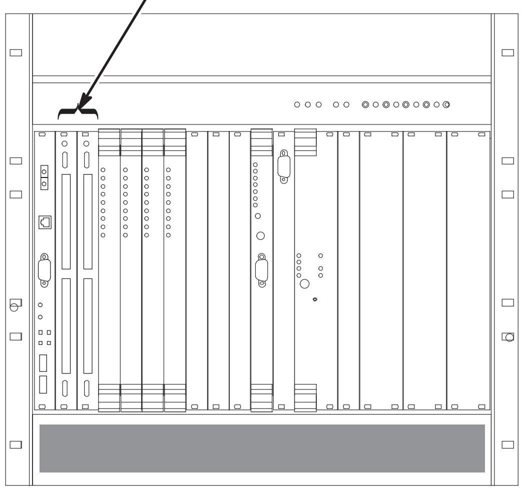
8–8–1 INSTALL REFLEX 400 AP BOARDS

Upgrade to an 8 Channel system requires the installation of ReFlex 400 AP boards, M3000RM option. Refer to Direction 5121980, also part of M3000RM, for instructions on installing AP boards.

Note: If site already has ReFlex 400 AP boards installed, then do not perform this procedure.

1 Remove blank panels or existing non–ReFlex 400B boards.

2 Install new Master (2325947–12) and Slave (2325847–13). These boards are part of M3000RM.

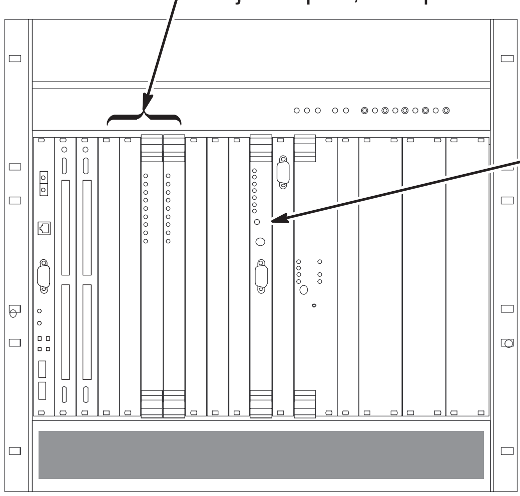


Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

8–8–2 REMOVE EXISTING DRF1 AND IRF1 BOARDS IN MGD CHASSIS

Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

1 Remove the two (or four) DRF1 boards in slots 4 thru 7. Loosen the screws in the ejector pins, then push the top and bottom tabs to eject the boards from the chassis.

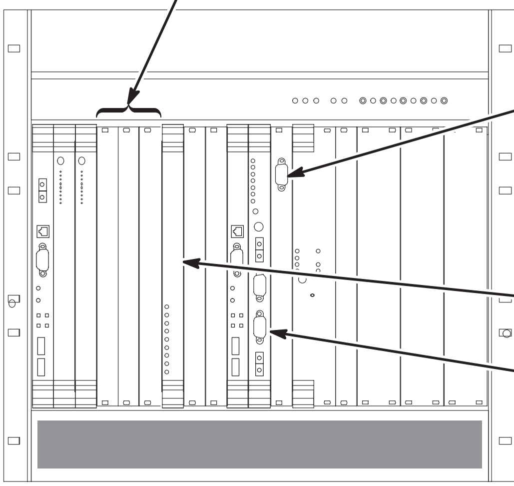


2 Remove the IRF1 board from slot 11. Loosen the screws in the ejector pins, then push the top and bottom tabs to eject the boards from the chassis.

8–8–3 INSTALL CAN TERMINATOR, DRF2, AND IRF2 BOARDS IN MGD CHASSIS

Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

1. Install Blank Filler Panels (2319152) over slots 4 to 6.

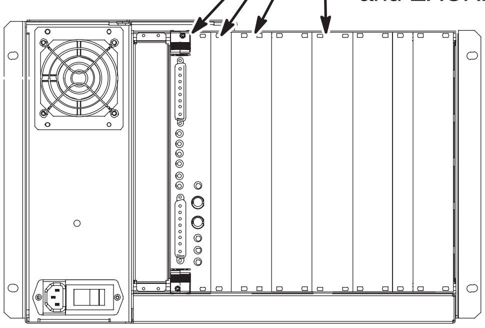


2. At slot 12, install Series CAN Terminator (5117168) on SCP2 between board and cable at connector PMC2.
   OR
   If SCP1, remove SCP1 and install SCP2 Board (2294300–12, part of M3000PK) with Series CAN Terminator between board and cables at connector PMC2.
3. Install DRF2 Module (2298332) in slot 7.
4. Install IRF2 Module (2358122–2) in slot11.

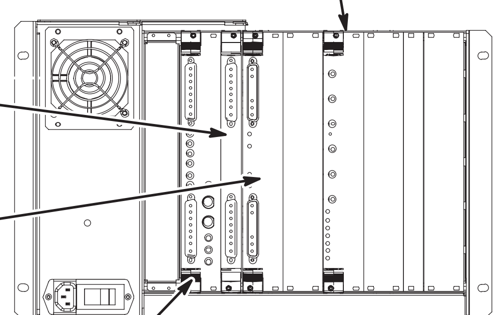
8–8–4 INSTALL UTNS3, RCVR2, MUX, AND EXCR2 BOARDS

Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

1. If system being upgraded has MUX, UTNS2, RCVR1, and EXCR1 boards at these four slots, then these four boards have to be replaced with UTNS3, RCVR2, MUX, and EXCR2 boards (part of M3000PH).



2. Remove existing Exciter board and install Remote RF Master Exciter 2 board (2361470–2) in slots 6 & 7.
3. Remove UTNS 2 board and install UTNS 3 board (2366218–2) in slot 12.
4. Remove existing RCVR board and install RCVR 2 board (2349808–2) in slots 10 & 11.



5. Remove existing MUX board and install new MUX board (5250106) in slots 13 & 14.

8–9 PROCEDURES FOR UPGRADING TO 16 CHANNEL

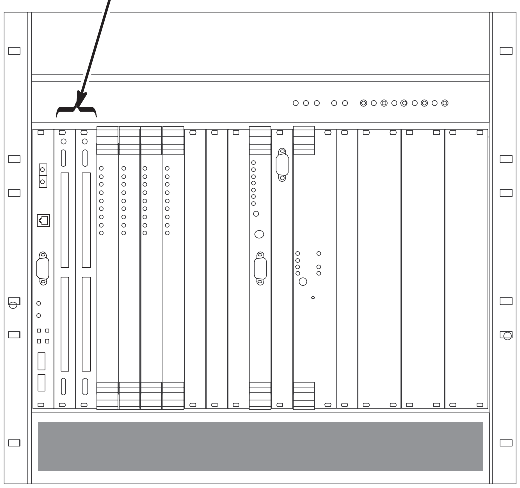
After upgrade, all systems will have one DRF2 board and IRF2 board.

Do NOT perform this procedure (Section 8–9) if site is not upgrading to 16 channels.

8–9–1 INSTALL REFLEX 800 AP BOARDS

Upgrade to a 16 Channel system requires the installation of ReFlex 800 AP boards, M3000RN option. Refer to Direction 5121980, also part of M3000RN, for instructions on installing AP boards.

1. Remove blank panels or existing non–ReFlex 800 boards.
2. Install new Master (2325947–15) and Slave (2325847–16). These boards are part of M3000RN.

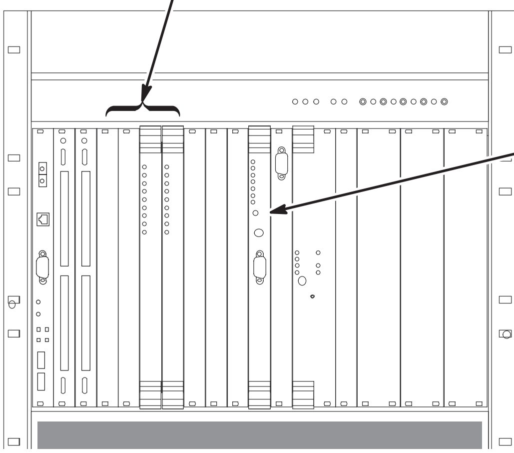


Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

8–9–2 REMOVE EXISTING DRF1 AND IRF1 BOARDS IN MGD CHASSIS

Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

1. Remove the two (or four) DRF1 boards in slote 4 thru 7. Loosen the screws in the ejector pins, then push the top and bottom tabs to eject the boards from the chassis.

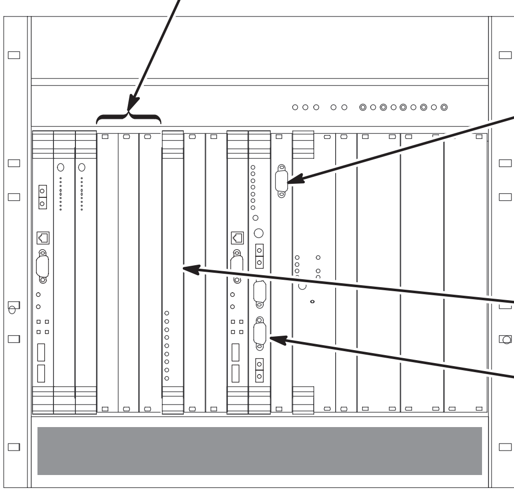


2. Remove the IRF1 board from slot 11. Loosen the screws in the ejector pins, then push the top and bottom tabs to eject the boards from the chassis.

8–9–3 INSTALL CAN TERMINATOR, DRF2, AND IRF2 BOARDS IN MGD CHASSIS

Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

1. Install Blank Filler Panels (2319152) over slots 4 to 6.



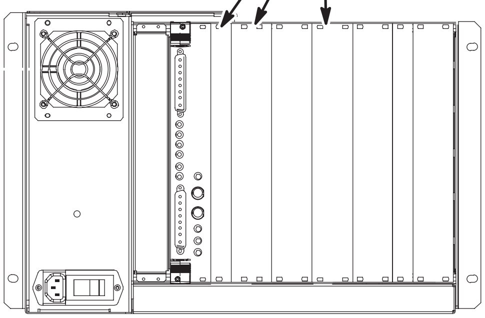
2. At slot 12, install Series CAN Terminator (5117168) on SCP2 between board and cable at connector PMC2.
   OR
   If SCP1, remove SCP1 and install SCP2 Board (2294300–12, part of M3000PK) with Series CAN Terminator between board and cables at connector PMC2.
3. Install DRF2 Module (2298332) in slot 7.
4. Install IRF2 Module (2358122–2) in slot 11.

8–9–4 INSTALL UTNS3, RCVR2, MUX, AND EXCR2 BOARDS (If Required)

Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

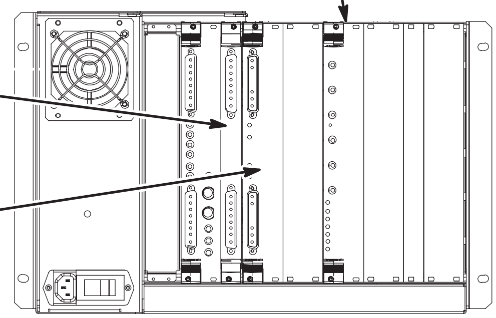
① If system being upgraded has UTNS2, RCVR1, and EXCR1 boards at these slots, then these three boards have to be replaced with UTNS3, RCVR2, and EXCR2 boards (part of M3000PH).

② Remove existing Exciter board and install Remote RF Master Exciter 2 board (2361470–2) in slots 6 & 7.



③ Remove UTNS 2 board and install UTNS 3 board (2366218–2) in slot 12.

④ Remove existing RCVR board and install RCVR 2 board (2349808–2) in slots 10 & 11.



8–9–5 REMOVE BLANK PANELS AND POWER SUPPLY FROM RRF CHASSIS (16 Channel Only)

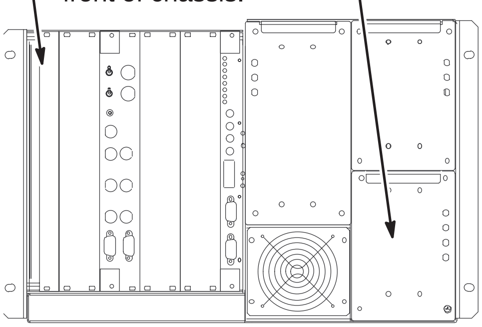
Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

Note: It is recommended that the existing MUX board be at the same part number and revision level as the new MUX board being installed in slots 4 & 5.

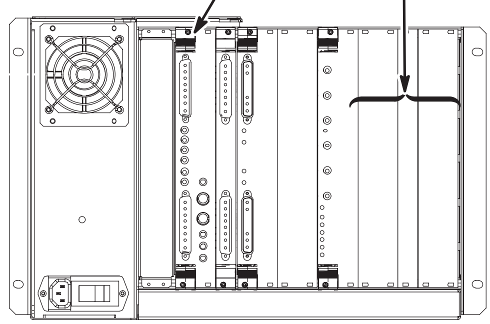
① De–install blank panel or board from slot 1 at front of chassis.

② The 5 Volt Power Supply will be replaced. Unfasten the four screw at bottom of power supply. The module should easily slide out.

③ De–install blank panels or boards from slots 1 to 5 at rear of chassis.



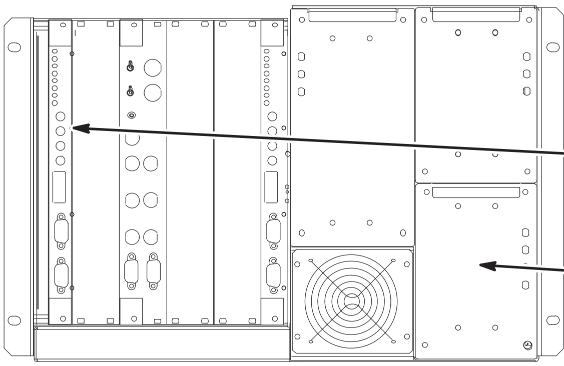
FRONT VIEW



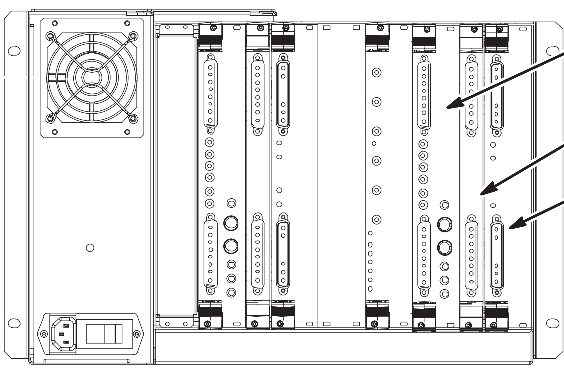
REAR VIEW

Note: All de–installed boards are to be returned for recycling.

8–9–6 INSTALL MODULES IN RRF CHASSIS



FRONT

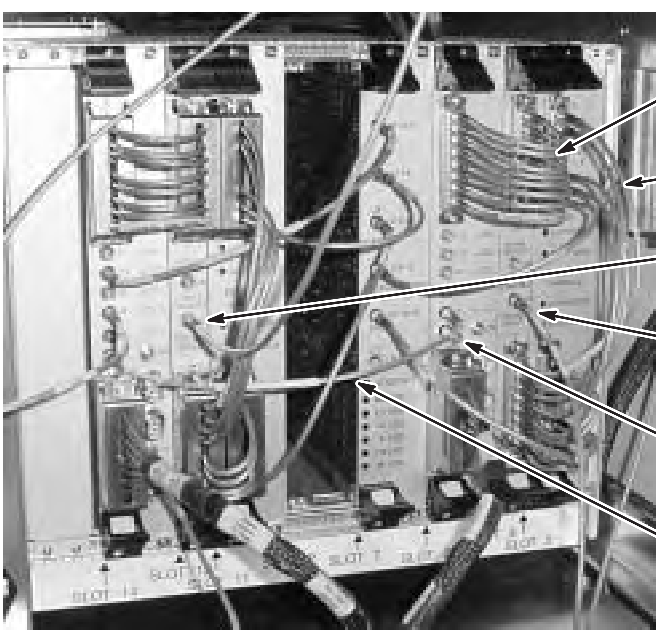


REAR

Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

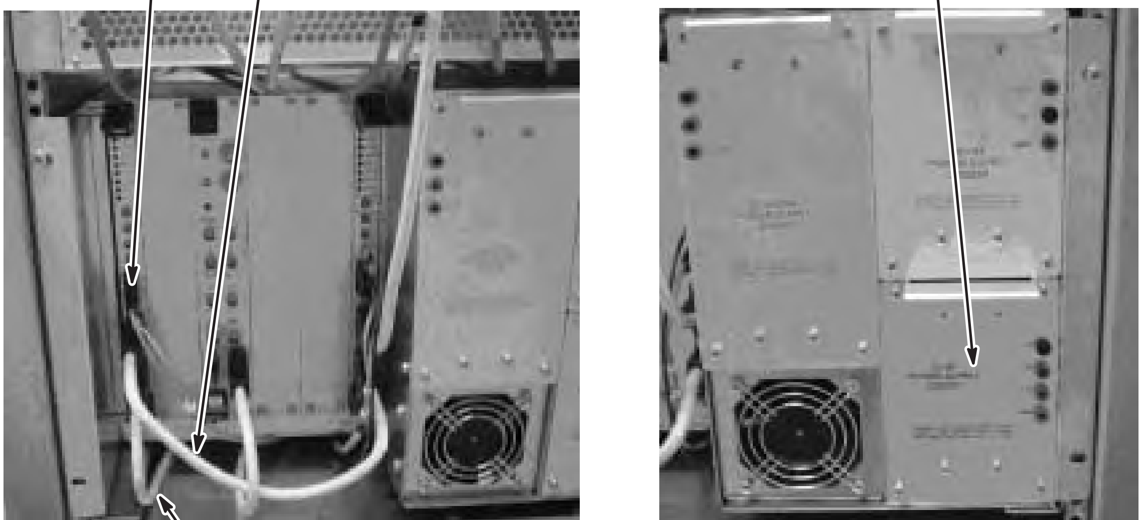
1. Install RF–DIF Board (2274184 or 5250028) in slot 1.
2. Slide in new 5V RF Module RF Power Supply and secure with four screws previously removed.
3. Install MUX Board (5250106) in slots 4 and 5.
4. Install UTNS3 Board (2366218–2) in slot3.
5. Install RCVR II Board (2349808–2) in slots 1 and 2.

8–9–7 INSTALL CABLES AT REAR OF RRF MODULE

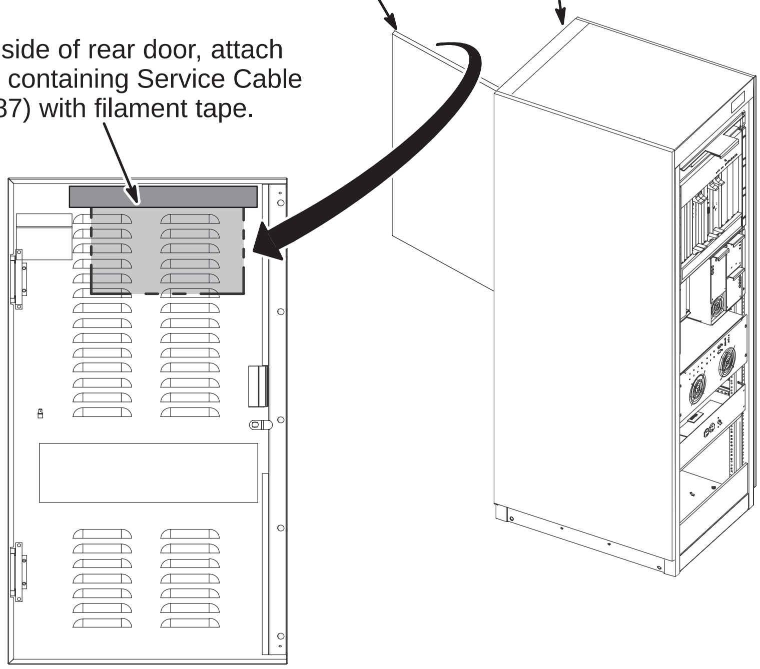


1. Connect cable 5123706 as marked.
2. Connect cable 5113193–2 as marked.
3. Connect cable 5115417 slot 12R J31 as marked. Route and connect to Coax Power Splitter “1”.
4. Connect cable 5115417–2 as marked. Route and connect to Coax Power Splitter “2”.
5. Move 50 Ohm Terminator from Mux1 J6 (slots 13 & 14) to Mux2 J6 (slots 4 & 5).
6. Connect cable 5115630 as marked.

8–9–8 INSTALL CABLES AND POWER SUPPLY AT FRONT OF RRF MODULE (5116741 COLLECTOR)



8–10 INSTALL NEW REAR DOOR AND RATING PLATE



SECTION 9 – SRF OR SRFD2 CABINET

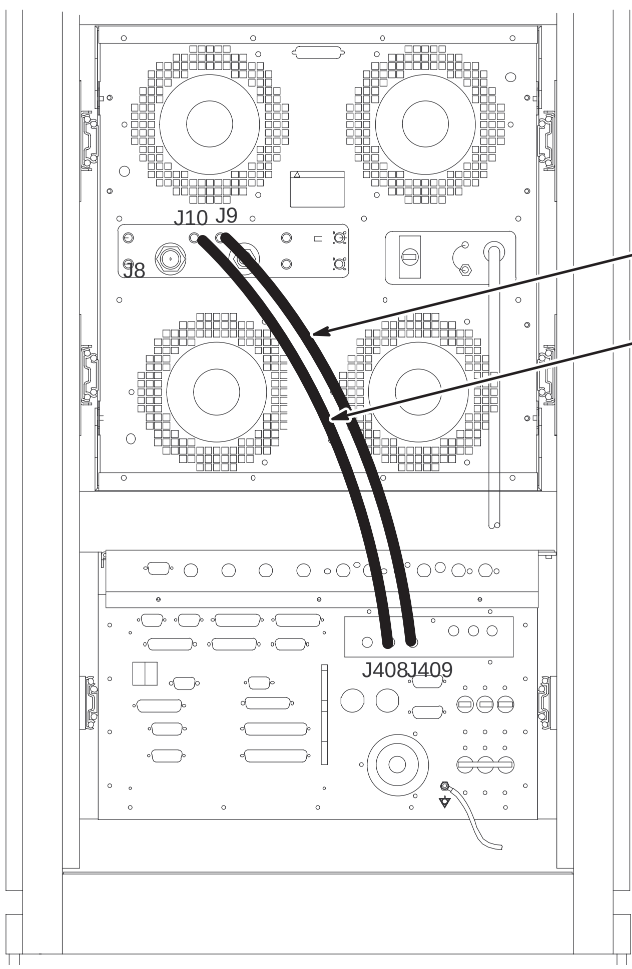
CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

9–1 INTRODUCTION

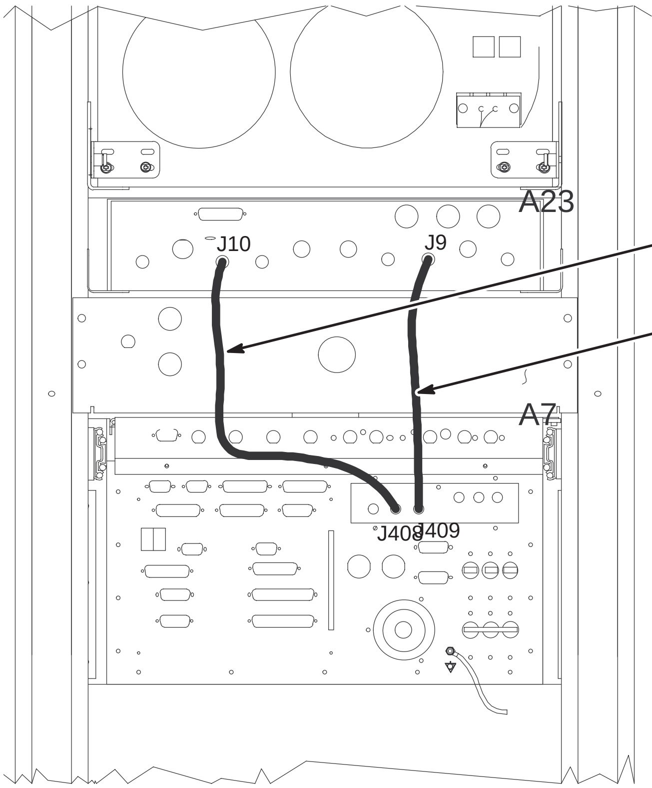
The existing SRF Cabinet will have only a few changes. Some of the functionality of the SSM Module is being transferred to the new Driver Module in the System Cabinet. The following procedures describe the steps needed to accomplish this change.

9–2 DEINSTALL CABLES IN SRFD2 CABINET



1. Deinstall cable between SSM J409 and RFI J9.
2. Deinstall cable between SSM J408 and RFI J10.
3. Return cables to GoldSeal for recycling.

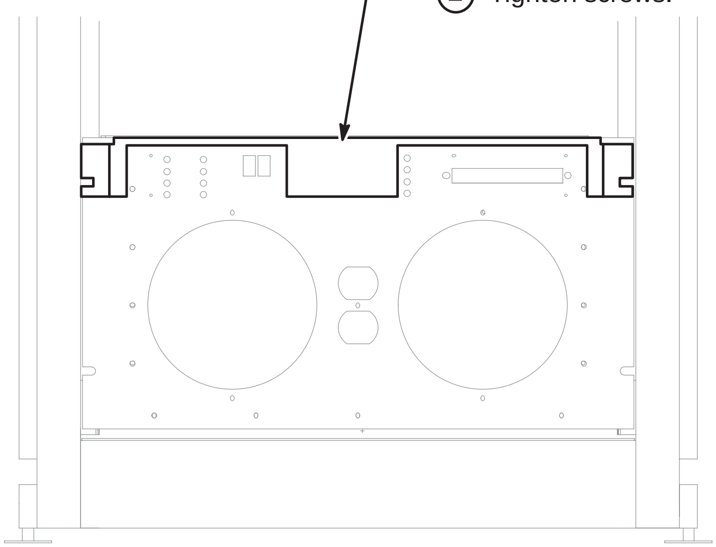
9–3 DEINSTALL CABLES IN SRF CABINET

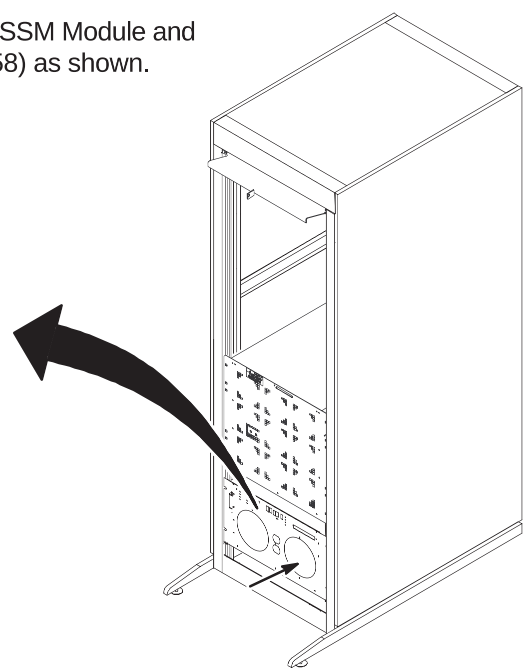


1. Deinstall cable between SSM J408 and RFI J10.
2. Deinstall cable between SSM J409 and RFI J9.
3. Return cables to GoldSeal for recycling.

9–4 INSTALL SWITCH COVER ON FRONT OF SSM MODULE

1. Loosen two screws at top of SSM Module and install Switch Cover (5120058) as shown.
2. Tighten screws.





SECTION 10 – ACGD/HFD CABINET

CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

10–1 INTRODUCTION

The ACGD/PDU Cabinet **WILL NOT BE** replaced but **WILL BE** upgraded to HFD–S when upgrading a site from EXCITE 11.x to 12.x. Instruction for the upgrade are found in Direction 5131778, *ACGD to HFD–S Upgrade Installation,* which is shipped with the M3000PT or M3000PW collector of parts.

The ACGD–Lite/PDU cabinet **WILL BE** replaced by an HFD Cabinet if the site is an MR/i SmartSpeed with an ACGD–Lite Cabinet. Refer to Section 4 for additional information and instructions on disconnecting, removing, relabeling, and/or reconnecting of cables when upgrading to EXCITE.

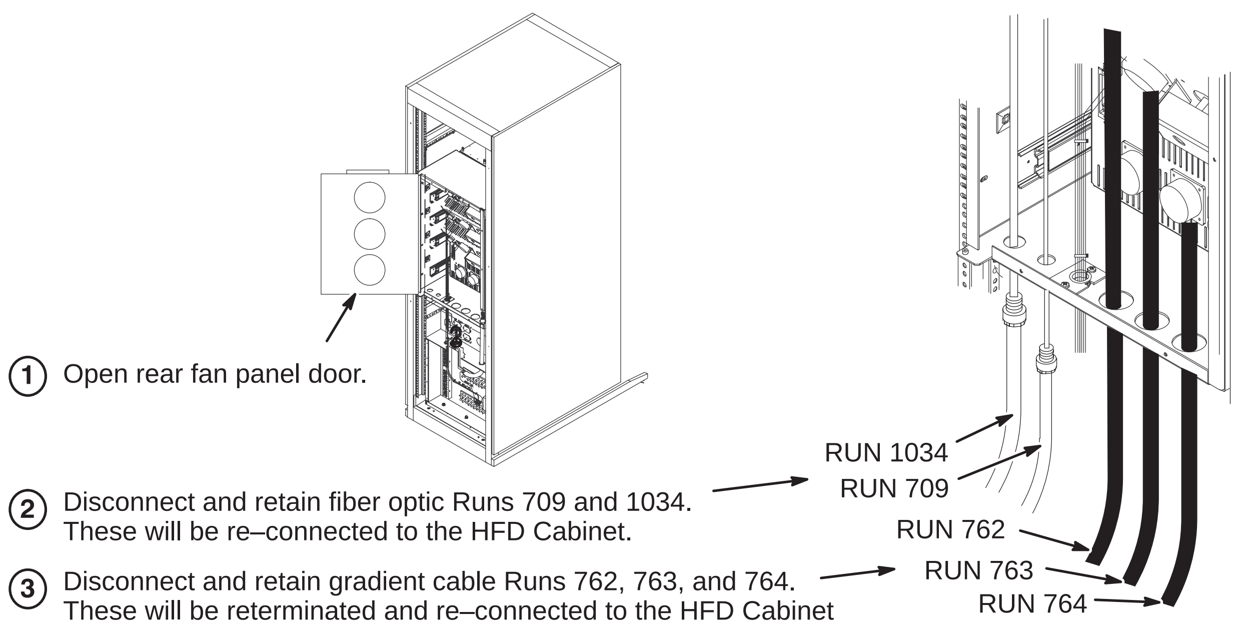
Refer to the following sub–sections for removal of existing ACGD–Lite Cabinet and installation of the new HFD Cabinet.

10–2 DEINSTALL ACGD–Lite CABINET

DANGER!!

LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. MAKE SURE THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

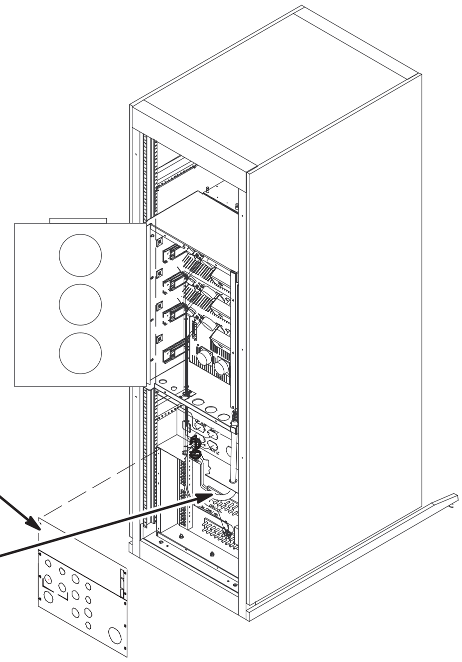
10–2–1 Disconnect Fiber Optic and Gradient Cables



10–2–2 Disconnect Power and Data Cables

NOTE: Refer to Section 4 for cable deinstallation information.

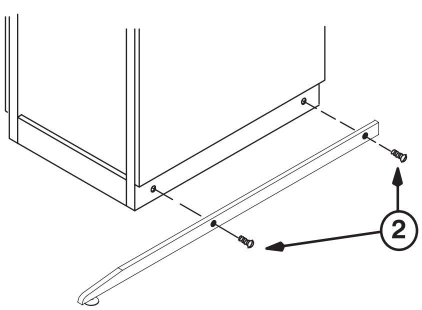
1. Disconnect data cables at top Interface panel. Loosen clamps around power cables at bottom Interface panel. Unfasten Interface panel screws and remove panels.
2. Disconnect and retain all PDU power cables. These will be re–connected to the HFD Cabinet.
3. Reinstall Interface panel and close all doors.



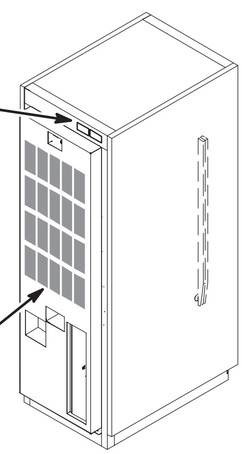
10–2–3 Remove ACGD–Lite Cabinet

WARNING!

DUE TO WEIGHT OF CABINET, AT LEAST TWO PEOPLE ARE REQUIRED TO MOVE CABINET



1. Remove and save cabinet magnets for transfer to new HFD Cabinet.
2. Unfasten Anti–tip bars (if installed) and place inside cabinet.
3. Return cabinet to GoldSeal. Refer to instructions in Section 1.



10–3 RETERMINATE GRADIENT CABLE RUNS 762, 763, AND 764

The Gradient cables will be reused but have to be reterminated. The HFD Cabinet uses pressure connectors. Refer to the follwing steps for the alteration to the “ONE” and “TWO” wires.

10–3–1 Remove Existing Ring Terminals

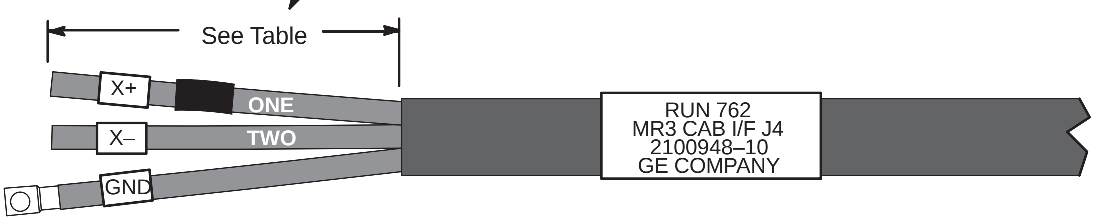
1 Cut off ring terminals on X+ and X– wires. **Do NOT cut ring terminal on Ground wire.** Ground wire can be reused as is.

2 Repeat Step 1 for Runs 763 and 764

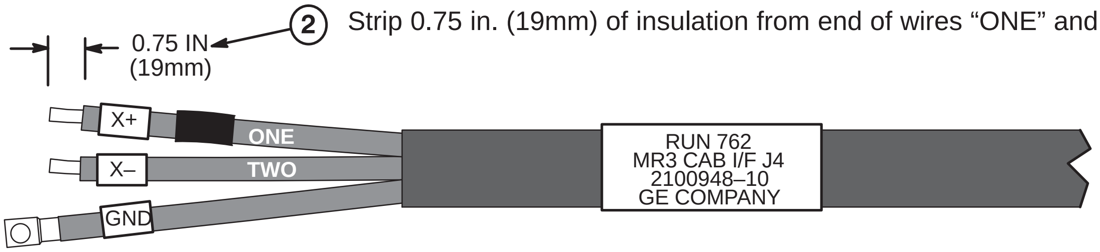
10–3–2 Terminate Runs 762, 763, and 764 for HFD Cabinet

	Run 762	Run 763	Run 764
Recommended Jacket Length to be Removed	8.0 IN (203mm)	11.0 IN (280mm)	14.0 IN (356mm)

1 Refer to above table to determine length of jacket and foil to be removed from end of the “ONE” and “TWO” wires for cable Runs 762 (X), 763 (Y), and 764 (Z).

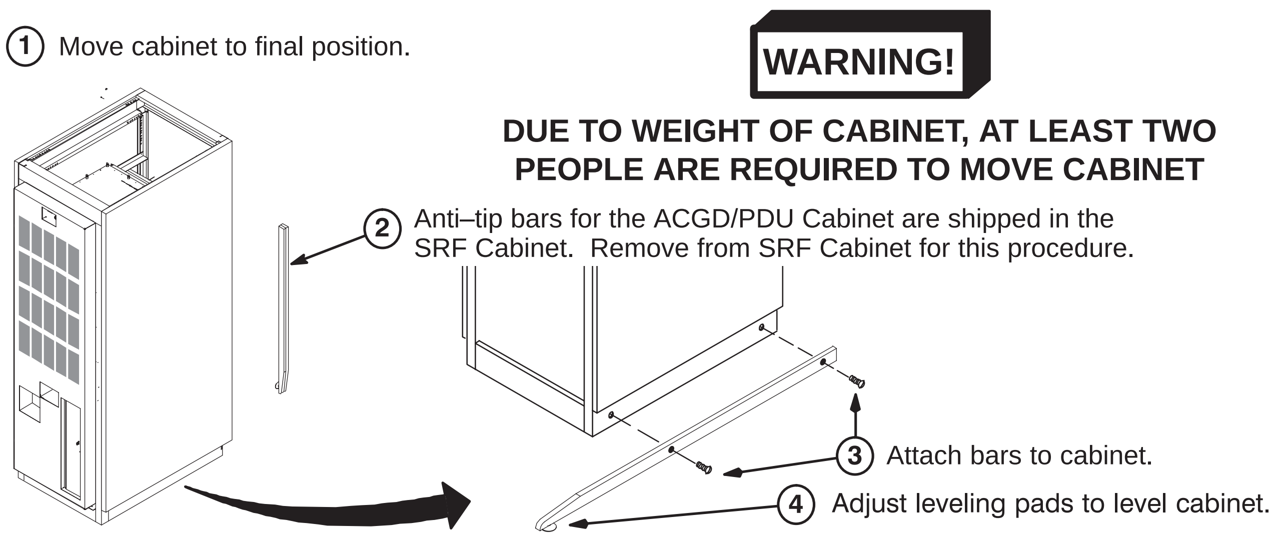


2 Strip 0.75 in. (19mm) of insulation from end of wires “ONE” and “TWO”.

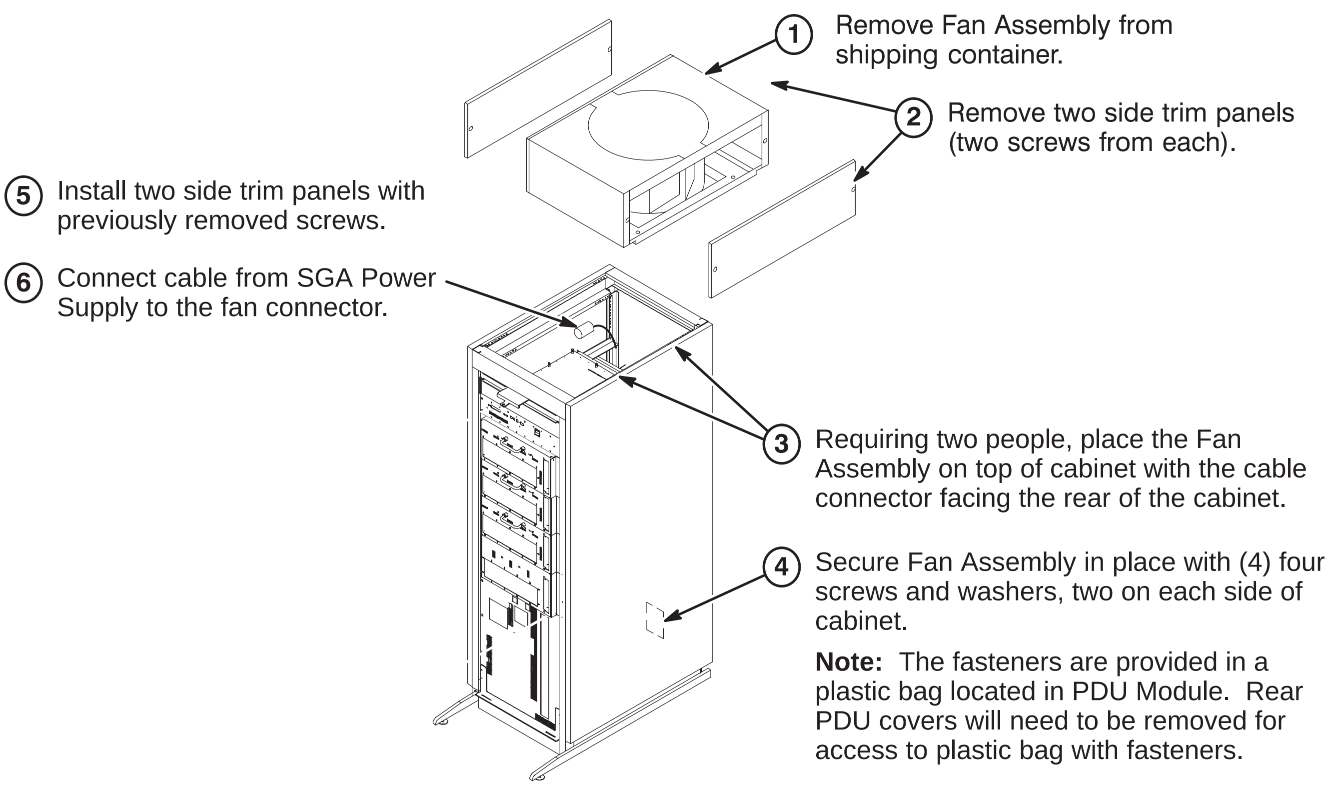


10–4 INSTALLATION OF HFD CABINET

10–4–1 POSITION HFD CABINET



10–4–2 INSTALL FAN ASSEMBLY AND ATTACH INSITE MAGNET

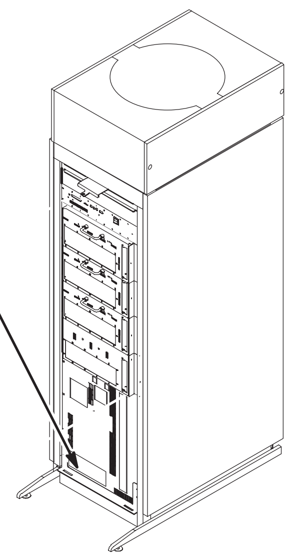


DANGER!!

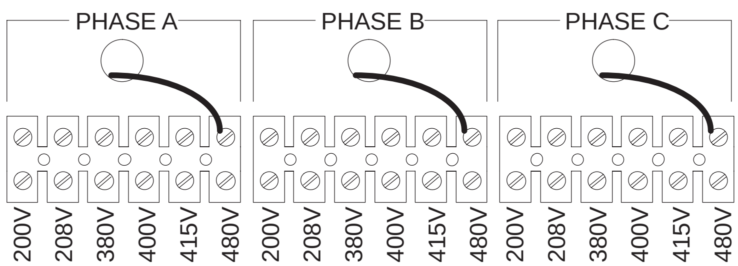
LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. MAKE SURE THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

10–4–3 INPUT VOLTAGE SELECTION

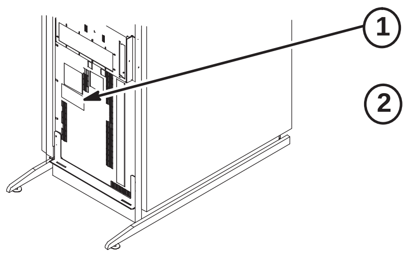
1. Using a voltmeter or other voltage indicating device, check to be sure that no voltage is being applied to the input of the PDU.
2. Remove the plate from the front of the PDU marked INPUT CIRCUIT BREAKER CURRENT SELECT / INPUT VOLTAGE SELECT.
3. For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.



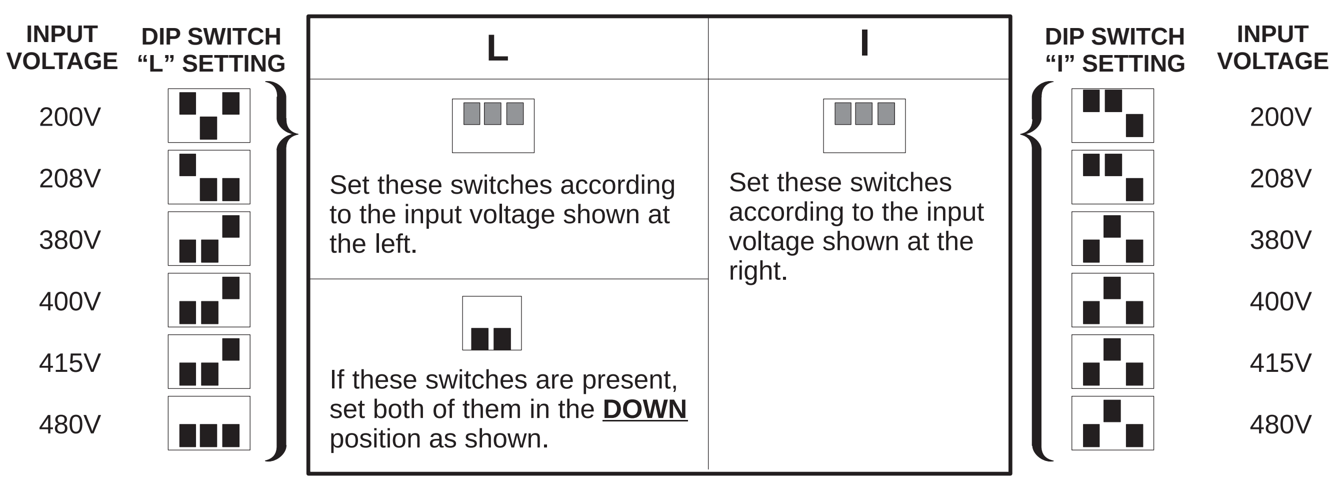
(SHOWN WIRED FOR 480V)



10–4–4 CIRCUIT BREAKER DIP SWITCH SETTINGS



1. Remove small cover plate under the main circuit breaker to access DIP switches controlling the circuit breaker operation.
2. Table below shows the position of the DIP switches for each input voltage selection. The switches immediately under the letter “L” and “I” are the only ones that should be changed.

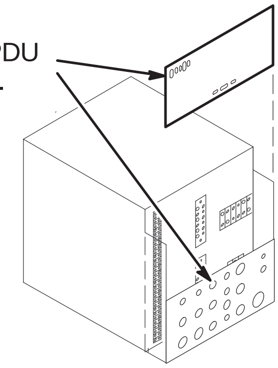


10–4–5 FACILITY POWER AND GROUND CONNECTION

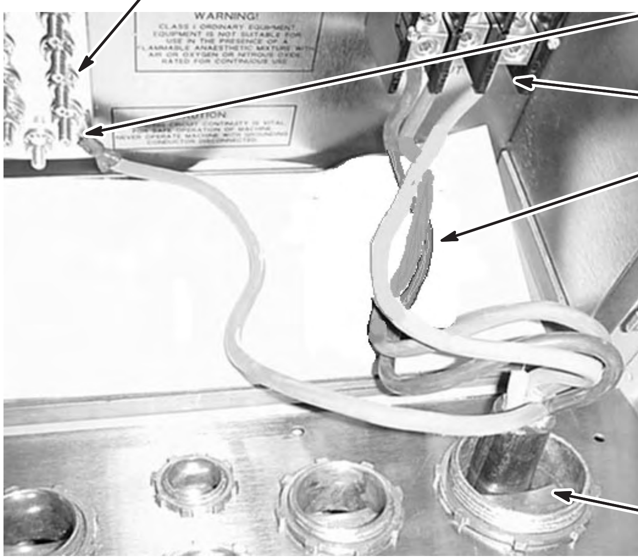
WARNING!

IF 3 PHASE WYE WITH NEUTRAL AND GROUND (5 WIRE SYSTEM) INPUT IS USED THE NEUTRAL MUST BE TERMINATED INSIDE THE MAIN DISCONNECT CONTROL AND <u>NOT</u> BROUGHT TO THE ACGD/PDU CABINET.

1 Remove rear PDU module covers.



6 Connect RF screen room ground conductor to Ground Bus Bar.



5 Connect facility input dedicated <u>ground</u> conductor to Ground (GRN/YEL) Bus Bar.

4 Connect L1–L3 conductors to terminal block.

3 **NOTE:** Air flow thru cabinet must exit thru top of cabinet. If not, fan at top of cabinet is rotating in wrong direction. To reverse rotation of fan, switch any two of the three incoming L1–L3 power wires.

DANGER!!

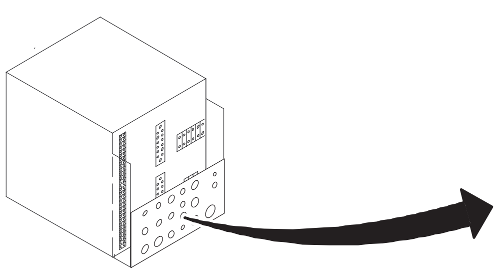
MAKE SURE ALL POWER TO CABINET IS OFF, LOCKED, & TAGGED BEFORE SWITCHING ANY ABOVE WIRES.

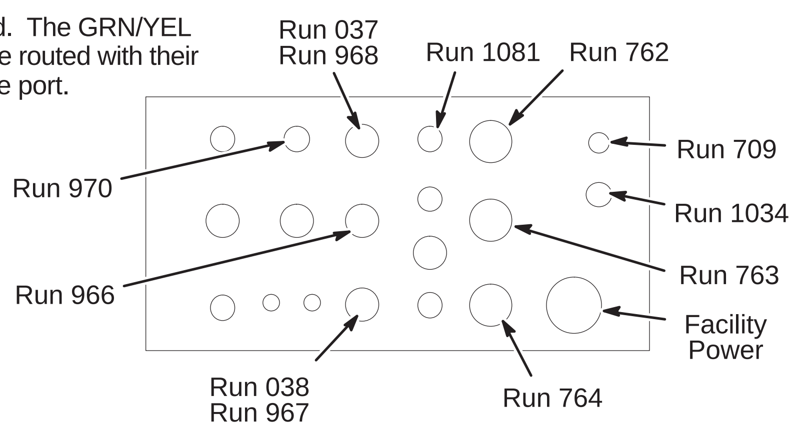
2 Route facility input power cable and ground wire thru interface panel.

10–4–6 ROUTE POWER CABLES THRU INTERFACE PANEL

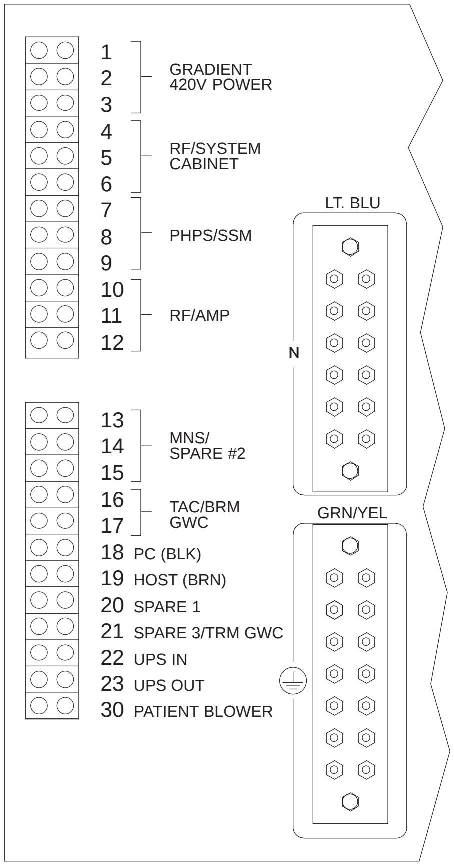
1 Route Runs thru interface panel as marked. The GRN/YEL ground wires (Runs 037 and 038) **MUST** be routed with their respective power cable Runs thru the same port.

2 Route any option Runs as marked.



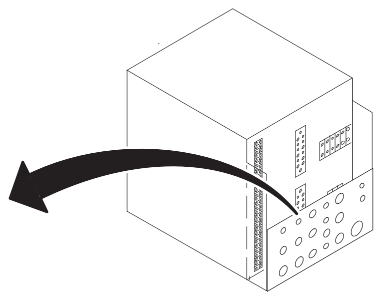


10–4–7 CONNECT POWER CABLES TO PDU MODULE



WARNING!

ALL POWER CABLES LISTED BELOW MUST BE CONNECTED TO PDU MODULE LOCATED WITHIN ACGD/PDU CABINET.



1. Route/connect power cables to pressure connectors and studs on back of PDU module as shown in Table below.
2. Trim and connect ground wire cables (Runs 037 and 038) to ground bus connectors.
3. Tighten clamps of interface panel to secure cables in place.

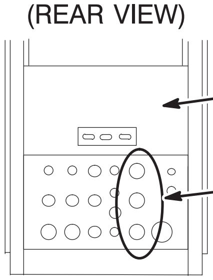
CONNECT TO	RUN	PHASE	DESTINATION
4,5,6	968	ϕ A, B, C	System Cabient 3ϕ, Neutral, Ground
7,8,9	966	ϕ A, B, C	SSM 3ϕ Neutral, Ground
10,11,12	967	ϕ A, B, C	RFI/AMP 3ϕ, Ground
16,17	970*	ϕ A,B	Water Chiller Single ϕ, Ground
18,19	1081	ϕ C,A	Operator Workspace Single ϕ, Neutral, GND
20	973*	ϕ A	Heat Exchanger Single ϕ, Neutral, GND
GRN/YEL	037		System Cabinet Ground Wire
GRN/YEL	038		SRF Cabinet Ground Wire

COLOR CODE: for remainder of cables: 3ϕ= Black, Red, Orange
Single ϕ = Brown, Black
Neutral = Light blue Gnd = Grn/Yel

* Non–WideOpen sites

10–4–8 GRADIENT OUTPUT CABLE INTERFACE PANEL ACCESS

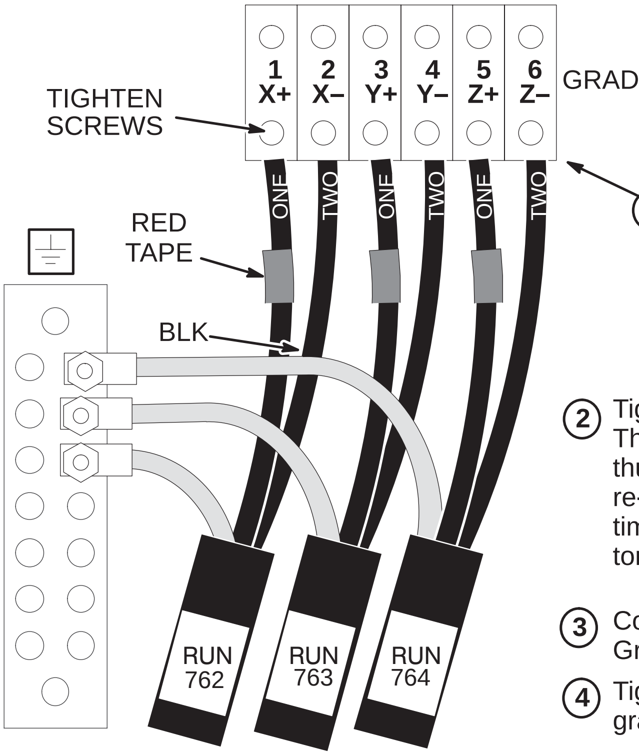
HFD CABINET
(REAR VIEW)



NOTE: Output cables (Runs 762, 763, and 764) were previously routed, cut to length, and terminated.

1. Remove Rear Cover Plate by removing seven screws that hold it in place.
2. Loosen screws on clamps for gradient output cables. Insert ends of gradient output cables into Cabinet Interface as marked. (Runs 762 into J3, Run 763 into J4, and Run 764 into J5)

10–4–9 GRADIENT OUTPUT CABLE ROUTING

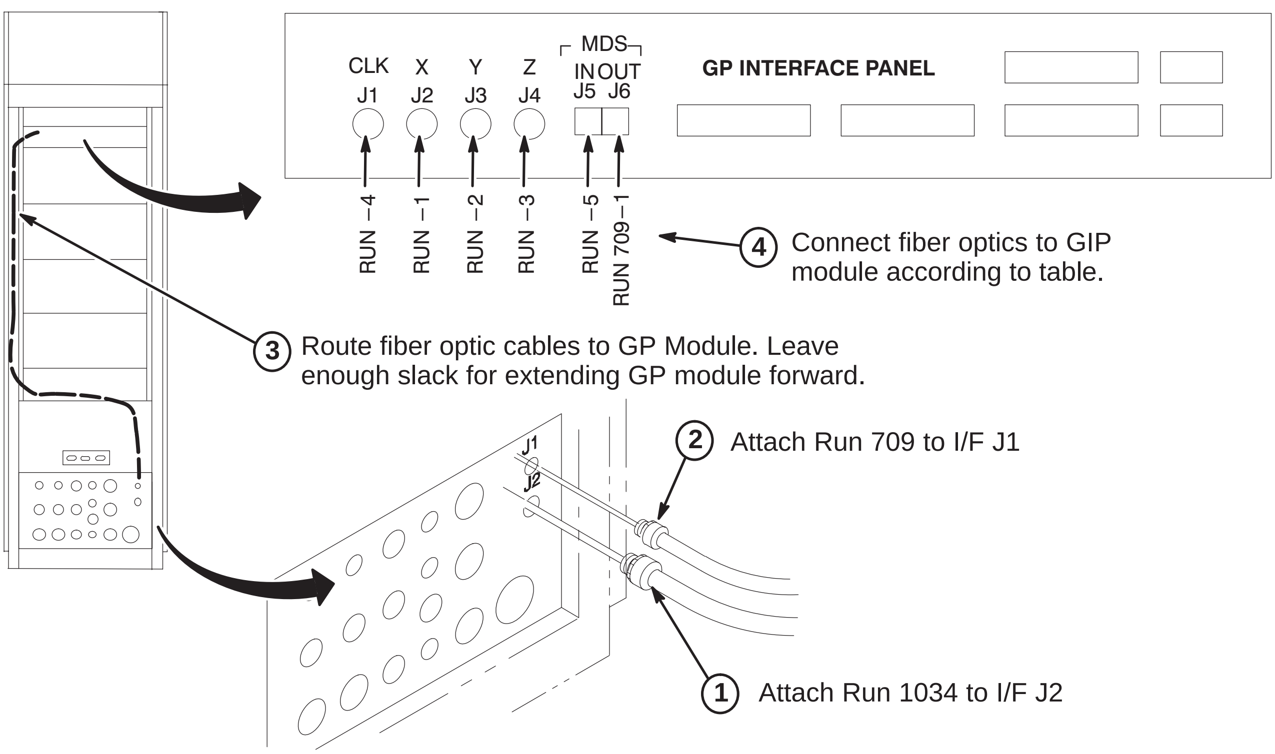


1. Connect gradient output cables with red–taped (+) #1 wires to + output terminals and black (–) #2 wires to – output terminals.

   NOTE: OVER TIGHTENING OF SCREWS IN STEP 2 CAN DAMAGE TERMINAL BLOCK.

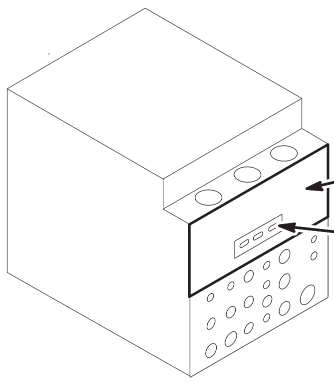
2. Tighten screws to <u>52–65 inch–lbs (5,88–7,34 N–M).</u> This is about what can be applied using only your thumb and little finger on wrench. Loosen 1/2 turn and re-tighten to the same torque. Repeat approximately 6 times or until wrench returns to the same spot after torquing.
3. Connect the GND leads of each run to the Ground Bus Bar studs.
4. Tighten clamps on Interface Panel to secure gradient output cables.

10–4–10 CONNECT FIBER OPTICS TO GP MODULE



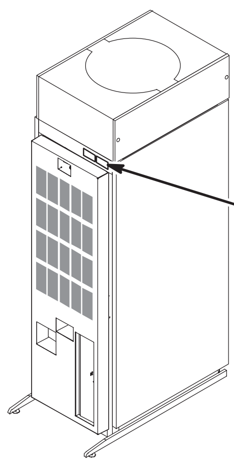
4. Connect fiber optics to GIP module according to table.
3. Route fiber optic cables to GP Module. Leave enough slack for extending GP module forward.
2. Attach Run 709 to I/F J1
1. Attach Run 1034 to I/F J2

10–4–11 CONNECT RUNS 971 TO PDU MODULE



1. Attach rear PDU module covers removed in Procedure **B1.**
2. Connect Run 971 to J2.

10–4–12 ATTACH INSITE CABINET MAGNETS



1. Attach front and rear covers.
2. Attach the “Gradient” cabinet magnet (46–320095P7) and the “PDU” cabinet magnet (46−320095P4) to front of cabinet as shown. These were transferred from the ACGD –Lite Cabinet.

SECTION 11 – COMPLETION

11–1 MECHANICAL INSTALLATION COMPLETION

The following steps need to be reviewed and completed by GE Field Engineer, especially if the system was installed by non–GE personnel. Check off boxes at the front of each item below when complete.

- ❑ Check mechanical flowchart to make sure all steps are complete (Section 2)
- ❑ Resolve shipment shortages
- ❑ Resolve ommissions by mechanical contractors
- ❑ Make sure all rating plates are installed
- ❑ Recheck all wiring connections using cable map (Section 4)
- ❑ Correct for wiring errors if necessary
- ❑ Complete Sections 11–2, 11–3, and 11–4.

11–2 RETURNING SHIPPING EQUIPMENT

The shipping material must be returned **as soon as possible.** Prompt return is required so that production shipments can continue with these labor saving dollies. These can be included with the obsolete equipment and materials being returned in Section 11–3. Recycling Operation will route them to appropriate location.

11–3 MATERIAL DISPOSITION GUIDANCE

Refer to Section 1–7. The equipment or material being disposed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

11–4 FINAL COORDINATION

1. Replace cabinet covers that have not been previously replaced.
2. Replace Magnet Enclosure covers and Rear Pedestal covers
3. Dock Patient Transport in position at front of bore.
4. Record and enter applicable data into applicable site configuration files and records.
5. Complete Product Locator information. See Section 1–8.

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

6. Leave the site's set of Manuals and CD–ROM at site. Organize and set up reference cabinet for the Manuals.
7. Locate any new Material Safety Data Sheets (MSDS). They must be retained on site. Customer is to be informed that material with MSDS information was brought into site and customer should know/decide where MSDS be retained.
8. Verify that coordination of application support for instruction of site users on Software Release 11.x has been made before returning system to the users. Turn site over to applications who will instruct users.

11–5 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Refer to the CD–ROM, Installation Calibration Flow, and perform the following:
   - PDU Voltage Checks Compact > PDU E–Stop Check
   - Startup Checks Compact
   - Transformer Taps Compact
4. Press the FULL ON button on the PDU front control panel.
5. Except for the ACGD Gradient power modules, switch all cabinet circuit breakers to ON. The ACGD Gradient power module circuit breakers should remain OFF for one hour after HVAC power up to allow room humidity to stabilize.
6. **Make sure the UPS circuit breaker is switched to the ON position.** This is required to feed power to the Operator Workspace.